BURDOCK GROUP
CONSULTANTS

NOTICE TO US FOOD AND DRUG ADMINISTRATION OF THE CONCLUSION THAT THE INTENDED USE OF A *PICHIA PASTORIS* BASED FOOD INGREDIENT CONTAINING SOY LEGH PROTEIN FOR HUMAN CONSUMPTION IS GENERALLY RECOGNIZED AS SAFE (GRAS)

Submitted by the Notifier:
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Prepared by the Agent of the Notifier:
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June 20, 2024

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Part 1: Signed Statements and Certification

1.1 Basis of Conclusion

Impossible Foods Inc. (hereinafter referred to as IFI), on the advice of a Panel of qualified experts, has concluded that soy leghemoglobin preparation (LegH Prep) is Generally Recognized As Safe (GRAS) in accordance with 21 CFR Part 170, Subpart E, and, therefore, exempt from the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act, under the conditions of its intended use as described below. The basis for this finding and supporting information is provided below.

1.2 Name and Address of the Notifier and Agent of the Notifier

Notifier

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Director, Regulatory Affairs

Impossible Foods Inc.

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Additional Agent Representative

1.3 Name of the Substance

The name of the substance of this GRAS conclusion is soy leghemoglobin preparation (LegH Prep). LegH Prep is produced from the root nodules of soy whose genus and species is *Glycine max*. The gene that encodes LegH protein is incorporated into the genome of the production organism, *Pichia pastoris* (now renamed *Komagataella pastoris*) (Karbalaei *et al.*, 2020), which produces commercially viable quantities of LegH in the form of LegH Prep. LegH Prep contains at least 65% LegH protein, with the remaining protein content being derived from *P. pastoris*.

1.4 Intended Conditions of Use

LegH Prep is an ingredient to be used as a flavor and nutrient component, which provides iron in plant-derived meat analogue products. This GRAS notification will discuss LegH Prep, which has comparable LegH protein and overall protein content, other macronutrient content, and is manufactured by the same methodology as described for LegH Prep in GRN No. 737 (FDA, 2017b), with improved process modifications. LegH Prep is a nutritive, plant-sourced flavor and nutrient component, intended to be used as an ingredient in meat analogue and meat analogue-containing products, including but not limited to beef (ground and unground), pork, lamb, goat, game, liver and organ meats, chicken, turkey, duck and other poultry, fish, cold cuts and cured meats, bacon, frankfurters, sausages, meat mixed dishes, poultry mixed dishes, seafood mixed dishes, egg rolls, dumplings, sushi, burritos and tacos, nachos, Mexican dishes, pizza, burgers, frankfurter sandwiches, chicken/turkey sandwiches, egg/breakfast sandwiches, seafood sandwiches, other sandwiches, soups, nutrition bars, protein and nutritional powders, and cookie and brownie food products, consistent with the uses provided in 21 CFR §170.3(n)(33) "Plant protein products, including the National Academy of Sciences/National Research Council "reconstituted vegetable protein" category, and meat, poultry, and fish substitutes, analogues, and extender products made from plant proteins" with the exception of infant formula or those foods with a standard of identity. IFI does not intend to add LegH Prep to any meat and/or poultry products that come under USDA jurisdiction. Therefore, 21 CFR §170.270 does not apply.

The proposed use is substitutive for existing meat analog products, rather than additive in use. LegH Prep is to be added to meat analogue and meat analogue-containing products identified herein (Table 5), such that the 90th *percentile* consumption from all categories may be up to and including 2,425.8 mg LegH Prep dry solids/day, approximately equivalent to 40.43 mg/kg bw/day.

1.5 Statutory Basis for GRAS conclusion

This GRAS conclusion is based on scientific procedures in accordance with 21 CFR §170.30(a) and (b).

1.6 Not Subject to Premarket Approval

LegH Prep is not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on the conclusion that the notified substance is GRAS under the conditions of its intended use.

1.7 Data and Information Availability Statement

The data and information that serve as the basis for this GRAS conclusion will be available for review and copying during customary business hours at the office of Burdock Group, 859 Outer Road, Orlando, FL 32814, or will be sent to FDA upon request.

1.8 Exemption from Disclosure under the Freedom of Information Act

None of the data and information in Parts 2 through 7 of this GRAS notice are considered exempt from disclosure under the Freedom of Information Act (FOIA). They do not contain any privileged or confidential information such as trade secrets and/or commercial or financial information and can be made publicly available.

1.9 Certification

The undersigned authors of this document— a notice of the GRAS conclusion for the use of soy leghemoglobin preparation (LegH Prep)—hereby certify that, to the best of their knowledge, this GRAS notice is based on a complete, representative, and balanced submission that includes both favorable and unfavorable information, that are pertinent to the evaluation of the safety and GRAS status of LegH Prep under the condition of its intended use.

Erik Hedrick, Ph.D. – Agent to the Notifier	**Date**

Associate Scientist, Burdock Group
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Orlando, FL 32814

Part 2: Identity, Method of Manufacture, Specifications, and Physical or Technical Effect

2.1 Identity of LegH Prep

LegH Prep is a preparation that contains the soy leghemoglobin protein (LegH), produced in the root nodules of soy (*Glycine max*). The LegH gene is incorporated into the genome of the production organism, *Pichia pastoris,* which produces commercially viable quantities of LegH in LegH Prep, the latter being the isolate of *P. pastoris,* containing the LegH. LegH Prep contains at least 65% LegH protein, with the remaining protein content being derived from *P. pastoris*. LegH Prep is an ingredient to be used as a flavor and nutrient component, which provides iron in plant-derived meat analogue products. This GRAS notification discusses a form of LegH Prep, which has comparable LegH protein and overall protein content, other macronutrient content, manufactured by the same methodology as described for LegH Prep in GRN No. 737 (FDA, 2017b), with improved process modifications. The purpose of the GRAS that is the subject of this notification is to establish the safety of LegH Prep when incorporated up to 2.0% into an expanded list of plant-based meat analogue products.

LegH Prep will be incorporated into plant-based meat analogue products and will impart a meat-like flavor by the incorporation of the LegH protein, by virtue of the LegH protein heme cofactor. Leghemoglobin proteins bind oxygen for respiration necessary in the process of nitrogen fixation, which is comparable to the function of myoglobin proteins (Becana *et al.*, 1995; Kosmachevskaya *et al.*, 2021). In 2017, FDA issued a 'no questions' letter in response to GRN 000737, "GRAS notification for soy leghemoglobin protein preparation derived from *Pichia pastoris*" (FDA, 2017b). The LegH Prep that is in the focus of this GRAS notification has a composition very comparable to the original LegH Prep; however, the amount used in food (up to 2.0% as compared to 0.8% for the LegH Prep in GRN No. 737) and to which foods LegH Prep is added has changed.

2.2 Production organism identification

P. pastoris is a facultative methylotrophic yeast that can utilize multiple energy sources, such as single-carbon substances (*e.g.*, methanol or methane) or multi-carbon substances with no carbon-carbon bonds (*e.g.*, dimethylamine or dimethyl ether). *P. pastoris* is also able to produce proteins with disulfide bonds and conduct glycosylation. Glycosylation is the addition of monosaccharides to the amide nitrogen of asparagine residues (*N*-linked glycosylation) or to the hydroxy group of threonine/serine residues (*O*-linked glycosylation). *P. pastoris* also expresses the protein disulfide isomerase (PDI), which preserves disulfide bonds (Prattipati *et al.*, 2020). These are common post-translational modifications that are vital to produce fully functional proteins in the production of protein-based fermentation products (Bretthauer and Castellino, 2000; Vogl *et al.*, 2013). *P. pastoris,* recently reclassified as *Komagataella pastoris* and *K. phaffii* (Kurtzman, 2009), is a cream-colored, spheroidal to ovoidal methylotrophic yeast, which can occur singly or in pairs, which can utilize methanol as a carbon source, and can grow at a high cell density (Prielhofer *et al.*, 2015; Karbalaei *et al.*, 2020). *P. pastoris* has been widely utilized in biochemical and biotechnical purposes as an expression system through expression vectors to produce heterologous proteins and as a model organism for genetic studies. *P. pastoris* is an excellent protein expression system due to the presence of an endoplasmic reticulum to ensure proper protein folding and proper secretions by the Kex2 signal peptidase (Karbalaei *et al.*, 2020). Moreover, *P. pastoris* produces a limited number of secretory proteins; therefore, the purification of the heterologous protein is simplified and more convenient. *P. pastoris* can reproduce asexually and sexually by budding or through an ascospore, respectively. The American Type Culture Collection (ATCC), an internationally recognized depository for microorganisms, has recognized *P. pastoris* as having a biosafety level 1, a category utilized for well-characterized cultures not known to cause disease in healthy human adults (ATCC, 2022). Additionally, the production strain of *P. pastoris* and the subject of this dossier, has been deposited into the ATCC.

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2.3 Description and specifications

The LegH Prep described in the GRAS that is the focus of this notification (the same as the LegH Prep in GRN No. 737) will be incorporated as an ingredient into IFI's beef, pork, poultry and other meat analogue and meat analogue containing products. LegH Prep is standardized to contain between two *percent* (2%)[1] and fifteen *percent* (15%) soy leghemoglobin protein and may be stabilized with suitable food-grade substances such as sodium ascorbate or sodium chloride. The LegH content within IFI LegH Prep has been increasing over time, which is due to continuous improvement efforts over more than ten years. These improvements in the manufacturing process, include better control of fermentation parameters, including aeration, mixing, pH, and temperature control. There have also been increases (~10%) in productivity which have been the direct result of slight modifications to fermentation media and adjustment of existing input levels without the addition of novel components to the media. In some cases, the removal of additional water by ultrafiltration can also increase LegH protein concentration. The LegH Prep is added into the final product process, adjusted for protein content to ensure that the minimum amount of LegH necessary for the intended effect is present in the final consumer food product, so the higher the titer, the less LegH Prep is required to achieve the desired LegH protein content in the selected food products.

IFI measured LegH protein levels using ultra performance liquid chromatography (UPLC). LegH protein purity was measured by IFI using sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE), Coomassie[2] staining, and gel densitometry. Proximate composition and heavy metal composition of LegH Prep were ascertained by Silliker Inc. (Salida, CA), Heavy metal analysis was conducted using an AOAC method (2015.01Mod<2232>) and were within stated specifications [lead (0.4 ppm), arsenic (0.05 ppm), mercury (0.05 ppm), and cadmium (0.2 ppm)]. Microbiological analysis was conducted by ascertaining the aerobic plate count (APC, < 10,000 CFU[3]/g) [AOAC method, MB073.06 (AOAC-RI PTM 051702)] and confirming the absence of *Salmonella* [AOAC method, MB217.01 (AOAC-RI # 100701)], *Listeria monocytogenes* [AOAC method, MB316.01 (AOAC-RI # 021201), and *E. coli* O157:H7 (AOAC method, MB217.01 (AOAC-RI # 100701)], which were all confirmed to be within required specifications. The average and range of listed specifications for LegH Prep are provided in Table 1. The specifications for the five lots of LegH Prep and their respective batch analyses, are provided in APPENDIX II confirming that the batches of LegH Prep meet stated specifications.

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[1] The original GRAS conclusion stated a specification for LegH protein was 6 – 9% in the LegH Prep.
[2] Also spelled "commassie".
[3] Colony forming units.

Table 1. Specifications for LegH Prep

Analysis	Method	Specification	AVG	Range
Soy LegH Protein (*w/w*)[1]	HPLC/UPLC*	6 – 15%	11.68%	8.0 - 10.8%
Soy LegH Protein Purity (*w/w*)[2]	SDS-PAGE*	≥65%	85.2%	79 - 92%
Total Protein (*w/w*)[3]	AOAC 991.20.I	> 10%	12.96%	11.40 - 14.78%
Fat (*w/w*)	AOAC 989.05	≤2%	0.10%	0.02 - 0.18%
Carbohydrates (*w/w*)	Calculation (Ref AOAC 986.25)	≤6%	2.41%	1.54 - 3.90%
Ash (*w/w*)	AOAC 945.46	≤4%	1.25%	0.9 - 1.70%
Solids (*w/w*)[4]	Calculated as 100 - Moisture	≤26%	16.92%	14.6 - 20.1%
Moisture (*w/w*)	AOAC 948.12	≥74%	83.10%	79.9 - 85.4%
pH	Internal method	6.0 – 9.7	7.7	6.5 - 9.4
Lead (ppm)	AOAC2015.01Mod<2232>	<0.4	<0.01	<0.01
Arsenic (ppm)	AOAC2015.01Mod<2232>	<0.05	0.012	<0.01-0.02
Mercury (ppm)	AOAC2015.01Mod<2232>	<0.05	<0.005	<0.005
Cadmium (ppm)	AOAC2015.01Mod<2232>	<0.2	<0.001	<0.001 - 0.001
E. coli 0157:H7 (negative *per* 25 g)	MB217.01 (AOAC-RI # 100701)	Absent by test	Absent	Absent
Salmonella spp. (negative *per* 25 g)	MB316.01 (AOAC-RI # 021201)	Absent by test	Absent	Absent
Listeria monocytogenes (negative *per* 25 g)	MB217.01 (AOAC-RI # 100701)	Absent by test	Absent	Absent

*Internal Method

[1]Soy leghemoglobin (LegH) protein may exceed 15% if additional water (moisture) is removed during the concentration step of the manufacturing process. Additional concentration (*i.e.*, less water) does not change the composition of the solids when dry.

[2]The balance of the proteins in the preparation is residual *Pichia* proteins.

[3]Estimated *via* Kjeldahl analysis for total nitrogen content.

[4]*Percent* solids specification is based on the sum of the maximum concentrations of total protein, fat, carbohydrates, and ash. Maximum total protein was calculated as the maximum soy leghemoglobin protein concentration divided by the minimum soy leghemoglobin protein purity.

; AVG = average; CFU = colony forming units; HPLC = high performance liquid chromatography; mM = millimolar; ppm = parts *per* million; SDS-PAGE = sodium dodecyl sulfate polyacrylamide gel electrophoresis; *w/w* = weight/weight; UPLC = ultra-performance liquid chromatography.

IFI reanalyzed the LegH Preparation prior to scaling up production in a large-scale facility in 2017, utilizing the Hazard Analysis and Critical Control Points (HACCP) (21 CFR §120[4]) approach to assess food safety risks associated with the manufacturing process and the product. The assessment was to evaluate any potential physical, chemical and biological hazards that may be associated with the LegH Prep using the material, process and equipment (including the environmental aspects) of the facility. The biological hazards assessment focused on the FDA's guidance concerning bacterial pathogens that may be associated with foods or food processing and equipment operation, which could cause consumer illness or disease. Other biological hazards, such as viruses and parasites, would generally be addressed by following current Good Manufacturing Practices (cGMP) implemented in the facility (21 CFR §112[5]). IFI has implemented several control steps to its preexisting programs to control bacterial pathogens, including *Salmonella* sp., *Listeria monocytogenes* and Shiga-toxin producing *Escherichia coli* (STEC) during the manufacturing process (Table 2).

[4] https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-120; site visited on January 20th, 2023.

[5] https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-112; site visited on January 20th, 2023.

Table 2. Process controls to produce LegH Prep by IFI

Ingredient Test	All suppliers have gone through the Impossible Foods Supplier Approval Program; All ingredients are tested for pathogenic bacteria on a lot basis; COAs are reviewed before each production usage.
Process Control	Processing temperature is generally maintained at <7°C downstream of fermentation. The post fermentation processing of the LegH protein uses suitable process controls to ensure microbial reduction. These controls can vary as necessary for site specific requirements, allowing for operational flexibility in sourcing the ingredient at scale. Process controls to reduce microbial levels may include steps such as microfiltration membranes, pasteurization or other suitable food grade technology. Temperature control such as freezing is maintained from the point of packaging to the point of usage.
Equipment and Facility Design	All direct food contact equipment is made of suitable food contact material, easy to clean, and meet the sanitary design requirements. Systems are designed based on risk assessment for food safety. The upper production stream, downstream and the final product packaging are designed to prevent cross contamination.
Environmental Control	Positive air flow is maintained in direct product exposure areas; environmental monitoring for pathogens is on a monthly basis. A CIP system is in place and has been verified for effectiveness. Daily sanitation is scheduled for non-CIP areas.
Employee Hygiene	All employees receive training on cGMPs, Food Defense, HACCP and Allergens on an annual basis; A background check program has been implemented.
Water Test	Quarterly water test for total coliform and *E. coli* has been instituted.
Final Product Testing	All final product is on positive release and the microbiological tests include assaying for *Salmonella*, EHEC, and *L. monocytogenes* on each lot.

cGMP = current Good Manufacturing Practice; CIP = Clean in place; COA = Certificate of Analysis; EHEC = enterohemorrhagic *Escherichia coli*; HACCP = Hazard analysis and critical control points.

As a result of the biological assessment under the Preventive Controls regulation (21 CFR §117[6]), IFI has production process controls in place that significantly reduce the potential for microbial contamination, in conjunction with testing for specific pathogens. Therefore, analysis for Aerobic Plate Count (APC) and Yeast/Mold counts were concluded by IFI as not needed as product release specifications. APC and Yeast/Mold counts are only evaluated as statistical process controls on an "as needed" basis.

The composition of the original LegH Prep (GRN No. 737) was compared with the LegH described in this GRAS notification (Table 3). Carbohydrate content was determined by subtracting *percent* protein, *percent* ash, and *percent* fat from the total *percent* solids for each lot. The average carbohydrate content that was calculated for the LegH Prep was within stated

[6] https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-117?toc=1; site visited on February 13th, 2023.

specifications (≤6%), with an average value of 2.41% ranging from 1.54 - 3. 90% (Table 1). The protein content (overall) that was analyzed for the LegH Prep was within stated specifications (≤16%), with an average value of 12.96% ranging from 11.40 - 14.78%, as determined by the Kjeldahl method, which determines total nitrogen and utilizes a standard factor (6.25) to calculate protein content (Table 1). The compositional analyses of this LegH Prep and the original LegH Prep (GRN No. 737) reveal that both versions of LegH Prep have similar compositions. As demonstrated, the fat, carbohydrate, ash and moisture contents are similar between all LegH Prep lots analyzed (Table 3). The proximate analysis results of the LegH Prep show that the average of the overall protein content (12.96%) and *P. pastoris* protein content (1.67%) is comparable to the original LegH Prep product (11.36%) and *P. pastoris* protein content (1.67%) (Table 3). The average fat content in the LegH Prep (0.10%) is over three times higher than the average fat content of the original LegH Prep product (0.03%); however, LegH Prep still meets the specification for less than 2% fat. While there were lots from the LegH Prep that exhibited higher carbohydrate content then the original LegH Prep, the average carbohydrate content (2.41%) was comparable to the original LegH Prep product (1.82%), with the average being reduced by approximately 0.6%. The ash content in the LegH Prep (1.25%) is approximately reduced by 0.25% as compared to the original LegH Prep Product (1.5%). These analyses show that the macronutrient content of this LegH Prep, described in this GRAS notification, is comparable to the macronutrient content of original LegH Prep product.

The protein determination *via* the Kjeldahl method correlates well with the amino acid analysis and the overall quantitative analysis of protein, carbohydrate, and fat. The standard factor of 6.25 used in the Kjeldahl methods is generally known to be a small overestimation of protein content in plants and some yeasts materials, compared to animal tissues. This is due to the fact some plants and yeasts contain higher levels of nonprotein nitrogen sources, including peptides too small to be precipitated and filtered, amides, free amino acids, nucleic acids and other non-polymeric nitrogen sources (Abdel-Hafez *et al.*, 1977; Imafidon and Sousulski, 1990). LegH protein purity was measured *via* SDS-PAGE, as SDS-PAGE is a widely accepted method (Oliveira and Domingues, 2018) to determine purity of specific proteins and, is consistent with IFI measurements reviewed by the FDA through the GRAS notification process and the color additive petition (CAP) that was approved by FDA[7], as well as other international filings. Overall, these studies indicate that the general compositional components contained within LegH Prep described within this GRAS notification are consistent with the compositional components found in the original LegH Prep product (GRN No. 737).

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[7] https://www.federalregister.gov/documents/2019/12/19/2019-27173/listing-of-color-additives-exempt-from-certification-soy-leghemoglobin; site visited December 8, 2023.

Table 3. Proximate mass balance of protein, ash, carbohydrates, fat, moisture, and solids in the original LegH Prep (2017) and LegH Prep (2022)

	PP-PGM2-16-015-101	PP-PGM2-16-088-101	PP-PGM2 16-102-101	PP-PGM2 16-144-101	PP-PGM2-16-200-101	LH-20-151-152-190FG	LH-20-163-164-190FG	LH-21-227-228-190FG	LH-22-313-314-190HF	LH-22-377-378-190HG
Analysis	**Original LegH Prep (GRN No. 737, 2017)**					**LegH Prep (2022)**				
Soy Leghemoglobin Protein (*w/w*) (%)	6.74	6.39	6.28	6.74	6.95	8.5	8.0	9.9	10.8	10.6
Soy Leghemoglobin Protein Purity (*w/w*) (%)	82	71	85	77	86	79	84	91	92	80
Pichia pastoris protein (*w/w*) (%)	1.48	2.61	1.11	2.01	1.13	2.26	1.52	0,98	0.94	2.65
Protein (*w/w*) (%)	11.21	10.89	10.62	12.65	12.89	11.40**	12.13**	12.00**	14.5**	14.78**
Fat (*w/w*) (%)	0.05	<0.01	<0.01	0.03	0.08	0.18	0.1	0.1	0.10	0.02
Total Carbohydrates (*w/w*) (%)	1.72#	0.99	1.67	2.01	2.73#	1.54#	3.2	1.6	1.8	3.9
Ash (*w/w*) (%)	1.87	0.67	2.63	2.62	2.74	1.17	1.7	0.9	1.1	1.4
Moisture (*w/w*) (%)	85.15	87.45	85.08	82.69	81.56	84.75	82.87	85.4	82.6	79.9
Solids[8] (*w/w*) (%)	14.85	12.55	14.92	17.31	18.44	15.25	17.13	14.6	17.5	20.1
Moisture + solids (summed) (%)	100.00	100.00	100.00	100.00	100.00	99.90	100.0	100.0	100.0	100.0

#Direct carbohydrate measurement.
*Moisture content determined *via* AOAC methods 925.09 and 926.08 (modified) by vacuum drying.
**Kjeldahl method utilized.
w/w = weight/weight.

[8] *Percent* solids specification is based on the summation of fat, carbohydrates, ash and protein.

2.4 Manufacturing Process

The production organism for LegH Prep is the yeast *Pichia pastoris*, the same production organism described in GRN No. 737 that received a "no questions" letter from FDA (FDA, 2017b). LegH Prep is prepared by construction of the production strain of *Pichia pastoris*, expression of the LegH protein when in submerged fermentation, enrichment, and stabilization of the expressed LegH protein. The genes utilized to express the LegH protein and the methodology to insert the genes into the *P. pastoris* yeast have not been altered from the original conclusion of GRAS (GRN No. 737) status that was notified to FDA (FDA, 2017b); therefore, a comprehensive description of the genetic modifications to the *P. pastoris* yeast to produce the LegH protein will not be described in detail here. To summarize the manufacturing process for the LegH Prep and the LegH protein, the production strain *P. pastoris* MXY0291 was constructed from *P. pastoris* BG11 (MXY0051) using a series of transformations to express the LegH protein (*Glycine max* gene LGB2). The genome of MXY0291 is fully sequenced and well-characterized (FDA, 2017b), and a reference genomic sequence of *P. pastoris* has been constructed that utilized in part strain BG10, a parental strain of the production organism (Sturmberger *et al.*, 2016). The *P. pastoris* production organism meets the criteria for a safe production microorganism (Pariza and Johnson, 2001) and has a qualified presumption of safety (QPS) for biological agents intentionally added to food (EFSA, 2017). Moreover, all raw materials used to produce LegH Prep are standard food grade or pharmaceutical grade ingredients or of a purity and quality suitable for their intended use (Aunstrep *et al.*, 1979; Enzyme Technical Association, 2005; Taylor and Baumert, 2013) and processed under conditions appropriate for food production under cGMP.

2.4.1 Raw materials

As *per* the original notification of GRAS status (FDA, 2017b), the raw materials that are used in the fermentation, production, and recovery process for LegH Prep are standard food grade or pharmaceutical grade, standard ingredients used in the food/enzyme industry, adhere to internal specifications [consistent with Foods Chemical Codex, 13th Edition requirements (United States Pharmacopeial Convention, 2022)], and of a purity and quality suitable for their intended use. The specifications for these raw materials include limits on lead and other pertinent heavy metals.

2.4.2 Fermentation

LegH protein from the *P. pastoris* MXY0291 production strain (or derivatives of MXY0291) is expressed during submerged fed-batch fermentation. Frozen cell banks for the production organism are maintained at -80°C in 20% *v/v*[9] glycerol, which are the source inoculum for LegH production. The master cell bank is stored at multiple locations. Working cell banks are prepared from the master cell bank and are tested for microbial purity, specific growth rate, and LegH yield prior to production fermentation. The fermentation broth is also periodically analyzed microscopically to ensure culture purity. Process parameters which include pH, temperature, agitation, dissolved oxygen, methanol concentration and glycerol concentration are routinely monitored throughout fermentation. Fermentations that incur microbial contamination and/or other

[9] *v/v* = volume/volume.

process deviations that affect safety and/or quality are sterilized by steam or chemical treated and discarded.

Raw materials used in the fermentation and recovery process for LegH Prep are standard ingredients used in the food/enzyme industry, and follow internal specifications, consistent with Food Chemical Codex (13th Edition). These specifications include limits on lead and other pertinent heavy metals. The raw materials are of a purity and quality suitable for their intended use (Aunstrup *et al*., 1979); they are food-grade, or high-quality chemical or pharmaceutical grades (United States Pharmacopeia [USP], National Formulary [NF], or American Chemical Society [ACS] grades) from approved suppliers.

2.4.3 Recovery process

The *P. pastoris* cells in the fermentation broth are lysed by mechanical shearing or high-pressure homogenization. Insoluble material within the lysate is removed by centrifugation and microfiltration. Ultrafiltration is used to concentrate the product and the LegH Prep manufacturing process has an added heat treatment step as a secondary step that further ensures the inactivation of any possible *P. pastoris* cells. The resulting concentrated sample is formulated with suitable food-grade substances (*i.e.*, sodium ascorbate and sodium chloride) and stored as a frozen liquid. A schematic overview of the manufacturing process is presented in Figure 1. Using an additional optional step, the LegH Prep could be subsequently spray dried within the drying chamber of an atomizer. The dried LegH powder is expelled into a cyclone and subsequently collected as dry LegH particles. Incorporation of this optional spray drying step would result in a LegH Prep that exhibits different compositional specifications, which include reduced moisture content that would subsequently increase the protein content of the LegH Prep. Therefore, less LegH Prep would be required to be added to foods to achieve the desired protein content in the selected food products. With a combination of improved LegH expression during fermentation and improved downstream processing yield, IFI increases the LegH titer on a *per*-batch basis.

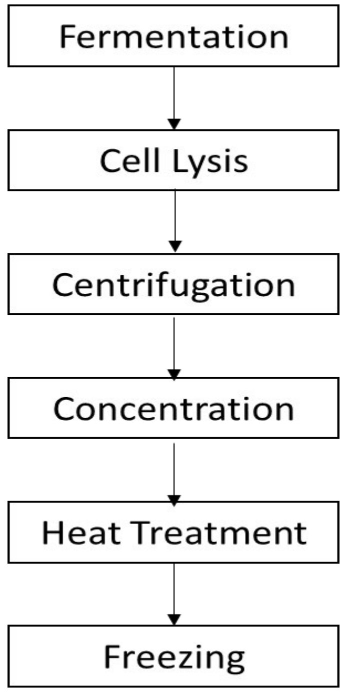

Figure 1. Schematic overview of the manufacturing process for LegH Prep

IFI tests each independent fermentation broth to ensure the absence of *Salmonella* (using MB217.01), *Listeria monocytogenes* (using MB316.01), and *E. coli* O157:H7 (using MB217.01). The final product from every LegH Prep production run is tested for *Salmonella*, *Listeria monocytogenes*, and *E. coli* O157:H7 as described above. Failure to comply with the specifications as outlined in Table 1, would result in the batch being evaluated and/or discarded, for instance in the case of the presence of a pathogen or toxic contaminant, the execution of additional sanitization standard operating procedures (SSOPs) in compliance with IFI's internal food-safety standards, and a root cause analysis.

2.5 Stability

Initial stability analyses found the LegH protein in the LegH Prep product is stable for six weeks at 4°C. The LegH protein has not been altered and the manufacturing includes an ultrafiltration and heat treatment step. The packaging conditions of this LegH Prep are equivalent to those for the LegH Prep in GRN No. 737. Therefore, it is assumed that the LegH Prep ingredient may be stored at -20°C as a frozen liquid for at least 24 months with no observable change in LegH protein stability or performance in ground beef analogue products.[10] IFI will continue to monitor stability of the LegH protein in the LegH Prep product over the stated storage life.

The LegH Prep that was the subject of the original conclusion of GRAS status notification (FDA, 2017b) was assessed for stability in a full meat analogue product. The stability of the leghemoglobin protein was evaluated over a ten-day period; the total protein was extracted and quantified from the analogue using denaturing gel electrophoresis and densitometry. No degradation products of the leghemoglobin were reported at any time point and there was no statistically significant decrease in any of the samples over the four-day period, indicating that the leghemoglobin protein is stable in a full meat analogue.

The stability of this LegH Prep is comparable to the original LegH Prep described in GRN No. 737. This LegH Prep was determined to be stable under the conditions of storage at Product Safety Laboratories (PSL) over the course of a 90-day rodent feeding study, to within an acceptable margin of variability. Neat test substance samples from the beginning of the study (Day 0), and again at the end of the study (Day 91) were analyzed for stability (Table 4). Detailed results of the extended stability of LegH Prep (30,000, 60,000, and 90,000 ppm) are in Table 4, which indicates that utilizing the highest concentration (90,000 ppm), the neat test article is stable across the 91-day storage period at -20°C. The highest concentration at the 91-day time point does pass, whereas the 30,000 and 60,000 ppm doses do not. The CV concentrations for 30,000 and 60,000 ppm at Week 13 (Day 91) did not meet the acceptance criteria of 85 – 115% (Table 4).

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[10] LegH protein is stable within Impossible Foods vacuum packed ground beef analogue product for at least 30 days at 4°C and at least 95 days at -20°C (FDA, 2017b).

Table 4. Extended stability of neat test substance

Day	Sample ID	% Titer recovery	Avg. % Titer recovery	SD / %RSD
0	NT 1 A	101.83% 100.94% 102.55%	101.8%	0.008/ 0.79%
46	NT 2 A	104.41% 100.06% 104.13%	102.9%	0.02/ 2.37%
91	NT 3 A	88.78% 85.64% 86.82%	87.1%	0.02/ 1.82%

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Part 3: Dietary Exposure

3.1 Estimated Daily Intake

The original LegH Prep ingredient was previously concluded as GRAS and received a "no questions" letter from FDA when notified (Keefe, 2018) for use in plant-derived ground beef analogue products that provide consumers an alternative to beef-containing products. For the current notification of conclusion of GRAS status for this LegH Prep, the intake profile (amount and frequency) by individuals in USDA's What We Eat in America (WWEIA) Continuing Survey of Food Intakes by Individuals 2017-2018 (CDC, 2017; 2020) was used to calculate the estimated daily intake (EDI) of LegH Protein for individuals consuming the food groups selected for the addition of LegH Prep *per* this GRAS evaluation, that is the subject of this notification. This current LegH Prep, which is the subject of this notification, is a nutritive, plant-sourced flavor and nutrient component, intended to be used as an ingredient in meat analogue and meat analogue-containing products, including but not limited to beef (ground and unground), pork, lamb, goat, game, liver and organ meats, chicken, turkey, duck and other poultry, fish, cold cuts and cured meats, bacon, frankfurters, sausages, meat mixed dishes, poultry mixed dishes, seafood mixed dishes, egg rolls, dumplings, sushi, burritos and tacos, nachos, Mexican dishes, pizza, burgers, frankfurter sandwiches, chicken/turkey sandwiches, egg/breakfast sandwiches, seafood sandwiches, other sandwiches, soups, nutrition bars, protein and nutritional powders, and cookie and brownie food products (Table 5), consistent with the uses provided in 21 CFR §170.3(n)(33) "Plant protein products, including the National Academy of Sciences/National Research Council "reconstituted vegetable protein" category, and meat, poultry, and fish substitutes, analogues, and extender products made from plant proteins" with the exception of infant formula or those foods with a standard of identity. The addition of LegH protein to the selected categories (which represent meat analogue and meat analogue-containing products) and their respective amounts are provided in Table 5.

Table 5. Food categories (meat analogue and meat analogue containing products) selected for the addition of LegH protein

Food categories	Minimum dosage level (% final product)	Maximum dosage level (% final product)	Maximum intended use level (ppm)	Maximum intended use level (mg/g)
Beef, excludes ground	0.5	1.0	10,000	10
Ground beef	0.35	1.0	10,000	10
Pork	0.1	0.4	4,000	4
Lamb, goat, game	0.7	2.0	20,000	20
Liver and organ meats	0.5	2.0	20,000	20
Chicken, whole pieces	0	0.1	1,000	1
Turkey, duck, other poultry	0.1	0.5	5,000	5
Fish	0	0.4	4,000	4
Eggs and omelets	0	0.12	1,200	1.2
Cold cuts and cured meats	0	0.5	5,000	5
Bacon	0.2	0.5	5,000	5
Frankfurters	0.2	0.4	4,000	4
Sausages	0	0.6	6,000	6
Meat mixed dishes	0.5	1.0	10,000	10
Poultry mixed dishes	0	0.1	1000	1.0
Seafood mixed dishes	0	0.4	4000	4

Food categories	Minimum dosage level (% final product)	Maximum dosage level (% final product)	Maximum intended use level (ppm)	Maximum intended use level (mg/g)
Egg rolls, dumplings, sushi	0	0.20	2000	2
Burritos and tacos	0	0.20	2000	2
Nachos	0	0.20	2000	2
Other Mexican mixed dishes	0	0.20	2000	2
Pizza	0	0.17	1,700	1.7
Burgers (single code)	0	0.60	6,000	6
Frankfurter sandwiches (single code)	0	0.24	2,400	2.4
Chicken/turkey sandwiches (single code)	0	0.3	3,000	3
Egg/breakfast sandwiches (single code)	0	0.0375	375	0.375
Other sandwiches (single code)	0	0.45	4,500	4.5
Seafood sandwiches (single code)	0	0.24	2,400	2.4
Soups	0.1	0.55	5,500	5.5
Nutrition bars	0	2.0	20,000	20
Cookies and brownies	0	2.0	20,000	20
Protein and nutritional powders	0.15	2.0	20,000	20

*The food categories correspond to those listed in the WWEIA 2017-2018 database: https://www.ars.usda.gov/ARSUserFiles/80400530/pdf/1718/Key%20Points%20Using%20WWEIA%20NHANES%202017-2018.pdf; site visited October 3rd , 2023.
ppm = parts *per* million.

The mean and 90th *percentile* EDIs were calculated at 757.75 g/day and 1788.0 g/day, respectively, for LegH Prep intake following addition of the LegH Prep to the selected categories described in Table 5. On a body weight basis, mean and 90th *percentile* weighted LegH protein was calculated at 12.01 and 27.82 mg/kg bw/day, respectively (Table 6).

Table 6. Estimated Daily Intake (EDI) of LegH protein following incorporation of LegH Prep into selected foods

	mg/day	mg/kg bw/day
Mean (weighted)	757.75	12.01
90th *Percentile* (weighted)	1788.0	27.82

Table 6 represents the mean and 90th *percentile* EDI values of LegH protein, following the incorporation of hydrated LegH Prep, to achieve the desired LegH protein inclusion levels in the selected food categories (Table 5) and foods (Appendix III). To determine the amount of LegH Prep required for incorporation into the selected categories to achieve the levels of LegH protein in the specified categories, the *percentage* of LegH protein within the LegH Prep must be considered. The LegH protein represents approximately 11.7% of the total LegH Prep; therefore, the mean and the 90th *percentile* EDI for the LegH Prep, which would have to be incorporated to achieve the levels of LegH protein in the specified food categories (as 11.7% of LegH protein is contained in LegH Prep), would be 102.65 and 237.8 mg *hydrated* LegH Prep/kg bw/day, respectively. However, these EDI values represent the *hydrated* LegH Prep, and the NOAEL derived from the 90-day study utilized the *dried* LegH Prep (see Section 6.3 below). Therefore, to achieve the desired amounts of LegH protein within the selected foods provided in Table 5, and to get an accurate estimation of the EDI for the LegH Prep, the *percentage* of LegH protein within the dried solids (which is approximately 17% of the LegH Prep) must be considered. This would result in a new mean and 90th *percentile* EDI of LegH Prep at 17.45 and 40.43 mg/kg bw/day, which corresponds to a mean and 90th *percentile* LegH Prep intake of 1,047 and 2,425.8 mg/day, respectively, for a 60-kg individual (Table 7).

Table 7. Estimated Daily Intake (EDI) of LegH Prep following incorporation of LegH Prep (from *dried solids*) in selected foods

Product	**mg/kg bw/day**	
	Mean	**90th *Percentile***
LegH Prep (dried solids)	17.45	40.43

3.2 History of use of LegH Prep and *P. pastoris*

LegH Prep has a history of use as a flavor and nutrient component, and as an ingredient in plant-based meat analogue products to provide a meat like flavor and a source of iron. Iron is a subcomponent of the heme cofactor, which is present in LegH protein, and released when cooked. FDA has issued a 'no questions' letter in response to GRN 000737, "GRAS notification for soy leghemoglobin protein preparation derived from *Pichia pastoris*" (FDA, 2017b). The production organism, *P. pastoris,* was discovered in the 1960s and was first isolated from the exudates of a chestnut tree in France (originally named *Zygosaccharomyces pastoris* by Guilliermond) (Karbalaei *et al.*, 2020). The organism was then later categorized to the genus *Komagataella* or *Pichia* (*P. pastoris*) and has been used extensively for decades as a production organism in the commercial production of disulfide and glycosylated heterologous proteins. *P. pastoris* is commonly used in the production of recombinant proteins for biopharmaceutical, industrial and environmental applications (Balamurugan *et al.*, 2007; Ahmad *et al.*, 2014; Safder, *et al.*, 2018), such as trypsin and phytase (Ahmad *et al.*, 2014). Commercial use was first initiated by Phillips Petroleum for production of a single cell protein as an animal feed additive (Ahmad *et al.*, 2014). *P. pastoris* dried yeast was approved as a feed ingredient in 1993 for use in broiler chicken feed formulations as a source of protein and other nutrients when used at not more than 10% by weight of the total ration (FDA, 1993). *P. pastoris* has been approved by FDA *via* a food additive petition (FAP) for use at up to 10% in the diet of broiler chicken feed (FDA, 1992) to provide nutritive value to the feed. LegH Protein is also approved as a color additive (21 CFR 73.520)[11] and dried *P. pastoris* is approved up to 10% weight for use in feed formulations of broilers (21 CFR 573.750)[12].

[11] https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-73/subpart-A/section-73.520; site visited December 28th, 2023

[12] https://www.ecfr.gov/current/title-21/chapter-I/subchapter-E/part-573/subpart-B/section-573.750; site visited December 28th, 2023.

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Part 4: Self-limiting Levels of Use

The use of LegH Prep in foods is considered to be self-limiting for technological reasons, such as product texture and/or flavor profile, either of which could affect consumer acceptance.

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Part 5: Experience Based on Common Use in Food Prior to 1958

The statutory basis for the conclusion of the GRAS status of LegH Prep in this document is not based on common use in food before 1958. The GRAS assessment is based on scientific procedures**.**

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Part 6: Narrative

6.1 Absorption, distribution, metabolism, and elimination (ADME)

P. pastoris utilized in the production of the LegH Prep for this GRAS notification belongs to the family Saccharomycetaceae, which comprises of several yeast genera that are currently used in food, including *Saccharomyces*, *Torula*, *Yarrowia*, *Dekkera* and *Brettanomyces*. *P. pastoris* was also the organism utilized for the production of LegH Prep in GRN No. 737 (FDA, 2017b). *S. cerevisiae*, or Baker's yeast, is a common organism for winemaking, brewing, and baking, and *Torula* yeast (*i.e.*, *Candida utilis*) may also be safely added to food (21 CFR §172.896[13]) and is used as a fermentation starter and for the neutralization of cheese curd and galactose consumption. Moreover, *Yarrowia lipolytica* is an oleaginous yeast that is used to produce eicosapentaenoic acid (EPA) rich oil (GRN No. 355) (FDA, 2010), to produce rebaudioside A (GRN No. 632) (FDA, 2016), steviol glycosides (GRN No. 759) (FDA, 2018), and phospholipase A2 (GRN No. 940) (FDA, 2021a), all of which received a "no objection" letter from the FDA. The *P. pastoris* parent strain obtained from the ATCC (NRRL Y-11430) is classified as a "Biosafety Level 1" organism, a classification indicating that the organism is well-characterized and "not known to consistently cause disease in immunocompetent adult humans, and present minimal potential hazard to laboratory personnel and the environment" (CDC, 2009). The DNA contained within the LegH Prep product, which does not contain antibiotic resistance genes, is expected to be digested, as is typical of DNA consumed from foods (FDA, 1992; Carver and Walker, 1995; Liu *et al*., 2015). All other components (carbohydrates, fats, and proteins) contained in the LegH Prep from *P. pastoris* will exhibit comparable absorption, metabolism, and excretion that is observed for typical carbohydrates, fats, and proteins.

França *et al.* (2015) investigated *P. pastoris* (X-33) gastrointestinal resistance and metabolism in mice. Male BALB/c mice (6 - 8 weeks of age; 15 - 20 g) were inoculated with a single oral dose (by gavage) of *P. pastoris* X-33 (7 log CFU) in mice fed an antibiotic-free diet. To evaluate the resistance of *P. pastoris* X-33 to the BALB/c mouse gastrointestinal tract, feces were collected (24, 48, and 72 hours post *P. pastoris* administration), dilutions of the feces were plated on YPD agar and counts were determined. Mice that did not receive yeast served as negative controls. The authors reported that *P. pastoris* survived in the mouse gastrointestinal tract, with 6.84 log CFU/g *P. pastoris* surviving in the feces 24 hours post administration. However, the amount of surviving *P. pastoris* significantly decreased (2.87 CFU/g) after 48 hours post administration, and no growth 72 hours post administration (França *et al.*, 2015). This study demonstrates that even though *P. pastoris* does survive within the GI tract for a short amount of time, it is gradually degraded and is unable to colonize within the GI tract. Therefore, *P. pastoris* is degraded and its fragments are metabolized like other proteins, fats, and carbohydrates.

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[13] https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-172/subpart-I/section-172.896; site visited December 10, 2022.

6.2 Palatability/Subacute studies

Reyes *et al*. (2023) conducted a 14-day feeding study to evaluate the palatability and general toxicity of LegH Prep in CRL Sprague-Dawley CD® IGS rats (20 males and 20 females; 5/sex/group; body weight range within ± 20% of the mean body weight). This study was conducted in compliance with OECD 407[14] Guidelines for the Testing of Chemicals and U.S. FDA Redbook guidelines. Rats were randomly assigned to the following groups: Group 1: basal control diet; Group 2: Low Dose LegH Prep [50,000 ppm (4167 mg/kg bw/day), 5% of diet]; Group 3: Intermediate Dose LegH Prep [100,000 ppm (8333 mg/kg bw/day), 10% of the diet]; Group 4: High Dose LegH Prep [150,000 ppm (12,500 mg/kg bw/day), 15% of the diet]. The diet and filtered tap water were supplied *ad libitum.* The neat test substance was stable under the conditions of storage over the course of this study. Based on the dietary stability and homogeneity analyses, LegH Prep was homogeneously distributed and stable in the diet and animals were determined to have received their target dietary concentrations over the course of the study. Animals were observed at least twice daily for viability and abnormal/adverse events or behavior. Animals were also observed for changes in skin, fur, eyes, and mucous membranes, occurrence of secretions and excretions and autonomic activity (*e.g*., lacrimation, piloerection, pupil size, unusual respiratory pattern), changes in gait, posture, and response to handling, presence of clonic or tonic movements, stereotypies (*e.g*., excessive grooming, repetitive circling), or bizarre behavior (*e.g*., self-mutilation, walking backwards). Individual animal body weights were recorded twice during the acclimation period and then on Day 0 (prior to study start), then test Days 3, 7, 10, and 14. Individual food consumption was measured and recorded to coincide with body weight measurements. Food efficiency and dietary intake of the test substance (mg/kg/day) were also calculated and reported. Hematology parameters that were measured included hematocrit (HCT), hemoglobin (HGB), mean corpuscular hemoglobin (MCH), mean corpuscular volume (MCV), platelet count (PC), red blood cell count (RBC), red cell distribution width (RDW), reticulocyte count (RC), white blood cell count (WBC), and differential leukocyte count (DLC). Clinical chemistry parameters that were measured included albumin (ALB), alkaline phosphatase (ALP), bilirubin (BIL), creatinine (CRE), calcium (Ca), chloride (Cl), total cholesterol (TC), fasting glucose (GLU), globulin (GLO), inorganic phosphorous (P), high density lipoprotein (HDL), low density lipoprotein (LDL), potassium (K), alanine transaminase (ALT), aspartate transaminase (AST), total protein (TP), sodium (Na), sorbitol dehydrogenase (SDH), triglycerides (TG), blood urea nitrogen (BUN), and creatine phosphokinase (CP). All surviving animals were euthanized at the termination of the study and a full necropsy was performed and were then analyzed for clinical pathology. Animals were fasted at the end of the study, and then blood was collected for clinical chemistry and hematology under isoflurane anesthesia (Reyes *et al*., 2023).

There were no mortalities throughout the study. There were no adverse test substance-related clinical observations, body weight, food consumption, or food efficiency changes, or necropsy observations attributable to LegH Prep administration. The mean dietary intakes were calculated to be 4646.9, 8843.5, and 13035.9 mg/kg/day for males and 4175.9, 8686.0, and

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[14] https://www.oecd.org/env/test-no-407-repeated-dose-28-day-oral-toxicity-study-in-rodents-9789264070684-en.htm; site visited on February 6th, 2023.

12401.5 mg/kg/day for females, respectively. There were significant ($p < 0.05$) changes observed in Group 3 and 4 male daily body weight gains, which were attributed to slightly lower food consumption for these corresponding intervals. However, body weight gain changes did not affect absolute body weight and were considered not toxicologically significant. Exposure to LegH Prep in either sex for the entire 14-day test period did not induce any adverse changes in clinical chemistry nor hematological parameters. There were significant ($p < 0.05$) changes observed in mean phosphorous levels in Group 4 males and potassium levels in males in Group 3 and 4 as compared to the basal diet controls. However, these changes are not of toxicological concern as these changes were not dose-dependent nor were there clinically adverse changes or pathological manifestations associated with these significant changes. Under the study conditions utilized, the study director concluded "based on the toxicological endpoints evaluated, there were no adverse changes in this study, attributable to the dietary administration of Soy Leghemoglobin Preparation" (Reyes *et al*., 2023). Therefore, up to 12,500 mg/kg/day LegH Prep was well tolerated, palatable, and did not induce any clinically adverse effects in Sprague-Dawley rats (Reyes *et al*., 2023).

Fraser *et al*. (2018) conducted a 14-day study in 48 Sprague-Dawley rats (male and female; six/sex/group) in which the freeze-dried LegH Prep (the same LegH Prep that is the focus of this notification of the conclusion of GRAS status) was administered in the diet at concentrations (0, 3156, 6312, and 12,612 ppm), which corresponded to target concentrations of 0, 263, 525 and 1050 mg/kg bw/day consumption levels of the LegH Prep. This study was conducted in compliance with OECD 407 Guidelines for the Testing of Chemicals and Food Ingredients and U.S. FDA Toxicological Principles for the Safety Assessment of Food Ingredients[15] (IV.C.4.a) and was approved by the Institutional Animal Care and Use Committees (IACUC). This study was conducted to ascertain the feasibility of the oral administration of freeze-dried LegH Prep in the diet and to establish the dose range for the 28-day dietary study. The concentrations of LegH Prep in the diet corresponded to active LegH doses of 0 (Group 1, control), 125 (Group 2, low), 250 (Group 3, medium), and 500 (Group 4, high) mg/kg bw/day, respectively. A significant amount (approximately 50%) of the LegH Prep dry solids is *P. pastoris* (based on 6.5% LegH protein, 0.58% NaCl and the LegH Prep solids being 13.7% of the non-dried LegH Prep. The animals were targeted to consume approximately 0, 132, 263, and 525 mg/kg bw/day *P. pastoris* solids. The feed formulation was held constant throughout the study, and filtered tap water and animal feed were supplied *ad libitum*. The calculated dietary intake levels were 0, 134, 269, and 531 mg/kg/day for Groups 1 - 4 male rats and 0, 148, 296, and 592 mg/kg/d for Groups 1 - 4 female rats, respectively.

The animals were observed for individual food consumption (based on total food consumption divided by two because of paired housing) and was recorded along with individual body weights. The animals were also observed for viability, signs of gross toxicity, and behavioral changes at least once daily during the study and weekly for a battery of detailed observations. At the end of the study, blood samples were obtained for clinical chemistry and hematological analyses. The clinical chemistry parameters included AST, ALT, SDH, ALP, BIL, BUN,

[15] https://www.fda.gov/files/food/published/Toxicological-Principles-for-the-Safety-Assessment-of-Food-Ingredients.pdf; site visited on January 18th, 2023.

creatinine, TC, TG, GLU, TP, ALB, GLO, Ca, P, Na, K, and Cl. The hematology parameters included RBC, HGB, HCT, MCV, MCH, MCHC, RDW, PC, WBC, absolute neutrophil (ANEU), absolute lymphocytes (ALYM), absolute monocytes (AMON), absolute eosinophils (AEOS), absolute basophils (ABAS), absolute leukocytes (ALUC), absolute reticulocytes (ARET), prothrombin time (PTT), and activated partial thromboplastin time (APTT). On Day 15, animals were euthanized and gross necropsies were conducted and evaluated for any macroscopic changes. Histological examinations were performed on the liver, spleen, and bone marrow of the animals from all four groups (Fraser *et al.*, 2018).The authors reported no mortalities during the 14-day study nor clinical observations attributable to LegH Prep administration. There were in-life clinical signs reported, including discoloration of the urine in Group 4 males (6/6) and Group 4 females (5/6). However, this was deemed not clinically significant, and the authors concluded:

> [I]ts single-day occurrence and lack of occurrence in any other in-life study, this observation is likely due to rehydration of the test substance by animal urine, which would result in the formation of a red/brown COL (color). Without a correlation to clinical hematology or any other parameter, these findings are interpreted to be of no toxicological significance (Fraser *et al.*, 2018).

There were no changes in mean food consumption, mean food efficiency, mean body weight, and mean daily body weight gain attributable to LegH Prep administration. There were also no changes in any of the clinical chemistry or hematological parameters nor macroscopic or microscopic findings that were attributable to LegH Prep administration. The mean and absolute relative organ-to-body weights for Groups 2 - 4 (LegH Prep groups) were not statistically different than those from Group 1 (control). The authors concluded that the LegH Prep was "well tolerated" in the 14-day study and palatable at all levels tested and stated, "these results suggest that LegH Prep would be well tolerated in a study of longer duration" (Fraser *et al.*, 2018). Due to the successful palatability of LegH Prep in the diet (500 mg/kg bw/day), the maximum dose in the subsequent 28-day study was increased to 750 mg/ kg bw/day LegH.

Frasier *et al.* (2018) conducted a 28-day dietary toxicity study in which 80 male and female Sprague-Dawley rats (10/sex/group) were administered freeze-dried LegH Prep. This study was conducted in compliance with OECD 408[16] Guidelines for Testing of Chemicals, Section 4 Health Effects: Repeated Dose 90-day Oral Toxicity Study in rodents and the U.S. FDA Toxicological Principles for the Safety Assessment of Food Ingredients, IV.C.4.a.[17] and was approved by the IACUC. LegH Prep was administered in the diet (0, 512, 1024, and 1536 mg/ kg bw/day) to rats to achieve active concentrations of LegH protein (0, 250, 500, and 750 mg/kg bw/day, respectively) which corresponded to the following four groups: Group 1 (control), Group 2 (low), Group 3 (medium), and Group 4 (high), respectively. The mean overall daily intake of the test substance in Groups 1 - 4 male rats was 0, 234, 466, and 702 mg/kg bw/day LegH, respectively. The mean overall daily intake in Groups 1 - 4 female rats was 0, 243, 480, and 718 mg/kg bw/day LegH, respectively. A significant amount (44.5%) of the LegH Prep dry solids is *P. pastoris* [based

[16] https://www.oecd-ilibrary.org/environment/test-no-408-repeated-dose-90-day-oral-toxicity-study-in-rodents_9789264070707-en; site visited on January 23rd, 2023.
[17] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/redbook-2000-ivc4a-subchronic-toxicity-studies-rodents; site visited on January 23rd, 2023.

on 6.39% LegH protein, 0.58% NaCl and the LegH Prep solids being 12.55% of the non-dried LegH Prep (Fraser *et al*., 2018)], which corresponds to consumption of approximately 0, 228, 456 and 684 mg/kg bw/day *P. pastoris* solids. This corresponds to an intake of 213.1, 424.8 and 640.0 mg *P. pastoris* solids/kg bw/day, respectively in males, and 222, 438 and 654.3 mg *P. pastoris* solids/kg bw/day, respectively in females.

Prior to study initiation and again on Day 23, the eyes of all rats were examined. Clinical evaluation, food consumption, body weight, and food efficiency were evaluated utilizing the same procedures as described in the 14-day dietary palatability study (Frasier *et al.*, 2018). On Day 22, blood and urine samples were collected for clinical chemistry, hematology, and urinalysis evaluation. Clinical chemistry parameters included AST, ALT, ALP, SDH, BIL, creatinine, BUN, TG, TC, TP, GLU, GLO, P, K, Ca, Na, and Cl. Hematological parameters included RBC, HCT, MCH, WBC, DLC, MCHC, HGB, MCV, RDW, ARET, ANEU, ALYM, AMON, AEOS, ABAS, ALUC, and PC. Urinalysis parameters included quality, color, clarity, urine volume (UVOL), microscopic urine sediment examination, pH, GLU, specific gravity, protein, ketone, bilirubin, blood, and urobilinogen (URO). At the end of the study, animals were euthanized, and blood was collected for coagulation parameters (PC, PTT, and APTT). All animals also underwent a full necropsy, which included examination of the external surface of the body, all orifices, and the thoracic, abdominal, and cranial cavities and their content. For histopathological examination, tissues/organs "representing systems" (Fraser *et al.*, 2018) were collected from animals from Group 1 and Group 4, in accordance with OECD 408 guidelines, except for the female reproductive organs, which involved histopathological examinations for the specified organs for all groups. Tissues/organs that were weighed included adrenal glands, kidneys, spleen, brain, liver, thymus, testes, epididymides, ovaries with oviducts, uterus, and heart (Fraser *et al.*, 2018).

The authors reported no mortalities, clinical observations, or adverse effects attributable to LegH Prep administration. Moreover, there were no body weight, body weight gain, food consumption, or food efficacy changes in all LegH Prep groups as compared to the control group that were attributable to LegH Prep administration. The authors reported a statistically significant ($p < 0.01$) decrease in the female Group 2 mean daily body weight on Days 14 - 21. There were also statistically significant ($p < 0.05$ - 0.01) increases in the mean daily food consumption in Group 3 males on Days 7 - 14 and in Group 4 males on Days 7 - 10. However, these changes were transient in nature and therefore deemed not toxicologically significant (Fraser *et al.*, 2018). The mean food efficiency for all female rats in Group 2 - 4 was comparable to those of the female control group throughout the study, except for statistically significant ($p < 0.01$) increases in Group 2 on Days 14 - 21; however, these changes were transient and did not affect body weight and were therefore deemed toxicologically insignificant. The authors concluded "[T]hese small but significant changes were all considered to be non-toxicologically relevant and nontest substance dependent" (Fraser *et al.*, 2018).

There were no LegH Prep-attributable changes in any of the clinical chemistry parameters for male rats in any group. There was a small, but statistically significant ($p < 0.05$), decrease in ALP in Group 2 and 4 females. However, this decrease did not exhibit a dose-dependent response, nor was it associated with clinical pathology or histopathology changes, and therefore deemed to lack toxicological significance. There were also statistically significant ($p < 0.05$) increases in male

Group 3 ALB and K values, decreases in GLU and Cl levels in Groups 2 and 3 females, increases in GLO levels in Group 3 females, and increases in Ca in Groups 2 and 3 females. The authors determined that due to the generally small magnitude of these changes, the lack of a dose-dependent response, being within biological variation, and not being associated with clinically adverse events of histopathological changes, these changes were of "no toxicological relevance and non-test substance dependent" (Fraser *et al.*, 2018). The authors reported there were no changes in hematological parameters attributable to LegH Prep administration for male rats of any group. In females, there were statistically significant ($p < 0.05$) increases in RBC, HCT, HGB, and ABAS values for Group 2 and decreases in ARET values in Group 3. These changes did not exhibit a dose-dependent response, were within biological variation, and were not associated with clinically adverse events of histopathological changes; therefore, they are "not toxicologically relevant and not test substance dependent" (Fraser *et al.*, 2018). There were no changes in coagulation parameters for female rats of any group attributable to LegH Prep administration. There were small, statistically significant ($p < 0.05$), increases in APTT exhibited in Group 3 and 4 males that lacked a dose-dependent response or were associated clinical or pathological changes, which were therefore deemed not toxicologically significant. Urinalysis evaluations revealed male and female groups administered LegH Prep were comparable to the control and did not exhibit significant changes (Fraser *et al.*, 2018).

There were no changes in organ weight, microscopic evaluations, or macroscopic evaluations, that were observed during necropsy that were attributable to LegH Prep administration. The only exception was female rats in Groups 2 and 4, which exhibited a "distinct estrous cycle stage distribution" (Fraser *et al.*, 2018). These female groups had an increased incidence of metestrus and decreased incidence of animals in estrus, as compared to the control and Group 3. However, the authors stated that single-day variations account for these differing estrous cycle distributions:

> Despite intrinsically normal estrous cycles, different estrous cycle stage distributions were observed between groups on any given day. This highlights the importance of longitudinal estrous cycle monitoring to evaluate estrous cyclicity. For example, if the estrous cycle stages were only monitored on a single day, a completely different conclusion would have been drawn regarding the test substance effect on estrous cycle if the animals had been analyzed, for example, on Day 18 of the dosing period compared to Day 21 (Fraser *et al.*, 2018).

Females in Group 2 and 4 also exhibited decreased presence of fluid-filled uteri and dilated uterine lumens and decreased uterine weights, compared to control and Group 3 females, which was "[c]onsistent with the estrous cycle stage distribution" observed in these rats (Fraser *et al.*, 2018). This 28-day study indicates and corroborates the results of the 14-day palatability study, that LegH Prep was well tolerated, did not exhibit any adverse effects, and is not a toxicological concern.

The results of the subacute/palatability studies are summarized in Table 8. The LegH Prep ingredient described in the 14-day and 28-day dietary studies conducted by Fraser *et al*. (2018) contains *P. pastoris* cellular components and is very similar in composition to the LegH Prep described in this GRAS notification; therefore, the safety study conducted on the LegH Prep described in Fraser *et al*. (2018) contributes to evaluating the safety of the LegH Prep ingredient described in this GRAS notification. Overall, studies from Reyes *et al*. (2023) and Frasier *et al.* (2018) resulted in LegH Prep not exhibiting mortalities, adverse events, signs of clinical toxicity attributable to LegH Prep. Therefore, the studies conducted by Reyes *et al*. (2023) and Frasier *et al.* (2018) corroborate that LegH Prep is palatable and well tolerated in rats and does not exhibit clinically toxicological or adverse effects.

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Table 8. Summary table of subacute toxicity/palatability studies with LegH Prep

Substance	Species, Sex and Number of Groups	Dose, Route and Duration	Results	Comments	Reference
LegH Prep	CRL Sprague-Dawley CD® IGS rats (20 males and 20 females; 5/sex/group	50,000 ppm (4167 mg/kg/day), 100,000 ppm (8333 mg/kg/day), 150,000 ppm (12,500 mg/kg/day), in the diet, 14 days	No mortalities throughout the study. There were no adverse test substance-related clinical observations, body weight, food consumption, or food efficiency changes, clinical chemistry, hematology, or necropsy observations attributable to LegH Prep administration.	LegH Prep (≤12,500 mg/kg/day) was well tolerated, palatable, and did not induce any clinically adverse effects in Sprague-Dawley rats.	Reyes *et al.*, 2023
LegH Prep	Sprague-Dawley rats; 48 (male and female; 6/sex/group)	0, 3156, 6312, and 12,612 ppm; in the diet; 14 days	No mortalities during the 14-day study nor clinical observations, changes in mean food consumption, mean food efficiency, mean body weight, and mean daily body weight gain, clinical chemistry, hematology, macroscopic findings, attributable to LegH Prep administration. The mean and absolute relative organ-to-body weights for LegH groups were not statistically different than those from control group.	The authors concluded that the LegH Prep was well tolerated in the 14-day study and palatable at all levels tested.	Frasier *et al.*, 2018
LegH Prep	Sprague-Dawley rats; 80 (male and female); 10/sex/group	0, 512, 1024, and 1536 mg/ kg bw/day; in the diet; 28 days	No mortalities, clinical observations, or adverse effects; no changes in body weight, body weight gain, food consumption, or food efficacy, clinical chemistry, hematology, coagulation, or urinalysis attributable to LegH Prep administration; No changes in organ weight, microscopic or macroscopic evaluation, observed during necropsy attributed to LegH Prep administration. Groups 2 and 4 exhibited a "distinct estrous cycle stage distribution" (increased incidence of rats in metestrus and decreased incidence of animals in estrus), compared to Control and Group 3. Females in Group 2 and 4 exhibited decreased presence of fluid-filled uteri, dilated uterine lumens and decreased uterine weights, compared to control and Group 3 females, consistent with estrous cycle stage distribution" observed in these rats).	This 28-day study indicates LegH Prep was well tolerated, did not exhibit any adverse effects, and is not a toxicological concern.	Frasier *et al.*, 2018

bw = body weight; CRL = Charles River Laboratories; Encap-UFO = encapsulated unfiltered; LegH Prep = Soy Leghemoglobin Preparation; MF = microfiltered; ppm = parts *per* million; Uncap-MF = unencapsulated microfiltered.

6.3 Subchronic studies

Reyes *et al*. (2023) conducted a 90-day dietary study in CRL Sprague-Dawley CD® IGS rats (80 rats; four groups; 20/group; ten/sex) to evaluate the safety of LegH Prep. The groups were assigned as basal diet control (Group 1), low dose (Group 2), intermediate dose (Group 3), and high dose (Group 4) (Table 9). An additional ten male and ten female test animals were randomly assigned to each of Group 1 and Group 4 (five/sex) as recovery animals for a 28-day recovery period. Animals were selected for the study based on adequate body weight gain, lack of injury or clinical signs of disease, and a body weight that was within ± 20% of the mean within a sex. Animals that were selected for the study were randomly distributed according to stratification by body weight so that there was no statistically significant difference among group body weight means within a sex. The study was conducted under Good Laboratory Practice (GLP): 21 CFR § 58[18] and in accordance with OECD 408[19] guidelines. Body weights, food consumption, clinical observations, estrus determination, ophthalmological observations, hematological and clinical chemistry parameters, coagulation, thyroid hormones, urinalysis, histopathology, organ weights, and individual necropsy records were recorded. Individual body weights were measured weekly and body weight gain was calculated for selected intervals and the overall study. Individual food consumption was measured at the same time as body weight measurements as well as food efficiency and dietary intake of LegH Prep. At terminal sacrifice, vaginal smears were collected from all female rats on the day of terminal sacrifice to determine the stage of estrus. Ophthalmological examinations included focal illumination, slit lamp biomicroscopy, and indirect ophthalmoscopy prior to study initiation and on Day 87 of the study (for all surviving animals).

Table 9. Study Design for 90-day dietary study on LegH Prep in rats (Reyes *et al.*, 2023)

Group	Number of Animals/Group (male/female)	Dietary concentration (ppm)[a]	Target Exposure (mg/kg/day)[b]	% in Diet
1	10/10	0	0	0
2	10/10	30,000	1875	3
3	10/10	60,000	3750	6
4	10/10	90,000	5625	9

[a] Concentration of LegH Prep as mixed in the diet.
[b] Target mg/kg/day are estimates based on a 400-gram rat consuming 25 G diet/day.
ppm = parts *per* million.

[18] https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-58; site visited on January 4th, 2023.
[19] https://www.oecd-ilibrary.org/environment/test-no-408-repeated-dose-90-day-oral-toxicity-study-in-rodents_9789264070707-en; site visited on January 4th, 2023.

The following hematology parameters were measured: HCT, HGB, MCH, MCV, PC, RBC, RCDW, reticulocyte count (RC), WBC, and leukocyte count (LC). The following clinical chemistry parameters were measured: ALB, ALP, BIL, creatinine, BUN, creatinine kinase (CK), Ca, Cl, Na, K, TC, GLUC, GLO, P, TG, high density lipoprotein (HDL), low density lipoprotein (LDL), ALT, AST, TP, and SDH. Coagulation parameters that were measured included APTT and PTT. Thyroid hormones that were measured included thyroid stimulating hormone (TSH), triiodothyronine (T3), and thyroxine (T4). Urinalysis parameters that were measured included bilirubin, ketones, quality, blood, microscopic urine sediment, specific gravity, color, pH, volume, clarity, total protein, urobilinogen, and glucose (Reyes *et al.*, 2023).

All animals within the study (including decedents) were subjected to a gross necropsy, which includes examination of the external surface of the body, all orifices, musculoskeletal system, and the cranial, thoracic, abdominal and pelvic cavities, with their associated organs and tissues and all gross lesions were recorded. The following organs were evaluated by histological examination in all animals of all groups: accessory genital organs (prostate and seminal vesicles), ileum with Peyer's patches, rectum, jejunum, salivary glands (sublingual, adrenals, kidneys, submandibular, and parotid), larynx, skeletal muscle, aorta, liver, skin, bone (femur), lungs, spinal cord (three levels: cervical, mid-thoracic, and lumbar), bone marrow (from femur and sternum), lymph node mandibular, lymph node mesenteric, spleen, brain (sections including medulla/pons, cerebellar, and cerebral cortex), mammary gland, sternum, nasal turbinates, stomach, nose, thymus, cecum, ovaries, thyroid, cervix, oviducts, trachea, colon, pancreas (with islets), urinary bladder, duodenum, parathyroid, uterus, esophagus, peripheral nerve (sciatic), vagina, Harderian gland, pharynx, heart, pituitary gland, eyes, optic nerve, testes, and epididymides. The following organs were weighed after dissection: adrenals (combined), kidneys, testes (combined), brain, liver, thymus, epididymis (combined), ovaries with oviducts (combined), uterus, heart, spleen, prostate, and seminal vesicles with coagulating gland (combined), thyroid, parathyroid, and pituitary gland (Reyes *et al.*, 2023).

There were no mortalities nor clinical observations attributable to LegH Prep administration observed in this study. All observations noted were considered incidental and of no toxicological significance, as there were no trends in observations that increased with dietary level. There were no changes in body weight, body weight gain, or food consumption in male and female rats attributed to the dietary administration of LegH Prep. Mean weekly body weights and mean daily body weight gains for male and female rats in Groups 2 - 4 were comparable to control Group 1 throughout the main test and the recovery period. Dietary administration of LegH Prep, for a period of at least 90 days in male and female rats, at target dietary levels of 30,000, 60,000, and 90,000 ppm, did not induce any changes in hematology coagulation, clinical chemistry, thyroid hormones, and urinalysis parameters. At the end of the study there were significant increases ($p < 0.01$) in MCV and MCH in the male Group 4 as compared to Group 1 (Table 10).

There were no statistically significant changes in any of the clinical chemistry parameters for any of the groups receiving LegH Prep as compared to the control (Table 11). Analysis of the urinalysis parameters during the main study phase revealed a significant increase in urinary ketones in male Group 3 ($p < 0.01$) and Group 4 ($p < 0.001$) as compared to the control in males (Table 12). However, during the recovery phase there were no significant changes observed between any of the groups (Table 13). Analysis of thyroid hormones revealed there was a significant increase in TSH in Group 3 ($p < 0.01$) and Group 4 ($p < 0.05$) and a significant increase in T3 levels in all three LegH Prep groups ($p < 0.01$) as compared to the control in males. In the recovery males, TSH and T4 were significantly elevated in Group 4 males as compared to control males (Table 14 and Table 15). For females, there was a significant decrease in T4 in Group 4 as compared to the control (Table 14 and Table 15). However, all statistically significant changes were within historical control ranges and not associated with histopathological changes; therefore, they were not considered toxicologically significant. Dietary exposure to LegH Prep at levels of up to 90,000 ppm for at least 90 days resulted in no test article-related macroscopic observations, organ weight changes, or microscopic findings. Compared to the control, significant ($p < 0.05$) decreases in the absolute and relative thymus-to-body weight and significant ($p < 0.05$) increases in epididymides-to-body weight, and kidney-to-brain weights (data not shown) were observed in the main study a non-dose-dependent manner (Table 16), which led to the conclusion these changes were not attributable to LegH Prep administration and deemed clinically insignificant. In the recovery phase, there was a significant ($p < 0.05$) increase in the epididymis-to-body weight and decrease in the pituitary-to-bodyweight as compared to the control in males (Table 17). In females, there was a significant ($p < 0.001$) increase in the ovaries with oviducts-to-bodyweight as compared to the control (Table 17); however, these changes were deemed incidental and not toxicologically significant. Under the conditions of the study and based on the observation of the clinical endpoint evaluated, the authors concluded, "the no-observed-adverse-effect level (NOAEL) for administration of Soy Leghemoglobin Preparation in the diet was determined to be 90,000 ppm for male and female Sprague-Dawley rats, a level calculated to provide dose levels of 4798.3 and 5761.5 mg/kg/day, the highest level tested in male and female rats, respectively, in this study" (Reyes *et al*., 2023).

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Table 10. Hematology and coagulation –90-day dietary study (Mean ± SD, *N* = 10/sex/group) (Reyes *et al.*, 2023)

Relative to Start Date (92)	Statistical Term		Group 1 (0 ppm)		Group 2 (30,000 ppm)		Group 3 (60,000 ppm)		Group 4 (90,000 ppm)	
Parameter	**M**	**F**	**M**	**F**	**M**	**F**	**M**	**F**	**M**	**F**
ABAS ($\times 10^3/\mu l$)	f	a	0.128 ± 0.057	0.071 ± 0.04	0.112 ± 0.071	0.083 ± 0.038	0.096 ± 0.044	0.061 ± 0.046	0.131 ± 0.05	0.051 ± 0.019
AEOS ($\times 10^3/\mu l$)	a	a	0.189 ± 0.088	0.112 ± 0.06	0.191 ± 0.068	0.145 ± 0.052	0.17 ± 0.071	0.131 ± 0.051	0.171 ± 0.07	0.114 ± 0.059
ALUC ($\times 10^3/\mu l$)	a	a	0.06 ± 0.015	0.061 ± 0.024	0.053 ± 0.026	0.065 ± 0.032	0.076 ± 0.02	0.041 ± 0.018	0.055 ± 0.026	0.038 ± 0.016
ALYM ($\times 10^3/\mu l$)	g	g	7.812 ± 1.624	4.938 ± 1.889	7.241 ± 2.334	4.819 ± 1.237	7.548 ± 2.311	4.166 ± 1.03	7.47 ± 2.266	3.9 ± 0.603
AMON ($\times 10^3/\mu l$)	f	a	0.357 ± 0.093	0.248 ± 0.145	0.312 ± 0.132	0.273 ± 0.071	0.432 ± 0.24	0.171 ± 0.074	0.31 ± 0.13	0.172 ± 0.07
ANEU ($\times 10^3/\mu l$)	g	g	1.517 ± 0.631	0.989 ± 0.598	1.546 ± 0.503	0.958 ± 0.324	2.252 ± 2.161	0.854 ± 0.245	1.424 ± 0.342	1.131 ± 0.48
ARET ($\times 10^3/\mu l$)	f	a	186.73 ± 44.046	167.23 ± 40.967	168.52 ± 31.949	141.02 ± 31.649	177.93 ± 39.972	148.76 ± 32.58	179.76 ± 24.208	161.7 ± 37.028
HCT (%)	a	g	47.2 ± 2.16	43.99 ± 5.53	47.96 ± 1.37	45.67 ± 1.15	47.54 ± 1.65	46.4 ± 1.74	48.25 ± 1.3	46.43 ± 1.91
HGB (g/dL)	f	g	15.57 ± 0.69	14.84 ± 1.81	16.01 ± 0.28	15.37 ± 0.25	15.7 ± 0.6	15.76 ± 0.63	15.91 ± 0.38	15.71 ± 0.66
MCV (fL)	a	a	54.06 ± 2.27	54.39 ± 1.47	54.72 ± 1.31	54.7 ± 1.18	55.09 ± 2.05	54.62 ± 0.79	56.99 ± 1.94	55.68 ± 1.05
MCH (pg)	a	a	17.88 ± 0.69	18.36 ± 0.44	18.29 ± 0.34	18.44 ± 0.4	18.19 ± 0.73	18.54 ± 0.37	18.79 ± 0.61	18.83 ± 0.35
MCHC (g/dL)	a	g	33.06 ± 0.43	33.76 ± 0.36	33.42 ± 0.52	33.7 ± 0.63	33 ± 0.29	33.96 ± 0.25	32.97 ± 0.31	33.81 ± 0.21
PLT ($\times 10^3/\mu l$)	a	g	1071.9 ± 122.9	917.6 ± 334.34	1074.8 ± 118.09	1016.3 ± 145.61	1070.5 ± 153.07	958 ± 114.32	1050.6 ± 90.98	923 ± 230.41
RBC ($\times 10^3/\mu l$)	a	g	8.717 ± 0.432	8.09 ± 1.018	8.769 ± 0.21	8.349 ± 0.204	8.633 ± 0.316	8.496 ± 0.323	8.47 ± 0.238	8.343 ± 0.385
RDW (%)	g	a	13.44 ± 0.85	0.48 ± 10	13.13 ± 0.49	12.05 ± 0.5	13.56 ± 0.8	11.96 ± 0.48	13.26 ± 0.38	12.17 ± 0.31

Relative to Start Date (92)	Statistical Term		Group 1 (0 ppm)		Group 2 (30,000 ppm)		Group 3 (60,000 ppm)		Group 4 (90,000 ppm)	
Parameter	**M**	**F**	**M**	**F**	**M**	**F**	**M**	**F**	**M**	**F**
WBC (× 10^3/µl)	f	g	10.06 ± 2.258	6.42 ± 2.37	9.464 ± 2.813	6.344 ± 1.359	10.573 ± 3.491	5.423 ± 1.213	9.561 ± 2.479	5.406 ± 0.832
APTT (seconds)	a	g	15.67 ± 0.72	14.94 ± 1.87	15.39 ± 1.28	14.01 ± 1.65	17.03 ± 1.62	15.64 ± 4.09	15.89 ± 1.75	14.33 ± 1.8
PT (seconds)	a	a	9.76 ± 0.38	9.27 ± 0.22	9.98 ± 0.32	9.19 ± 0.16	10.21 ± 0.52	9.34 ± 0.2	9.88 ± 0.25	9.32 ± 0.16

SD = standard deviation; a = Anova & Dunnett; f = Anova & Dunnett (Log); g = Anova & Dunnett (Rank); M = male; F = female; ABAS = absolute basophil; AEOS = absolute eosinophil; ALUC = absolute large unstained cell; ALYM = absolute lymphocyte; AMON = absolute monocyte; ANEU = absolute neutrophil (all forms); ARET = absolute reticulocyte; b = Kruskal-Wallis & Dunn, c = Anova & Dunnett (Log): * = $p < 0.05$; d = Anova & Dunnett ** = $p < 0.01$; e = Anova & Dunnett* = $p < 0.05$; HCT = hematocrit; HGB = hemoglobin; MCH = mean corpuscular (cell) hemoglobin; MCHC = mean corpuscular (cell) hemoglobin concentration; LegH = leghemoglobin protein; MCV = mean corpuscular (cell) volume; PLT = platelet count; PT = prothrombin time; RBC = red blood cell count; RDW = red cell distribution width; WBC = white blood cell count.

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Table 11. Summary of clinical chemistry parameters (Mean ± SD) (Reyes *et al.*, 2023)

Day(s) Relative to Start Date M(92), F(93)	Statistical Term		Group 1 (0 ppm)		Group 2 (30,000 ppm)		Group 3 (60,000 ppm)		Group 4 (90,000 ppm)	
Parameter	**M**	**F**	**M**	**F**	**M#**	**F**	**M**	**F**	**M**	**F**
ALT (U/L)	a	a	25.4 ± 6.2	21.2 ± 9	27.3 ± 4.4	17.2 ± 3.9	42.3 ± 51.9	16.4 ± 3.2	32.6 ± 24.7	19.3 ± 4.3
ALB (g/dL)	a1	a1	4.19 ± 0.24	5.39 ± 0.65	4.31 ± 0.15	5.31 ± 0.5	4.33 ± 0.3	5.13 ± 0.55	4.38 ± 0.28	5.47 ± 0.37
ALKP (U/L)	a1	a2	67.3 ± 9.2	29.1 ± 8.7	67.6 ± 10.3	30.6 ± 9.6	66.4 ± 11.3	36 ± 9.8	59 ± 16.1	29.6 ± 5.4
AST (U/L)	a	a1	77.8 ± 17.1	71.9 ± 15.9	75.2 ± 6.9	67.8 ± 18.4	88.3 ± 48.6	68.6 ± 15.4	85.7 ± 30.1	70.2 ± 10.2
CALC (mg/dL)	a1	a1	11.47 ± 1.09	12.41 ± 1.13	11.44 ± 0.47	12.23 ± 0.92	11.68 ± 0.58	11.88 ± 1.11	11.57 ± 0.67	12.17 ± 0.57
CL (mmol/L)	a1	a1	101.57 ± 1.55	104.6 ± 1.88	101.63 ± 1.7	105.83 ± 1.49	100.43 ± 1.57	104.6 ± 2.05	101.98 ± 1.71	103.75 ± 2.35
CHOL (mg/dL)	a1	a1	63.4 ± 15.4	80.2 ± 17.7	65.7 ± 16.5	79.7 ± 14.7	78.7 ± 20.8	64.3 ± 26.9	72.6 ± 16.5	70.2 ± 18.3
CK (U/L)	a	a	212.2 ± 173.914	153.4 ± 51.728	145.556 ± 44.136	209.2 ± 200.261	151.7 ± 50.197	188 ± 116.466	230.1 ± 167.781	182.6 ± 77.879
CREA (mg/dL)	a	a3	0.276 ± 0.038	0.304 ± 0.053	0.259 ± 0.031	0.33 ± 0.071	0.265 ± 0.049	0.317 ± 0.062	0.242 ± 0.03	0.273 ± 0.047
GLOB (g/dL)	a	a3	2.38 ± 0.19	2.07 ± 0.24	2.5 ± 0.14	2.07 ± 0.19	2.33 ± 0.23	2.02 ± 0.25	2.38 ± 0.13	2.1 ± 0.21
GLUC (g/dL)	a1	a3	243.1 ± 45	208.5 ± 23.9	232.8 ± 35.3	200.2 ± 37	299.8 ± 76	216.8 ± 41.9	211.9 ± 33.3	223.7 ± 32.6
HDL (mmol/L)	a	a3	0.98 ± 0.23	1.587 ± 0.345	1.056 ± 0.27	1.594 ± 0.258	1.27 ± 0.313	1.296 ± 0.498	1.15 ± 0.31	1.43 ± 0.333
IPHS (mg/dL)	a	a3	8.47 ± 1.18	8.21 ± 1.31	9.3 ± 0.93	8.76 ± 0.78	9.62 ± 1.4	8.87 ± 1.59	9.13 ± 1.55	9.36 ± 1.38
LDL (mmol/L)	a	a4	0.21 ± 0.074	0.159 ± 0.057	0.267 ± 0.122	0.159 ± 0.053	0.35* ± 0.127	0.121 ± 0.033	0.25 ± 0.071	0.141 ± 0.058

Day(s) Relative to Start Date M(92), F(93)	Statistical Term		Group 1 (0 ppm)		Group 2 (30,000 ppm)		Group 3 (60,000 ppm)		Group 4 (90,000 ppm)	
Parameter	**M**	**F**	**M**	**F**	**M#**	**F**	**M**	**F**	**M**	**F**
K (mmol/L)	a1	a5	7.166 ± 1.441	6.533 ± 1.403	8.223 ± 1.652	6.852 ± 1.425	8.693 ± 2.121	7.494 ± 1.435	8.078 ± 2.585	8.173 ± 2.302
NA (mmol/L)	a	a3	141.4 ± 3.03	142.3 ± 1.77	141.33 ± 1.41	143.4 ± 2.01	140.6 ± 1.9	141.6 ± 2.01	141.5 ± 2.27	141.7 ± 2.91
SDH (U/L)	a2	a1	15.52 ± 4.33	15.39 ± 5.1	19.33 ± 7.29	11.13 ± 2.26	24.9 ± 23.23	12.33 ± 5.12	23.59 ± 28.37	13.33 ± 3.96
BILI (mg/dL)	a3	a1	0.054 ± 0.018	0.075 ± 0.034	0.068 ± 0.031	0.072 ± 0.025	0.072 ± 0.023	0.064 ± 0.025	0.078 ± 0.024	0.066 ± 0.016
TP (g/dL)	a3	a1	6.57 ± 0.36	7.46 ± 0.75	6.81 ± 0.2	7.38 ± 0.52	6.66 ± 0.4	7.15 ± 0.67	6.76 ± 0.33	7.57 ± 0.48
TRIG (mg/dL)	a3	a2	85.3 ± 37	75.3 ± 42.6	82.2 ± 31.6	57 ± 24.8	99.8 ± 33.9	65.4 ± 41.1	99.2 ± 45.2	76.5 ± 38.2
BUN (mg/dL)	a3	a4	13.1 ± 1.7	17.6 ± 3.4	13.1 ± 1.5	18 ± 2.1	14 ± 1.4	16.4 ± 1.6	13.4 ± 1.4	16.8 ± 1.9

Statistical terms males: a = Anova & Dunnett* = $p < 0.05$; a1 = Anova & Dunnett (Log); a2 = Anova & Dunnett (Rank); a3 = Anova & Dunnett; for females: a = Anova & Dunnett (Rank), a1 = Anova & Dunnett, a2 = Anova & Dunnett (Log); a3 = Anova & Dunnett; a4 = Anova & Dunnett (Rank); a5 = Anova & Dunnett (Log); #N=9/group; ALB = albumin, ALKP = alkaline phosphatase; ALT = alanine aminotransferase; AST = aspartate aminotransferase; BILI = total bilirubin; BUN = blood urea nitrogen; CK = creatinine phosphokinase; CALC = calcium; CHOL = cholesterol; CL = chloride; CREA = creatinine; ELISA = enzyme-linked immunosorbent assay; F = female; GLOB = globulin; GGT = gamma glutamyl transferase; GLUC = glucose; HDL = high density lipoprotein cholesterol; IPHS = inorganic phosphorous; K = potassium; LDL = low density lipoprotein cholesterol; LegH = Leghemoglobin protein; M = male; NA = sodium; SDH = sorbitol dehydrogenase; T3 = triiodothyronine; T4 = thyroxine; TP = total protein; TRIG = triglycerides; TSH = thyroid stimulating hormone; U/L = units/liter.

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Table 12. Urinalysis–main study phase 90-day dietary study (Mean ± SD) (Reyes *et al.*, 2023)

	Group 1 (0 ppm)		Group 2 (30,000 ppm)		Group 3 (60,000 ppm)		Group 4 (90,000 ppm)	
Parameter	**M**	**F**	**M**	**F#**	**M**	**F**	**M#**	**F**
Urine Volume (ml) [a]	4 ± 3.36	1.62 ± 1.09	5.05 ± 2.98	1.52 ± 1.29	4.4 ± 1.32	3.93 ± 5.32	4.15 ± 3.54	3.26 ± 3.25
pH [a1]	6.4 ± 0.39	6 ± 0	6.45 ± 0.28	6.33 ± 0.5	6.2 ± 0.26	6.35 ± 0.47	6.39 ± 0.42	6.3 ± 0.35
Urine Glucose (mg/dL) [a1]	0 ± 0	0 ± 0	0 ± 0 n	0 ± 0n	0 ± 0 n	0 ± 0n	0 ± 0 n	0 ± 0n
Urine Ketone (mmol/L) [a1]	5 ± 4.1	0 ± 0	8 ± 4.8	3.3 ± 5	16 ± 13.5**	2 ± 2.6	17.8 ± 8.3***	1.5 ± 2.4
Urine Protein (mg/dL) [a1]	65 ± 36.9	56.5 ± 37.7	41 ± 31.7	51.7 ± 36.6	39.5 ± 33.4	57.5 ± 45.9	51.7 ± 36.6	46.5 ± 37.5
Specific Gravity [a1]	1.026 ± 0.0052	1.03 ± 0	1.026 ± 0.0052	1.0272 ± 0.0057	1.026 ± 0.007	1.027 ± 0.0067	1.0267 ± 0.0071	1.029 ± 0.0032
Urobilinogen (EU/dL) [a1]	0.2 ± 0	0.2 n ± 0	0.2 ± 0	0.2 ± 0n	0.2 ± 0	0.2 ± 0 n	0.2 ± 0	0.2 ± 0n

[1]Urine obtained on Day 92 relative to start date. #N = 9/group; Statistical Terms Males: a = Anova & Dunnett (Log), a1 = Anova & Dunnett (Rank): ** = p<0.01, *** = p<0.001; n = inappropriate for statistics; Statistical Terms Females: a = Anova & Dunnett (Log), a1 = Anova & Dunnett (Rank): n = inappropriate for statistics; F = Female; LegH = leghemoglobin protein; M = Male; SD = standard deviation.

Table 13. Urinalysis–recovery phase-90-day dietary study (Mean ± SD) (Reyes *et al.*, 2023)

Day(s) Relative to Start Date (120)	Statistical Term		Group 1 (0 ppm)		Group 4 (90,000 ppm)	
Parameter	**M**	**F**	**M**	**F**	**M**	**F**
Urine Volume (ml)	a	a	4.2 ± 1.6	3.9 ± 2.9	3.7 ± 3.55	2.64 ± 2.58
pH	a1	a	6.4 ± 0.22	6.7 ± 0.76	6.3 ± 0.45	6.3 ± 0.76
Urine Glucose (mg/dL)	a2	a1	0	0	0	0
Urine Ketone (mmol/L)	a1	a1	5 ± 6.1	0 ± 0	2 ± 2.7	0 ± 0
Urine Protein (mg/dL)	a	a2	98 ± 116.9	35 ± 38.4	95 ± 119.3	89 ± 124
Specific Gravity	a2	a	1.028 ± 0.0027	1.021 ± 0.0089	1.028 ± 0.0045	1.027 ± 0.0045
Urobilinogen (EU/dL)	a2	a1	0.2 ± 0	0.2 ± 0	0.2 ± 0	0.2 ± 0

*N = 5/sex/group; Statistical Terms Males: a = Anova & Dunnett (Log); a1 = Anova & Dunnett, a2 = Anova & Dunnett (Rank); Statistical Terms Females: a = Anova & Dunnett, a1 = Anova & Dunnett (Rank), a2 = Anova & Dunnett (Log); F = Female; LegH = leghemoglobin protein; M = Male; SD = standard deviation.

Table 14. Thyroid hormone profile–90-day dietary study (Mean ± SD) (Reyes *et al.*, 2023)

Day(s) Relative to Start Date M(92), F(93)			Group 1 (0 ppm)		Group 2 (30,000 ppm)		Group 3 (60,000 ppm)		Group 4 (90,000 ppm)	
Parameter	**Statistical Term**		**M**	**F**	**M**	**F**	**M**	**F**	**M**	**F**
	M	**F**								
TSH (ng/mL)	a	c	2.9548 ± 0.2143#	3.1087 ± 0.3449	2.75 ± 0.229	2.8384 ± 0.0967*	3.2692 ± 0.1678**	3.2968 ± 0.1975	3.2333 ± 0.2427*	3.2296 ± 0.3588#
T4 (ng/mL)	a	d	49.2826 ± 3.0133	40.6217 ± 4.6983	50.5529 ± 5.6793	42.5739 ± 4.7898	45.6746 ± 5.5237	39.9237 ± 2.8535	44.4091 ± 3.18	34.9737 ± 2.7091**
T3 (ng/mL)	b	d	1.2892 ± 0.0356	1.611 ± 0.1804	1.4169 ± 0.0828	1.5371 ± 0.1623#	1.4500 ± 0.1323**	1.502 ± 0.1838	1.4199 ± 0.0854**	1.4716 ± 0.1557

#N = 9/sex/group; Statistical Terms Males: a = Anova & Dunnett: *= p < 0.05; **= p < 0.01, b = Anova & Dunnett (Rank): **=p < 0.01;Statistical Terms Females: c = Anova & Dunnet (log)*= p < 0.05, d = Anova & Dunnett(log): = p < 0.01; F = female; LegH = leghemoglobin protein; M = male; SD = standard deviation; TSH = thyroid stimulating hormone; T4 = thyroxine; T3 = triiodothyronine.

Table 15. Thyroid hormone profile–recovery phase (Mean ± SD) (Reyes *et al.*, 2023)

Day(s) Relative to Start Date (120)			Group 1 (0 ppm)		Group 4 (90,000 ppm)	
Parameter	**Statistical Term**		**M**	**F**	**M**	**F**
	M	**F**				
TSH (ng/mL)	a	b	3.7228 ± 0.229	3.6678 ± 0.2411	4.3748* ± 0.5011	3.796 ± 0.2964
T4 (ng/mL)	a	c	39.9522 ± 2.2033	34.9662 ± 3.2876	44.9994* ± 3.5658	48.2854 ± 18.2472
T3 (ng/mL)	a	b	1.3516 ± 0.1549	1.6262 ± 0.2093	1.4092 ± 0.1117	1.7252 ± 0.1338

Statistical Terms Male: a = Anova & Dunnett: *= p<0.05; **= p <0.01; Statistical Terms Female: b = Anova & Dunnett, c = Anova & Dunnett (Rank); F = female; LegH = leghemoglobin protein; M = male; SD = standard deviation; TSH = thyroid stimulating hormone; T4 = thyroxine; T3 = triiodothyronine.

Table 16. Mean relative organ-to-body weights (g) –main study-90-day dietary study (Reyes *et al.*, 2023)

	Statistical Term			Group 1 (0 ppm)		Group 2 (30,000 ppm)		Group 3 (60,000 ppm)		Group 4 (90,000 ppm)	
Parameter	**M**	**F**		**M**	**F**	**M**	**F**	**M**	**F**	**M**	**F**
Adrenal/TBW (Ratio)	a	a	Mean±SD	0.1021 ± 0.0301	0.2027 ± 0.0349	0.0998 ± 0.023	0.1996 ± 0.0364	0.1043± 0.0261	0.2319± 0.038	0.092± 0.0237	0.2151± 0.0342
Brain/TBW (Ratio)	a	a	Mean±SD	3.941 ± 0.175	6.602 ± 1.063	3.78 ± 0.388	6.484 ± 0.632	3.847± 0.366	6.583± 0.603	3.879± 0.239	6.458± 0.819
Epididymis/ TBW (Ratio)	a	-	Mean±SD	2.6253 ± 0.2339	-	2.7169 ± 0.2619	-	2.7345± 0.3059	-	2.5785± 0.3439	-
Heart/TBW (Ratio)	a	c	Mean±SD	2.808 ± 0.166	3.285 ± 0.3	2.714 ± 0.128	3.141 ± 0.26	2.716± 0.217	3.276± 0.186	2.676± 0.29	3.333± 0.49
Kidneys/TBW (Ratio)	f	b	Mean±SD	5.812 ± 0.6	6.721 ± 0.608	6.34 ± 0.581	6.419 ± 0.407	6.293± 0.724	6.71± 0.405	6.036± 0.487	7.062± 1.02
Liver/TBW (Ratio)	f	a	Mean±SD	23.792 ± 1.933	26.358 ± 2.605	24.902 ± 2.209	24.299 ± 1.328	25.092± 1.902	25.182± 1.992	23.131± 2.692	26.488± 2.755
Pituitary/TBW (Ratio)	a	c	Mean±SD	0.0027 ± 0.0005	0.0093 ± 0.0037	0.0034 ± 0.0009	0.0075 ± 0.0023	0.0027± 0.0008	0.009± 0.0036	0.0031± 0.001	0.0083± 0.0031
Pro, SV, CG (combined)/ TBW (Ratio)	a	-	Mean±SD	0.006 ± 0.001	-	0.006 ± 0.001	-	0.005±0.0 01	-	0.006±0.0 01	-
			N	10	-	9	-	10	-	10	-
Spleen/TBW (Ratio)	a	a	Mean±SD	1.566 ± 0.145	1.814 ± 0.308	1.559 ± 0.114	1.539 ± 0.137	1.427± 0.197	1.664± 0.306	1.43± 0.277	1.643± 0.286
Testes/TBW (Ratio)	f	-	Mean±SD	6.154 ± 0.741	-	6.245 ± 0.602	-	6.221± 0.624	-	6.293± 0.585	-
Thymus/TBW (Ratio)	e	c	Mean±SD	0.5904 ± 0.1547	1.0478 ± 0.3276	0.4483* ± 0.077	0.8559 ± 0.2851	0.5467± 0.1056	0.7308*± 0.1056	0.4462*± 0.118	0.7896± 0.1872
Thyroid-Parathyroid/ TBW (Ratio)	e	b	Mean±SD	0.4464 ± 0.12201	0.75609 ± 0.30197	0.55585 ± 0.18021	0.81656 ± 0.23839	0.52261± 0.09894	0.85358± 0.0576	0.59005± 0.08934	0.9797± 0.17293
Ovaries with Oviducts/ TBW (Ratio)	-	a	Mean±SD	-	0.3571 ± 0.0559	-	0.3377 ± 0.0708	-	0.3177± 0.1	-	0.3687± 0.069
Uterus/TBW (Ratio)	-	f	Mean±SD	-	2.263 ± 0.736	-	2.103 ± 0.749	-	2.279± 0.532	-	2.012± 0.708

*N = 10/sex/group unless indicated otherwise.

Abbreviations: M = Male; F = Female; LegH = leghemoglobin protein; SD = standard deviation.

	Statistical Term		Group 1 (0 ppm)		Group 2 (30,000 ppm)		Group 3 (60,000 ppm)		Group 4 (90,000 ppm)	
Parameter	**M**	**F**	**M**	**F**	**M**	**F**	**M**	**F**	**M**	**F**

Statistical Terms: a = Anova & Dunnett; b = Kruskal-Wallis & Dunn, c = Anova & Dunnett (Log): * = $p < 0.05$, d = Anova & Dunnett ** = $p < 0.01$; e = Anova & Dunnett* = $p < 0.05$; f = Anova & Dunnett (Log).

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Table 17. Mean relative organ-to-body weights (g) –recovery phase* (Mean ± SD) (Reyes *et al.*, 2023)

LegH Dose Levels	**Statistical Terms**		**Group 1 (0 ppm)**		**Group 4 (90,000 ppm)**	
Parameter	**M**	**F**	M	F	M	F
Adrenal/TBW (Ratio)	e	a	0.0738 ± 0.0284	0.1993 ± 0.0365	0.0917 ± 0.0145	0.2073 ± 0.585
Brain/TBW (Ratio)	e	a	3.518 ± 0.345	6.006 ± 0.652	3.516 ± 0.324	6.103 ± 0.445
Epididymis/TBW (Ratio)	e	-	2.3404 ± 0.2462	-	2.8179* ± 0.3609	-
Heart/TBW (Ratio)	e	c	2.615 ± 0.311	3.175 ± 0.21	2.582 ± 0.141	3.025 ± 0.252
Kidneys/TBW (Ratio)	b	b	5.392 ± 0.356	6.286 ± 0.668	4.629 ± 2.509	6.038 ± 0.41
Liver/TBW (Ratio)	e	a	22.674 ± 2.414	25.462 ± 2.292	24.199 ± 1.347	23.529 ± 1.35
Pituitary/TBW (Ratio)	c	c	0.0034 ± 0.0008	0.0073 ± 0.0027	0.0026* ± 0.0003	0.007 ± 0.0015
Pro, SV, CG (combined)/TBW (Ratio)	b	-	0.005 ± 0	-	0.006 ± 0.002	-
Spleen/TBW (Ratio)	c	a	1.425 ± 0.241	1.55 ± 0.264	1.42 ± 0.184	1.666 ± 0.146
Testes/TBW (Ratio)	e	-	5.721 ± 0.404	-	6.337 ± 0.904	-
Thymus/TBW (Ratio)	a	c	0.3361 ± 0.0696	0.6767 ± 0.1184	0.3061 ± 0.1146	0.5967 ± 0.111
Thyroid-Parathyroid/TBW (Ratio)	a	b	0.59223 ± 0.13084	1.13666 ± 0.16291	0.68185 ± 0.08464	1.01281 ± 0.1466
Ovaries with Oviducts/TBW (Ratio)	-	a	-	0.2831 ± 0.0309	-	0.4004** ± 0.0696
Uterus/TBW (Ratio)	-	f	-	2.179 ± 0.197	-	1.925 ± 0.514

[a]*N* = 5/sex/group; a = Anova & Dunnett; b = Kruskal-Wallis & Dunn, c = Anova & Dunnett (Log): * = p < 0.05, d = Anova & Dunnett ** = p < 0.01; e = Anova & Dunnett* = p < 0.05; f = Anova & Dunnett (Log); F= female; LegH = leghemoglobin protein; M = male; SD = standard deviation.

6.4 Other studies

6.4.1 Potential toxigenicity/pathogenicity

Banerjee and Verma (2000) evaluated the ability of *P. pastoris* to produce secreted toxins and inhibit the growth of different yeast cells. The *P. pastoris* strain was obtained from the Research Culture Collection of Northern Research Laboratory (NRRL, Peoria, IL; strain NRRL No. Y 4290). Two cultures of *P. pastoris* strain (NRRL No. Y 4290) were grown (1×10^6 cell/ml) and cultured for 2, 3 and 14 days, after which the cell suspensions were centrifuged, and the supernatant filtered and concentrated (50X). The concentrated supernatant was then spotted on the surface of dextrose agar in a 24-well microliter dish and then overlaid with top agar containing viable indicator strains of pathogens and yeasts. As controls, toxins from *Kluveromyces lactis* and *Pichia acacia* were plated with the indicator strain *Dabaryomyces tamarii*. The authors reported that the *K. lactis* toxin produced a clear zone of growth inhibition, confirming the ability to show toxin effectiveness in this study. The *P. pastoris* concentrated supernatant did not show any toxigenic activity against any of the indicator strains, showing that the (NRRL No. Y 4290) strain of *P. pastoris* did not produce toxins against sensitive indicator strains (Banerjee and Verma, 2000).

França *et al.* (2015) evaluated the pathogenicity/toxigenicity of *P. pastoris* X-33[20], by the administration of viable *P. pastoris* to male BALB/c mice. To this end, two separate experiments were conducted: in the first study, mice (15/group) were administered a single dose of either *P. pastoris* X-33 [7 log CFU (1×10^7 CFU/day) in 1 ml 0.9% saline] or 1 mL 0.9% saline suspension (served as a control) daily by oral gavage for 20 days. Whereas, in the second experiment, male BALB/c mice (10/group) were either fed an antibiotic-free diet *ad libitum* containing *P. pastoris* X-33 (1×10^7 CFU/g), the antibiotic-free diet alone, or the diet containing *Saccharomyces boulardii* (1×10^7 CFU/g) for 20 days. In both experiments, animals were monitored for behavioral effects daily, and subsequently on the 20th day after the start of gavage administration; feces were collected 24, 48, and 72 hours post yeast administration. Mice from both experiments (5/group) were euthanized, tissues and organs were collected (specific tissues/organs not stated) and histopathological analysis was conducted; whereas the remaining ten mice, from both experiments, were challenged by oral gavage with one dose of *Salmonella enterica* serovar *Typhimurium* (strain 29630) at 5 log CFU/mouse, which is the LD_{50} dose. The survival rates were then calculated ten days later. Fresh feces from the surviving animals were collected on the 13th day after challenge and mice were subsequently euthanized. For histopathological analysis, the intestine, liver, and spleen were removed, specifically to evaluate intestinal colonization of *S. enterica Typhimurium* and possible translocation to the liver and spleen.

The authors reported that mice administered exclusively *P. pastoris* X-33 did not exhibit any reported behavioral changes nor signs of injury as evidenced by histopathological analysis. Moreover, there were no changes in the intestinal microvilli and lamina propria reported as compared to the saline control group. The group of mice that received *P. pastoris* X-33 in the diet and were subsequently challenged with an LD_{50} dose of *S. typhimurium* had a significantly increased ($p < 0.05$) survival *percentage* (80% survival) as compared to mice that were challenged

[20] *P. pastoris* strain X-33 was purchased from Invitrogen (California) (Invitrogen, 2006).

with *S. typhimurium* without *P. pastoris* within the diet (50% survival). The *percentage* of surviving mice with measurable levels of *S. typhimurium* in the intestine, liver and spleen was 80%, 60% and 80% in the control group (*S. typhimurium* without *P. pastoris* within the diet), respectively, compared to 12.5%, 0%, and 0% in the respective organs of the *P. pastoris* X-33 group. This study indicates that mice administered viable *P. pastoris* (1×10^7 CFU/day) for up to 20 days did not exhibit any adverse effects for the endpoints evaluated, behavioral changes, nor clinically adverse histopathological changes (França *et al.*, 2015).

Becerril-García *et al.* (2022) evaluated the potential pathogenicity of *P. pastoris* in BALB/c mice by intravenous administration of live cells. *P. pastoris* GS115 and *Candida albicans* (ATCC 66027) yeast cultures were maintained in Sabouraud dextrose agar at 30 and 37°C, respectively. *C. albicans* is a well-defined pathogen and was used as a positive control for pathogenicity in this study. Female BALB/c mice (5 - 7 weeks old; four groups; 20/group) were anesthetized with ketamine and xylazine (80 and 120 mg/g bw, respectively) inoculated intravenously (IV) with either *P. pastoris* or *C. albicans*, or PSS solution (200 µl) *via* the lateral tail vein. The fourth group was maintained as a housing control and did not receive any injection. Yeast cells were washed with physiological saline solution (PSS) (0.89%) and counted with a hemacytometer to be adjusted at 5.0×10^6 viable yeast cells/ml in the PSS injection. Animals were then monitored daily for adverse effects, behavior, and bodyweight changes. Subsequently, five mice of each group were euthanized at 4, 24, and 48 hours post injection (hpi) and six days post infection (dpi). The sera and tissue of the kidney, liver, brain, spleen, heart, and lung were collected. Only one mouse was euthanized at six dpi due to illness, according to humane end points stated by the local Institutional Animal Care and Use Committee (IACUC), whereas all the other survived until study termination. Fungal burden was measured in homogenized organs ($\log_{10}$ CFU/g tissue) after 48 hours of incubation at 30°C for *P. pastoris* and 37°C for *C. albicans*. Histological analysis included the identification of polymorphonuclear (PMN) and mononuclear (MN) inflammatory infiltrate sections stained with H-E according to their staining characteristics, cell size, and cell nucleus morphology (PMN cells: the nucleus contains multiple lobes, MN cells: a round nuclei). Periodic acid-Schiff base (PAS)-stained sections were used to visualize the presence of fungi in tissues.

The pathogenicity of *P. pastoris* was also evaluated utilizing an *in vivo* delayed-type hypersensitivity test (DTH), with *C. albicans* being utilized as a positive control. This involved a sensitization phase, which was done by the epicutaneous administration of *P. pastoris* or *C. albicans* (2×10^8 CFU) in 50 µl of PSS on exfoliated stratum corneum. The PSS control mice underwent the same procedure, except with just PSS alone, whereas the other control did not undergo exfoliation of the stratum corneum, which were used to assess the baseline status. Animals were monitored for seven days for adverse effects and behavioral changes. The efferent phase was then conducted by the inoculation of 10 µl with either heat-killed yeast species (1×10^7 CFU) and the subsequent swelling that was induced after 24 hours was calculated based off the baseline thickness was recorded. Mice were then euthanized, and the spleen was obtained for *in vitro* culture assays. The authors also utilized an *in vitro* assay, which used recovered splenocytes incubated with 1X erythrocyte lysis solution for 15 minutes at 37°C and washed with phosphate buffered saline (PBS). In a 96-well plate, 1×10^5 cells were seeded in RPMI-1640 and stimulated with yeast protein extract from either *P. pastoris* or *C. albicans* (5.0 µg/ml). A positive control was used, concanavalin A (ConA; 3.0 µg/ml). After three days with yeast extract treatment (monoclonal stimulation) or five days with ConA treatment (polyclonal activation), cell proliferation was measured using the (3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulfophenyl)-2H-tetrazolium) (MTS[21]) assay (Becerril-García *et al.*, 2022).

The authors reported that all mice inoculated with *P. pastoris* survived, maintained normal bodyweight, did not exhibit any adverse effects, and showed no changes in physical and locomotion characteristic, which was comparable to the saline control group and the positive control group. This contrasted with the positive control *C. albicans*, where mice exhibited "severe weight loss" and after four dpi, exhibited piloerection and a hunched posture; one mortality was reported at six dpi (Becerril-García *et al.*, 2022). The fungal burden assay revealed the presence of *P. pastoris* in the lung until 24 hpi and in the spleen, heart, liver, and kidney for 48 hpi, with the fungal burden of *P. pastoris* decreasing significantly ($p \leq 0.05$) to undetectable at 144 hpi. However, the authors reported there was no evidence of brain colonization of *P. pastoris* after the intravenous inoculation of mice. The authors reported that mice infected with *C. albicans* exhibited significantly ($p \leq 0.05$) higher and persistent fungal burden in all organs, which persisted in the spleen and the liver 144 hpi. *C. albicans* also exhibited progressive invasion of the brain and kidney (Becerril-García *et al.*, 2022).

Histological analysis revealed *P. pastoris* did not invade any brain structures, which contrasted with *C. albicans*, which exhibited systemic infection and the presence of fungal structures and inflammatory cell infiltrate during the first 48 hpi. Neither yeast-infected mice groups exhibited the presence of fungal structure nor inflammatory cell infiltrate; however, yeast cells were detected in the spleen and heart 24 hpi, which subsequently cleared at 38 hpi without the presence of cellular infiltrate. Yeast cells of both species also persisted at 48 hpi in the liver, with the presence of MN and PMN infiltrate; however, both yeasts were fully cleared at 144 hpi. In mice infected with *C. albicans*, inflammatory cellular infiltrate with fungal structures was observed at 24 hpi. The kidneys of mice infected with *C. albicans* exhibited progressive fungal structure invasion and "massive recruitment of cellular infiltrate as infection progressed" (Becerril-García *et al.*, 2022). Macroscopic analysis revealed the "typically extensive pathology in the kidneys" of the mice infected with *C. albicans*, which was characterized by lesions on the surface of the kidney and the presence of abscesses (Becerril-García *et al.*, 2022). The authors concluded "these results show that after intravenous inoculation with *P. pastoris*, this yeast did not cause pathology in mice, and its dissemination to some tissues is quickly eliminated during the first hours" (Becerril-García *et al.*, 2022). The results of the *in vivo* DTH assay confirmed that mice sensitized to *P. pastoris* did not exhibit pathogenicity, as the thickness of the ear did not significantly change; however, *C. albicans*, exhibited a significant ($p < 0.01$) increase in the thickness of the ears (0.5 - 0.6 mm), indicative of a cell-mediated sensitization. The *in vitro* splenocyte assay revealed that neither protein extracts of *P. pastoris* nor *C. albicans* induced splenocyte proliferation nor induction of IL-2 secretion. The authors concluded that "our results show that systemic intravenous administration of live *P. pastoris* cells in mice is unable to establish

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[21] A colorimetric assay that calculates cell proliferation by measuring cellular metabolic activity.

an invasive process, and that its early clearance from tissues does not induce either, humoral or cell-mediated immune response in immunocompetent mice" (Becerril-García *et al.*, 2022).

6.5 Genotoxicity

For the LegH Prep that is the subject of this GRAS notification, a bacterial reverse mutation assay was conducted (Reyes *et al*., 2023) to evaluate the potential for genotoxicity of LegH Prep. The assay was conducted under GLP, and the protocol was based on OECD Guideline 471[22] and ICH Guidelines S2A[23] (ICH, 1996) and S2B[24] (ICH, 1997). The LegH Prep was evaluated for mutagenicity in five bacterial strains using *Salmonella typhimurium* (TA98, TA100, TA1535, and TA1537) and *Escherichia coli* (WP2 uvrA) according to the plate incorporation (Experiment I) and preincubation (Experiment II) methods, in the presence and absence of a metabolic activation system (S9). The concentrations of LegH Prep with and without S9 activation were 0, 31.6, 100, 316, 1000, 2500 and 5000 µg/plate. Cell toxicity was ascertained by a clearing or a depletion of the background lawn or a reduction in the number of revertants down to a mutation factor of approximately ≤ 0.5 as compared to the negative (solvent) control. The mutation factor was calculated for each experiment by dividing the mean value of the revertant counts by the mean values of the negative control. LegH Prep would be considered mutagenic if a biologically significant positive response for at least one of the dose groups was observed in at least one tester strain with or without metabolic activation or if there is a clear and dose-related increase in the number of revertants. The authors reported no biologically relevant increases in revertant colony formation in any of the tester strains, neither in the absence or presence of metabolic activation, utilizing any concentration of LegH Prep. In experiment I (Table 18), LegH Prep treated plates exhibited toxic effects in tester strain TA1535 at a concentration of 5000 µg/plate (without metabolic activation). The positive controls for this experiment were 4-nitro-o-phenylenediamine (4-NOPD), 2-aminoanthracene (2-AA), sodium azide (NaN_3), and methyl methanesulfonate (MMS). In experiment II (Table 19), toxic effects of LegH Prep were noted in tester strain TA1537 at concentrations of ≥2500 µg/plate in the absence of metabolic activation. The mutation factors for experiment I (Table 20) and experiment II (Table 21) did not indicate signs of toxicity and were within historical ranges. The authors reported that no other toxic effects of LegH Prep were observed in experiment I or II. Therefore, the authors concluded, "LegH Prep is non-mutagenic in this bacterial reverse mutation assay" (Reyes *et al*., 2023). Therefore, under the conditions of the study, LegH Prep was non-mutagenic in this bacterial reverse mutation assay.

[22] https://www.oecd-ilibrary.org/environment/test-no-471-bacterial-reverse-mutation-test_9789264071247-en; site visited on January 18th, 2023.
[23] https://www.fda.gov/media/71959/download; site visited on January 18th, 2023.
[24] https://www.fda.gov/media/71971/download; site visited on January 18th, 2023.

Table 18. Experiment I: Summary of bacterial reverse mutation assay results (Mean ± SD revertant colonies) (Reyes *et al.*, 2023)

LegH Prep	TA98		TA100		TA1535		TA1537		WP2 uvrA	
µg/plate	**−S9**	**+S9**	**−S9**	**+S9**	**−S9**	**+S9**	**−S9**	**+S9**	**−S9**	**+S9**
Water	37 ± 19.2	40 ± 3.8	104 ± 5.1	104 ± 1	13 ± 0.6	9 ± 3	11 ± 2.9	13±1.7	254 ± 7	279 ± 23.6
31.6	34 ± 2	37 ± 8.5	108 ± 4.9	105 ± 13.2	15 ± 2.1	6 ± 3.2	18 ± 1.7	13±5	213 ± 36.1	242 ± 32.7
100	33 ± 5.7	35 ± 4.9	92 ± 9.6	108 ± 7.2	9 ± 3.2	6 ± 1.2	11 ± 1.2	10±1.7	228 ± 14.6	271 ± 15.6
316	33 ± 7	37 ± 6.4	101 ± 17.1	95 ± 13.2	12 ± 1.5P	5 ± 1P	9 ± 3.1P	10±2.5P	283 ± 1.85P	299 ± 18.6P
1000	42 ± 10.5P	42 ± 6.2P	90 ± 8.7P	108 ± 7.2P	9 ± 2.9P	9 ± 4.4P	11±1.2P	13±3.8P	277 ± 26.9P	300 ± 26.5P
2500	30 ± 10.4P	44 ± 8.7P	103 ± 1.7P	98 ± 7.6P	8 ± 1.2P	8 ± 1.2P	10±2.5P	14±5.6P	202 ± 2.6P	238 ± 30.6P
5000	24 ± 5.8P	45 ± 2.6P	89 ± 10P	93 ± 8.7P	5 ± 3P	6 ± 0.6P	9±3.1P	9±4.2P	222 ± 7.6P	265± 22.1P
4-NOPD (10)	542 ± 45	NA	432 ± 22	NA	918 ± 78.4	NA	60±15.5	NA	1255 ± 61.2	NA
2-AA (2.5)	NA	2057 ± 223.1	NA	1361 ± 19.2	NA	264 ± 10.1	NA	175±11.9	NA	693 ± 125

Data are shown as Mean ± SD revertants/plate for three replicates for each concentration in each experiment.
2-AA = 2-aminoanthracene; LegH = soy leghemoglobin protein; NA = data not applicable; 4-NOPD = 4-nitro-o-phenylenediamine; P = visible precipitate; SD = Standard deviation.

Table 19. Experiment II: Summary of bacterial reverse mutation assay results (Mean ± SD revertant colonies) (Reyes *et al.*, 2023)

LegH Prep	TA98		TA100		TA1535		TA1537		WP2 uvrA	
µg/plate	−S9	+S9	−S9	+S9	−S9	+S9	−S9	+S9	−S9	+S9
Water	48 ± 4.4	53 ± 2.1	92 ± 13	113 ± 6.8	9 ± 4.2	12 ± 5.5	19 ± 7.8	15 ± 1.2	236 ± 25.7	345 ± 11
31.6	48 ± 2.3	55 ± 5.3	123 ± 11.4	118 ± 10.2	10 ± 6.2	12 ± 3.6	18 ± 5.1	19 ± 3	196 ± 21.6	307 ± 23.7
100	50 ± 7.6	46 ± 5.6	106 ± 0.6	120 ± 5.2	13 ± 4.5	14 ± 9	12 ± 3.1	19 ± 1.2	236 ± 27	256 ± 28.5
316	45 ± 9.3P	37 ± 3.8P	119 ± 6.9P	107 ± 12.5P	9 ± 1.5P	13 ± 3.5P	13 ± 2P	16 ± 3.5P	265 ± 18.6P	277 ± 3.5P
1000	42 ± 17.5P	55 ± 3.8P	102 ± 8.5P	112 ± 9.2P	10 ± 5.7P	13 ± 3.5P	16 ± 2P	17 ± 5.7P	229 ± 35P	300 ± 11.2P
2500	46 ± 6.8P	48 ± 6.5P	106 ± 15.5P	112 ± 20.4P	9 ± 2.6P	12 ± 6.2P	10 ± 4P	20 ± 4.7P	228 ± 21.9P	285 ± 45.2P
5000	52 ± 3.2P	42 ± 5.7P	122 ± 13.5P	100 ± 13.2P	9 ± 2.6P	13 ± 4.5P	8 ± 2.6P	16 ± 1P	231 ± 15.1P	337 ± 27.5P
4-NOPD	545 ± 52	NA	891 ± 72	NA	1570 ± 13.1	NA	96 ± 6.4	NA	1912 ± 172.1	NA
2-AA	NA	1771 ± 130.6	NA	1054 ± 230.3	NA	173 ± 18	NA	120 ± 11.9	NA	1177 ± 142.7

Data are shown as Mean ± SD revertants/plate for three replicates for each concentration in each experiment.
2-AA = 2-aminoanthracene;4-NOPD = 4-nitro-o-phenylenediamine; LegH = soy leghemoglobin protein; NA = data not applicable; P = visible precipitate; SD = Standard deviation.

Table 20. Summary of bacterial reverse mutation assay results of LegH Prep (Experiment I; up to 5,000 µg/plate) – Mutation factor (Reyes *et al.*, 2023)

LegH Prep	TA98		TA100		TA1535		TA1537		WP2 uvrA	
µg/plate	−S9	+S9	−S9	+S9	−S9	+S9	−S9	+S9	−S9	+S9
Water	1	1	1	1	1	1	1	1	1	1
31.6	0.9	0.9	1	1	1.2	0.7	1.7	1	0.8	0.9
100	0.9	0.9	0.9	1	0.7	0.6	1.1	0.8	0.9	1
316	0.9	0.9	1	0.9	1	0.6	0.9	0.8	1.1	1.1
1000	1.1	1	0.9	1	0.7	1	1	1	1.1	1.1
2500	0.8	1.1	1	0.9	0.7	0.9	1	1.1	0.8	0.9
5000	0.7	1.1	0.8	0.9	0.4	0.7	0.9	0.7	0.9	0.9
4-NOPD	14.6	NA	4.1	NA	72.5	NA	5.7	NA	4.9	NA
2-AA	NA	51	NA	13.1	NA	29.3	NA	13.5	NA	2.5

2-AA = 2-aminoanthracene; LegH = soy leghemoglobin protein; NA = data not applicable; 4-NOPD = 4-nitro-*o*-phenylenediamine; SD = Standard deviation.
Data are shown as Mean ± SD revertants/plate for three replicates for each concentration in each experiment.

Table 21. Summary of bacterial reverse mutation assay results of LegH Prep (Experiment II; up to 5,000 µg/plate) – Mutation factor (Reyes *et al.*, 2023)

LegH Prep	TA98		TA100		TA1535		TA1537		WP2 uvrA	
µg/plate	−S9	+S9	−S9	+S9	−S9	+S9	−S9	+S9	−S9	+S9
Water	1	1	1	1	1	1	1	1	1	1
31.6	1	1	1.3	1	1.1	1	1	1.2	0.8	0.9
100	1	0.9	1.2	1.1	1.4	1.1	0.6	1.3	1	0.7
316	0.9	0.7	1.3	0.9	1	1	0.7	1.1	1.1	0.8
1000	0.9	1	1.1	1	1	1	0.8	1.1	1	0.9
2500	1	0.9	1.2	1	0.9	1	0.5	1.3	1	0.8
5000	1.1	0.8	1.3	0.9	1	1.1	0.4	1	1	1
4-NOPD	11.4	NA	9.7	NA	168.2	NA	5	NA	8.1	NA
2-AA	NA	33.6	NA	9.3	NA	14	NA	7.8	NA	3.4

2-AA = 2-aminoanthracene; LegH = soy leghemoglobin protein; NA = data not applicable; 4-NOPD = 4-nitro-*o*-phenylenediamine.

Reyes *et al*. (2023) also conducted an *in vitro* mammalian micronucleus assay in human lymphocytes using LegH Prep without S9 metabolic activation (162, 325, and 650 µg/mL) and with S9 metabolic activation (62.5, 125 and 250 µg/mL) for 4 hours, and for 44 hours without S9 metabolic activation (250, 500 and 750 µg/mL) (Table 22). This study was compliant with OECD Guidelines for Testing of Chemicals, Section 4 No. 487[25] "*In Vitro* Mammalian Cell Micronucleus Test" and Commission regulation (EU) 2017/735 B.49[26] "*In Vitro* Mammalian Cell Micronucleus Test". The authors reported precipitation of LegH Prep in the cultures at the end of treatment at ≥650 µg/mL without metabolic activation (Table 23) and at ≥250 µg/mL with metabolic activation in experiment I (Table 24) and at ≥750 µg/mL in experiment II (Table 25). MMS (50 and 65 µg/mL) and colchicine (Colc, 0.04 and 0.4 µg/mL) were used as positive control without metabolic activation whereas cyclophosphamide (CPA, 15 µg/mL) was used as a positive control in the presence of metabolic activation. Culture medium was used as a negative control. If cytotoxicity is observed, the highest concentration evaluated should not exceed the limit of 55% ± 5% cytotoxicity according to the OECD Guideline. Higher levels of cytotoxicity may induce chromosome damage as a secondary effect of cytotoxicity, with lower concentrations exhibiting intermediate to little or no toxicity.

Table 22. Study design for *in vitro* mammalian micronucleus assay in HBPLs using LegH Prep (Reyes *et al.*, 2023)

	Without S9		**With S9**
	Exp. I	**Exp. II**	**Exp. I**
Exposure period (LegH Prep)	4 h	44 h	4 h
Cytochalasin B exposure	40 h	43 h	40 h
Preparation interval	44 h	44 h	44 h
Total culture period*	92 h	92 h	92 h

*Exposure started 48 h after culture initiation.
Exp. = experiment.

To distinguish a limit between cytotoxic and noncytotoxic effects, a value of the relative cell growth of 70% compared to the negative/solvent control which corresponds to 30% of cytostasis. In experiment I and II, with and without metabolic activation, no increases of the cytostasis above 30% nor biologically relevant increases of the micronucleus frequency were reported after 4 and 44 hours LegH Prep (Table 25 and Table 26). All the positive controls distinct and statistically significant increases of the micronucleus frequency, and therefore worked as expected. The authors of the study concluded "it can be stated that during the study described and under the experimental conditions reported, the test item LegH Prep did not induce structural and/or numerical chromosomal damage in human lymphocytes" (Reyes *et al*. 2023). The authors concluded, "LegH Prep was found to be nonclastogenic/nonaneugenic in the *in vitro* mammalian micronucleus assay using human lymphocytes" (Reyes *et al*. 2023).

[25] https://www.oecd.org/chemicalsafety/test-no-487-in-vitro-mammalian-cell-micronucleus-test-9789264264861-en.htm; site visited on February 13th, 2023.
[26] https://www.legislation.gov.uk/eur/2017/735/annex/division/B.49/data.pdf; site visited on February 13th, 2023.

Table 23. Summary result table for Experiment I and II, micronucleus induction in human lymphocytes, 4 h treatment, 44 h fixation interval: without metabolic activation (Reyes *et al.*, 2023)

	Dose Group	Concentration (µg/mL)	No. of cells evaluated	Cytostasis (%)	Relative Cell Growth (%)	Micronucleated Cells Frequency (%)	Historical Control Limits Negative Control	P	Statistically Significant Increase [a]
Experiment I 4 h treatment, 44 h fixation interval	C	0	2000	0	100	0.60	0.17% - 1.17%	/	/
	2	162.5	1773	30	70	1.18		NO	-
	3	325	2000	0*	110	0.50		NO	-
	4	650	2000	0*	105	0.55		YES	-
	MMS	65	1260	25	75	3.30		NO	+
	Colc.	0.4	1096	53	47	1.85		NO	+
Experiment II 44 h treatment, 44 h fixation interval	C	0	2000	0	100	1.05	0.17% - 1.17%	/	/
	4	250	2000	25	75	1.15		NO	-
	5	500	2000	24	76	0.45		NO	(+)
	6	750	2000	21	79	0.45		YES	(+)
	MMS	50	2000	13	87	2.35		NO	+
	Colc.	0.04	1020	71	29	4.42		NO	+

+: significant increase; (+): significant decrease; -: not significant; *: the cytostasis is defined 0, when the relative cell growth exceeds 100%; a: statistically significant increase compared to negative control (χ^2 test, $p<0.05$); Colc. = Colchicine, Positive Control (without metabolic activation); CPA = Cyclophosphamide, Positive Control (with metabolic activation); Cytostasis [%] = 100- Relative Cell Growth [%];MMS = Methylmethanesulfonate, Positive Control (without metabolic activation); Negative Control (Culture medium); P = Precipitation (yes = precipitation, no = no precipitation); Relative Cell Growth = 100 x ((CBPI Test conc – 1) / (CBPI control -1))

Table 24. Summary result table for Experiment I, micronucleus induction in Human Lymphocytes, 4 h treatment, 44 h fixation interval; with metabolic activation (Reyes *et al.*, 2023)

	Dose Group	Concentration (µg/mL)	No. of Cells Evaluated	Cytostasis (%)	Relative Cell Growth (%)	Micronucleated Cells Frequency (%)	Historical Control Limits Negative Control	P	Statistically Significant Increase [a]
Experiment I 4 h treatment, 44 h fixation interval	C	0	1725	0	100	1.04	0.17% - 1.17%	/	/
	2	62.5	2000	0*	106	0.40		NO	(+)
	3	125	2000	0*	110	0.35		NO	(+)
	4	250	2000	0*	131	0.55		YES	-
	CPA	15	1700	26	74	3.46		NO	+

+: significant increase; (+): significant decrease; -: not significant; *: the cytostasis is defined 0, when the relative cell growth exceeds 100%; a: statistically significant increase compared to negative control (χ^2 test, $p<0.05$); C: Negative Control (Culture medium); Colc.: Colchicine, Positive Control (without metabolic activation); CPA: Cyclophosphamide, Positive Control (with metabolic activation); Cytostasis [%] = 100- Relative Cell Growth [%];MMS: Methylmethanesulfonate, Positive Control (without metabolic activation); P = Precipitation (yes: precipitation, no: no precipitation); Relative Cell Growth: 100 x ((CBPI Test conc – 1) / (CBPI control -1)).

Table 25. Summary of bacterial reverse mutation assay results of LegH Prep (Experiment I; up to 5,000 µg/plate) – Mutation factor (Reyes *et al.*, 2023)

LegH Prep	TA98		TA100		TA1535		TA1537		WP2 uvrA	
µg/plate	−S9	+S9	−S9	+S9	−S9	+S9	−S9	+S9	−S9	+S9
Water	1	1	1	1	1	1	1	1	1	1
31.6	0.9	0.9	1	1	1.2	0.7	1.7	1	0.8	0.9
100	0.9	0.9	0.9	1	0.7	0.6	1.1	0.8	0.9	1
316	0.9	0.9	1	0.9	1	0.6	0.9	0.8	1.1	1.1
1000	1.1	1	0.9	1	0.7	1	1	1	1.1	1.1
2500	0.8	1.1	1	0.9	0.7	0.9	1	1.1	0.8	0.9
5000	0.7	1.1	0.8	0.9	0.4	0.7	0.9	0.7	0.9	0.9
4-NOPD	14.6	NA	4.1	NA	72.5	NA	5.7	NA	4.9	NA
2-AA	NA	51	NA	13.1	NA	29.3	NA	13.5	NA	2.5

2-AA = 2-aminoanthracene; LegH = soy leghemoglobin protein; NA = data not applicable; 4-NOPD = 4-nitro-*o*-phenylenediamine; SD = Standard deviation. Data are shown as Mean ± SD revertants/plate for three replicates for each concentration in each experiment.

Table 26. Summary of bacterial reverse mutation assay results of LegH Prep (Experiment II; up to 5,000 µg/plate) – Mutation factor (Reyes *et al.*, 2023)

LegH Prep	TA98		TA100		TA1535		TA1537		WP2 uvrA	
µg/plate	**−S9**	**+S9**	**−S9**	**+S9**	**−S9**	**+S9**	**−S9**	**+S9**	**−S9**	**+S9**
Water	1	1	1	1	1	1	1	1	1	1
31.6	1	1	1.3	1	1.1	1	1	1.2	0.8	0.9
100	1	0.9	1.2	1.1	1.4	1.1	0.6	1.3	1	0.7
316	0.9	0.7	1.3	0.9	1	1	0.7	1.1	1.1	0.8
1000	0.9	1	1.1	1	1	1	0.8	1.1	1	0.9
2500	1	0.9	1.2	1	0.9	1	0.5	1.3	1	0.8
5000	1.1	0.8	1.3	0.9	1	1.1	0.4	1	1	1
4-NOPD	11.4	NA	9.7	NA	168.2	NA	5	NA	8.1	NA
2-AA	NA	33.6	NA	9.3	NA	14	NA	7.8	NA	3.4

2-AA = 2-aminoanthracene; LegH = soy leghemoglobin protein; NA = data not applicable; 4-NOPD = 4-nitro-*o*-phenylenediamine.

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Previously to confirm the lack of genotoxicity of LegH Prep in GRN No. 737, Fraser *et al.* (2018) conducted a bacterial reverse mutation assay to evaluate the genotoxicity of the LegH Prep. The assay was conducted under GLP, and the protocol was based on OECD Guideline 471 and ICH Guidelines S2A and S2B. Five bacterial strains: *Salmonella typhimurium* (TA98, TA100, TA1535 and TA1537) and *Escherichia coli* (WP2 uvrA) were used to evaluate the potential genotoxicity of LegH Prep according to the plate incorporation and reincubation methods, in the presence and absence of a metabolic activation system (S9). Sterile water was used as a negative control, whereas five known mutagens: NaN_3, ICR 191 acridine, daunomycin, MMS, and 2-AA were used as positive controls. The concentrations of LegH Prep with and without metabolic activation were 0, 23.384, 74, 233.84, 740, 2338.4, 7400, 23,384 and 74,000 µg/plate, corresponding to 1.58, 5.0, 15.8, 50, 158, 500, 1580, and 5000 µg LegH protein/plate, respectively. The authors reported that LegH Prep treatment did not increase the mean number of revertant colonies *per* plate in any of the strains tested, either in the absence or presence of the S9 metabolic activating system. The authors concluded that "LegH Prep was nonmutagenic in the bacterial reverse mutation test, which evaluated 5 strains of bacteria and 8 different concentrations of LegH Prep up to a maximum dose of 5000 µg LegH/plate" (Fraser *et al.*, 2018).

Fraser *et al.* (2018) also evaluated the ability of the LegH Prep to induce structural or numerical (polyploid or endoreduplicated) chromosomal aberrations, utilizing a chromosomal aberration study in human peripheral blood lymphocytes (HPBL). This test was in compliance with OECD 473[27] "*In Vitro* Mammalian Chromosome Aberration Test" and the European Commission Regulation (EC) No. 440/2008 B.10.[28] In accordance with the OECD guidelines, Experiment 1 utilized LegH protein (100 - 5000 µg/ml) which corresponds to HPBL cells being exposed to 0 – 74,000 µg/ml LegH Prep in the presence or absence of S9 for four hours. The positive controls utilized were ethyl methanesulfonate (EMS) without metabolic activation and cyclophosphamide with metabolic activation. The mitotic index was evaluated, as a decreased mitotic index can inhibit the ability to evaluate chromosomal aberrations. As compared to the negative control, the mitotic index did decrease below 70% with high concentrations of LegH (≥ 2500 µg/ml) without metabolic activation, but not with metabolic activation; however, there were no significant differences in the proliferation index with or without metabolic activation. Moreover, there were no significant increases in cells with structural or numerical chromosome aberrations with LegH protein treatment (≤ 5000 µg/ml), which corresponds to ≤ 74,000 µg/ml LegH Prep, with or without metabolic activations.

In Experiment 2, precipitation of LegH Prep occurred at concentrations of ≥500 µg/ml LegH Prep and the mitotic index values, relative to control, decreased below the 45% threshold at concentrations >1000 µg/ml LegH when the LegH Prep was incubated with the HPBL for 24 hours in the absence of metabolic activation. Therefore, only concentrations of ≤1000 µg/ml LegH were evaluated for chromosomal aberrations in HPBL when incubated for 24 hours in the absence of

[27] https://www.oecd.org/env/test-no-473-in-vitro-mammalian-chromosomal-aberration-test-9789264264649-en.htm; site visited on January 23rd, 2023.
[28] https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32008R0440&from=EN; site visited on January 23rd, 2023.

metabolic activation. In either Experiment 1 (with or without metabolic activation) or Experiment 2 (without metabolic activation), no significant increases in structural or numerical chromosomal aberrations were observed at any of the concentrations evaluated. The positive controls in both Experiment 1 and Experiment 2 worked as expected. The authors concluded that "LegH Prep is considered nonclastogenic in the *in vitro* mammalian chromosome aberration assay using HPBL" (Fraser *et al*., 2018).

6.6 Allergenicity

6.6.1 Allergenicity studies for LegH Prep

The allergenicity potential for LegH Prep and *P. pastoris* has been comprehensively evaluated. The LegH Prep that was the subject of GRN No. 737 was evaluated for potential allergenicity by a thorough literature review, bioinformatics sequence comparisons to known allergens or toxins, and *in vitro* pepsin digestion (Jin *et al.*, 2018). As stated earlier, the LegH Prep that is the subject of this GRAS notification is comparable in composition (*i.e*., the fat, carbohydrate, ash, and moisture content to the original LegH prep in GRN No. 737 (see Table 3). They, the two preparations, both contain comparable levels of LegH protein and other proteins [LegH Prep (2022) overall protein content (11.65%) is comparable to the original LegH Prep (GRN No. 737) product (11.36%)]. Moreover, both versions of LegH Prep utilize the same production organism, *P. pastoris*, and contain the same additional *P. pastoris* proteins (in both versions of LegH Prep, there is at least 65% LegH protein and the remainder of the LegH Prep contains *P. pastoris* proteins). As the fermentation media does not contain any protein, there are no proteins or peptides from the fermentation media that would be present. This LegH Prep has comparable LegH protein, overall protein content, and other macronutrient content as the original LegH Prep described in GRN No. 737 and the major difference is the amount of the LegH Prep that is to be added to food and the number of meat analogue foods in which LegH Prep is added. Therefore, allergenicity studies for the original LegH Prep in GRN No. 737 can demonstrate the lack of allergenic potential for the LegH Prep in this GRAS notification.

Another more recent allergenicity analysis was conducted by Reyes *et al.* (2021) to investigate the potential allergenicity and toxicity of *Pichia pastoris* proteins in IFI's LegH Prep 2.0. Although LegH Prep 2.0 is produced by the same production organism, *P. pastoris*, LegH Prep 2.0 is produced using a modified manufacturing process, yielding a higher *P. pastoris* content within the final formulation, and utilizes an additional heat treatment step to inactivate the potential growth of remaining *P. pastoris* cells. However, the rest of the manufacturing process utilizes the same raw materials and manufacturing steps as described in GRN No. 737. The authors conducted allergenicity studies, which included methods utilized by Jin *et al.* (2018), to compare proteins and composition of LegH Prep 2.0 to the original LegH Prep (GRN No. 737). Although it is a less purified version of the original LegH Prep and therefore contains more *P. pastoris* components, it is comparable in composition and macronutrient content, contains comparable levels of LegH protein (≥ 65%) and overall protein content, and utilizes the same production organism as described in GRN No. 737 (and as described for the LegH Prep that is the subject of this GRAS notification). Therefore, this study is relevant for establishing the lack of allergenic potential in this LegH Prep.

Jin *et al.* (2018) conducted a literature search in 2017 using four different digital public literature databases.[29] The literature searches utilizing the words/phrases "*Glycine max*", "allergy", and "heme" resulted in a publication from PubMed describing an impaired carotenoid and flavonoid biosynthetic pathway due to a mutation in *Arabidopsis* HY1 protein, which was discarded as the study was deemed irrelevant. A broader search indicated that soybean is a common food allergen in young children; however, it is rarely observed in adults. The recent recommendation from the FAO/WHO Expert Panel included removal of soybean from the list of priority allergenic foods for regulatory purposes based on modest prevalence and potency and very low severity (Joint FAO/WHO, 2021). There were also reports of allergy in seed-pod proteins, with several soy allergens having been identified exclusively in the seeds; however, LegH protein is only found in root nodules and is not present within seeds or seed pods and not known to be present in soy sprouts. Therefore, the authors concluded that people would not be allergic to this protein and there would be no need to conduct serum IgE binding studies. Moreover, there is no sensitive population to obtain serum to conduct such studies.

The authors also utilized the words "toxin" or "toxicity", which resulted in papers describing heme peroxidases and/or the heme oxygenase system that were not relevant to allergenicity for the LegH protein. When the authors utilized the terms "*Pichia pastoris*", the revised name "*Komagataella phaffii*", and the keyword "NOT recombinant" resulted in the authors concluding "the results demonstrated no link to allergy related to endogenous proteins from *Pichia pastoris*" (Jin *et al.*, 2018). Searches that involved a combination of "*Pichia pastoris*" and "toxin" yielded a vast number of references to publications describing expression genes for recombinant proteins, which were reduced when the word "recombinant" was excluded. One publication (Banerjee and Verma, 2000) tested 14 different strains of *Pichia pastoris* (*Komagataella phaffii*) for toxin expression, which reported there was no evidence of toxic activity in any of the strains examined. Other than a report evaluating the heterologous protein Kpkt, which is active in reducing microbial spoilage in wine production, no other publications were identified that were related to toxicity. The authors concluded "no evidence was found indicating that *P. pastoris* had endogenous toxic proteins" (Jin *et al.*, 2018).

Reyes *et al.* (2021) performed protein analysis using shotgun proteomics (which utilized LC-MS/MS), bioinformatics, and a pepsin *in vitro* digestion assay on leghemoglobin and residual host proteins. The top-down proteomics study was done to characterize all four isoforms of soy leghemoglobin. The mass spectrometric raw data were analyzed with the MaxQuant software, which is based on the high-quality genome sequence of *P. pastoris* CBS7435, with the sequence of LegH protein being manually added to the database. The false discovery rate (FDR) was set to 0.01 for protein and peptides, and all identified proteins were quantified using a label free quantification (LFQ) algorithm of MaxQuant. Intake protein mass spectrometry (IPMS) was conducted to assess whether the LegH protein manufactured with the modified process (LegH Prep 2.0) was equivalent to LegH protein manufactured from the original process (LegH Prep). The authors reported that all samples, for both processes, after deconvolution the mono-isotopic mass

[29] The four databases were: PubMed (https://www.ncbi.nlm.nih.gov/pubmed/); Web of Science™ Core Collection (v.5.24) (http://apps.webofknowledge.com/); Scopus® (https://www.scopus.com/); and National Agricultural Library Catalog (AGRICOLA, https://agricola.nal.usda.gov/) (Jin *et al.*, 2018).

of the only major peak was reported as 15384.04 Da, which was consistent with soy LegH C2 with N-terminal methionine excision (calculated mass = 15384.0399 Da) and showed consistency with the LegH C2 isolated from root soy nodule. The authors stated, "There is no other high-intensity peak observed within the mass range of 60 kD, though some minor peaks were observed in LegH Prep 2.0, which is consistent with the mass spectrometry analysis from the original LegH Prep" (Reyes *et al.,* 2021). The authors concluded that even though these minor peaks were not characterized, they were most likely protein fragments. The authors reported that glycosylation was not observed, nor any other post-translational modification was observed on LegH Prep 2.0, consistent with LegH Prep. Top-down proteomic analysis revealed all four isoforms were identified in each purified sample from the soy root nodule, and for each of them, the major proteoform identified was the unmodified versions, which was, "consistent with the result from the IPMS experiment" (Reyes *et al.*, 2021).

Residual *Pichia pastoris* host proteins in the LegH Prep 2.0 were characterized by mass spectrometry-based quantitative shotgun proteomics using three independent production lots. The relative abundance of each identified protein was estimated by using the total label-free quantification (LFQ[30]) intensity of each protein from a single sample and normalizing it to the total intensity of the identified proteins in the sample. To evaluate the allergenicity and toxicity of the host cell proteins, all the proteins from *P. pastoris* with the relative abundance greater than 1% were subjected to a detailed bioinformatics analysis, consistent with the methodology used in GRN No. 737, which received a "no objection" letter from the FDA (FDA, 2017b). A protein was included in further bioinformatics searches if it were present at an abundance higher than 1% in any batch based on the shot-gun proteomics data. To this end, the protein sequences were based on the genome sequence of *P. pastoris* CBS7435. To evaluate the consistency between the LegH protein yielded from LegH Prep 2.0 and the desired sequence, peptide mapping was performed on the identified peptides in the shotgun proteomics analysis. In total, there were 21 peptides from LegH identified and the overall sequence coverage was 97.9% with most of the residues covered, except the N-terminal methionine and two C-terminal residues following Lys143. The authors reported this, "showed good consistency with the designated sequence for the recombinant LegH expression, no mutation or deletions were observed" (Reyes *et al.,* 2021).

Bioinformatic and sequence database searches were conducted to analyze LegH and the LegH Prep (Jin *et al.*, 2018). The LegH Prep was subjected to a Coomassie stained SDS-PAGE electrophoresis, which resulted in ten visible bands (with each band representing ≥1% of the total protein fraction of LegH Prep as determined by densitometry) that were analyzed by LC-MS/MS. Proteomic analysis revealed that LegH Prep contained 17 additional *P. pastoris* proteins that are normally expressed by the yeast, excluding LegH protein. Each protein sequence was used to identify full-length matches with the NCBI Entrez Protein database. Sequences are annotated with publications or with likely protein types. Using the BLASTP[31] (Basic Local Alignment Search Tool Protein) search algorithm, the NCBI protein database was searched using keyword limits ("allergen", "toxin", and "toxic") to identify potential allergen cross-reactivity or similarity to

[30] A mass spectrometry method used to determine the relative amount of proteins in two or more biological samples.
[31] BLASTP (https://blast.ncbi.nlm.nih.gov/Blast.cgi) is a tool used to make protein to protein comparisons.

newly discovered allergens or toxins. The full-length sequence of LegH protein was compared to all sequences in the NCBI–Entrez database without any keyword limit. LegH protein has a high degree of similarity and is closely related to plant (predominately other legumes) hemoglobin proteins which bind oxygen and shares approximately 26% homology with some chordate hemoglobin proteins. When the BLASTP search included keywords "allergen" or "toxic" there were no matches; however, when the words "allergen" and "toxin" were used, low identity matches were made, which were unlikely to be homologous due to the low identities and short alignments. Similarly, there were no results provided by BLASTP searching the complete NCBI protein database, that indicated the 17 additional *P. pastoris* proteins would pose a significant risk of allergy or toxicity when consumed. The authors reported that the 17 *P. pastoris* proteins were enzymes or enzyme inhibitors that are ubiquitous in nature, and that sequences related to the 17 found in LegH Prep are also found in many molds and yeasts, and included *Saccharomyces cerevisiae* and *Saccharomyces bayanus*, and *Saccharomyces boulardii*, found in the production of food (*e.g.*, bread and wine) or consumed as a probiotic (*i.e., S. boulardii*).

The 17 *P. pastoris* proteins were assessed for potential allergenicity using the AllergenOnline (AOL) database[32] as the primary tool for matches to allergenic sequences. Three comparison methods were utilized: a sliding 80 amino acid window FASTA search (E-score of 10, identity >35%) and an exact word search tool looking for eight amino acid exact matches were used, along with an overall FASTA3 full-length search (using E-scores of 10 and 1.0). The E-scores were set to limit irrelevant matches but identify potential cross-reactivity alignments. There was a lack of significant alignments of seven of the 17 *P. pastoris* proteins with proteins in the AOL database or with known allergens or toxins in the NCBI BLASTP database. The authors used as a negative control, a comparison of the *P. pastoris* proteins of interest to all known sequences in the NCBI protein database *via* BLASTP. Ten of the 17 protein sequences had matches with most of the protein sequence similarities sequence being more closely related to enzymes found in commonly consumed plants, the widely consumed *S. cerevisiae* yeast, mushrooms, or morels, than to potentially allergenic proteins. The authors stated that "the negative-control analysis identified proteins expected to be present in commonly consumed human foods, with no known allergenicity, that had far higher sequence-identity to the LegH and *Pichia* proteins of interest than any of the matches we found between these proteins and known allergens" (Jin *et al.*, 2018).

Reyes *et al.* (2021) also utilized bioinformatic analysis of sequence homology to investigate the potential allergenicity and toxicity of *P. pastoris* proteins present in LegH Prep 2.0 at ≥1% of the total protein fraction, consistent with methodology used in GRN No. 737, which received a "no objection" letter from the FDA (FDA, 2017b). The Bioinformatics analysis utilized in this study is the same method used in GRN No. 737. The use of shotgun proteomics indicated that LegH Prep 2.0 revealed eleven additional proteins besides LegH present at concentration of ≥1.0% or above for at least one of the three independent batches that were used. From a previous evaluation, four of these proteins had already been demonstrated to pose an improbable risk of allergenicity or toxicity (Jin *et al.*, 2018); therefore, only seven of the proteins were characterized

[32] The AllergenOnline database (http://www.allergenonline.org/) is a peer-reviewed allergen list and sequence database that is updated annually.

(Table 27). The average relative abundance values were calculated from three technical replicates of each batch and include the standard deviation.

Table 27. LegH Prep proteins found at ≥1% relative abundance (Reyes *et al.*, 2021)

Protein ID	Annotation	Abundance in 18 - 318	Abundance in 19 - 045	Abundance in 19 - 056
XP_002493171.1	NAD(+)-dependent formate dehydrogenase	0.018 ± 0.0083*	0.005 ± 0.0004	0.017 ± 0.0005
XP_002490864.1	Translational elongation factor EF-1 *alpha*	0.013 ± 0.0025	0.011 ± 0.0006	0.011 ± 0.0017
XP_002493194.1	Uroporphyrinogen decarboxylase	0.008 ± 0.0018	0.011 ± 0.001	0.011 ± 0.0017
XP_002490697.1	Hexokinase-2	0.013 ± 0.0037	0.005 ± 0.0004	0.008 ± 0.0002
XP_002491701.1	Acetate-CoA ligase	0.011 ± 0.0013	0.007 ± 0.0009	0.006 ± 0.0004
XP_002491345.1	Glyceraldehyde-3-phosphate dehydrogenase (GAPDH), isozyme 3	0.006 ± 0.0022	0.003 ± 0.0005	0.006 ± 0.0005
XP_002493065.1	Transketolase	0.006 ± 0.0009	0.009 ± 0.0021	0.008 ± 0.0008

*Average relative abundance ± the standard deviation.

These seven proteins were assessed for potential allergenicity using the AOL database, which is a tool that analyzes protein sequences and makes matches to potential allergenic sequences. This includes a sliding 80 amino acid window FASTA search, which includes an E-score of 10 and an identity >35%, an exact word search tool which looks for eight amino acid exact matches, and an overall FASTA3 full-length search, which utilizes an E-scores of 10 and 1.0) (Table 28). The E-scores were set to concomitantly limit irrelevant matches and identify potential cross-reactivity alignments. The BLASTP search algorithm available on the NCBI website was also used to locate potential allergen cross-reactivity or similarity to newly discovered allergens or toxins, as the database is updated or modified every few days. The sequence homology had "very good alignments" to known homologous proteins found in *Saccharomyces* sp. and *Aspergillus oryzae* (koji mold[33]), which suggests the allergenicity/toxigenicity risk from exposure to these proteins would be minimal (Reyes *et al.*, 2021). AOL and bioinformatics searches, the authors reported, "data of the proteins showed that no significant matches were found" (Reyes *et al.*, 2021); therefore, these proteins are not presented in Table 28 or Table 29. Of the seven proteins that exhibited an abundance greater than 1%, GAPDH was a significant match to the highly evolutionarily conserved GAPDH from wheat (*Triticum aestivum*) protein (Table 29). GAPDH

[33] A species of mold used in the saccharification of rice, sweet potato, and barley for the production of alcoholic beverages such as sake and shōchū.

was one of the proteins present that were also found in toxic organisms (Table 29); however, it is a ubiquitous protein present in humans and in organisms that are used in commonly consumed food products such as *A. oryzae* and *P. roqueforti*. Moreover, the identity scores were even high for *S. cerevisiae* as compared to *Triticum aestivum*, and highly consumed foods which contain *S. cerevisiae*, that do not exhibit allergenicity. Therefore, the authors concluded, "the likely risk of allergenicity and toxicity associated with *Pichia* GAPDH is relatively low" (Reyes *et al.*, 2021).

GAPDH is an enzyme that catalyzes the sixth step of glycolysis and thus participates in the conversion of glucose to energy; therefore, all organisms in nature with the ability for glycolysis will possess GAPDH. Given that GAPDH is a common, ancestrally conserved, house-keeping protein, which also has homologues present in *Aspergillus oryzae*, *Penicillium roqueforti*, and humans; it is expected some degree of sequence identity would be present among GAPDH from various sources. AOL contains examples of highly conserved proteins, like GAPDH, that are identified as allergens from certain species but the likelihood of functional cross-reactivity to versions of the same protein found in other species is unlikely because, in that scenario, patients sensitized and reactive to GAPDH from one species would experience allergic reactions on exposure to a wide range of species that express GAPDH. Moreover, clinical evidence of widespread cross-reactivity attributed to GAPDH does not exist. The high sequence identity to human GAPDH leads to serious questions about how these highly conserved proteins could be allergens from some species without patients also being reactive to their own highly identical human GAPDH.

Two GAPDHs have been identified as IgE-binding proteins among patients with inhalant occupational allergy to wheat (Sander *et al.*, 2011) and ingested pangasius catfish (Ruethers *et al.*, 2020). The GAPDH from wheat was a minor IgE-binding protein with binding to serum IgE from only 5% of patients with baker's asthma (Sander *et al.*, 2011). Similarly, GAPDH was a minor IgE-binding protein showing binding to serum IgE from only 6% of patients with allergies to pangasius catfish. Neither of these proteins have been demonstrated to be functional allergens capable of releasing mediators from basophils or provoking wheal-and-flare responses in skin prick tests. Therefore, these two GAPDHs are identified as putative allergens, utilizing the AOL system, exclusively on IgE-binding. It should be noted that several other publications have identified GAPDH as IgE-binding proteins from other, rather diverse species including manioc (Santos *et al*., 2011), shrimp (Khanaruksombat *et al.*, 2014), Mediterranean silverside fish () (González-Mancebo *et al*., 2014), and mango (Cardona *et al.*, 2018). However, these GAPDHs are not yet recognized by the IUIS and are not among the allergens in the AOL database. The GAPDH putative allergens have been assigned identifying names by the International Union of Immunological Science (IUIS) Nomenclature Committee as Tri a 34 from wheat and Pan h 13 from pangasius catfish.

Since fish are common causes of food allergy, a comment on the clinical importance of GAPDH as food allergens is warranted. GAPDH has been identified as an IgE-binding protein in two species of fish, pangasius catfish and Mediterranean silversides (González-Mancebo *et al*., 2014; Ruethers *et al.*, 2020). With fish, the major pan-allergen responsible for most fish allergies is a muscle protein called parvalbumin (Ruethers *et al.*, 2018). Several minor allergens exist in fish (Ruethers *et al.*, 2018). Serum IgE-binding for fish-derived GAPDH has only been reported for these two species of fish and is observed in only 6% of patients in the study by Ruethers *et al.* (2020). Additionally, parvalbumin is expressed at much higher levels in fish compared to GAPDH (Ruethers *et al.*, 2020). GAPDH is also a heat-sensitive allergen that would be neutralized by many food manufacturing processes (Ruethers *et al.*, 2020). The major argument against the clinical importance of GAPDH as a novel allergen from *Pichia* is the ubiquity of this enzyme in nature, its highly conserved sequence, and the general lack of evidence that any human consumers are allergic to the wide variety of species expressing potentially cross-reactive GAPDHs. Therefore, the potential allergenicity of GAPDH from *Pichia* based exclusively on partial bioinformatic sequence identity does not indicate allergenicity nor toxigenicity based on the ubiquity of GAPDH proteins in nature, its ancestral nature with highly conserved sequences, and its similarity with human GAPDH.

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Table 28. Assessment of potential allergenicity of LegH Prep 2.0 proteins (Reyes *et al*., 2021)

Search method	Assessment of potential allergenicity
	The 333 AA GAPDH, isozyme 3 (XP_002491345.1)
AOL full FASTA3	One highly significant match with 3.2E-97 E-value and 70% identity over 332 AA was identified to a putative allergen Tri a 34.0101, a GAPDH from *T. aestivum*, identified as an allergen based binding IgE in baker's asthma. There was no published allergic reaction to ingestion of Tri a 34.0101. *P. camemberti* is widely consumed with brie and other soft-ripened cheeses and *A. oryzae* is widely used in Asia as koji in soy sauce and soy paste fermentation, both of which have a GAPDH that is closer to the allergenic ones than is the *Pichia* protein XP_002491345.1. XP_002491345.1 has a very high sequence similarity to the GAPDH in *S. cerevisiae* with alignment E-score of 0.0 and 81% sequence identity. Therefore, cross-reactivity between XP_002491345.1 and Tri a 34.0101 is highly unlikely and the risk of allergy is likely to be very low.
AOL 80mer FASTA3	One alignment was found with >35% identity over 80 AA. It was the same a GAPDH from *T. aestivum* as the full-length FASTA, with highest scoring 80mers of 87.5% identity.
AOL Exact 8 AA	Multiple 8 AA matches were identified to the same glyceraldehyde-3-phosphate dehydrogenase from *T. aestivum* as the full-length FASTA.
NCBI BLASTP (Allergen)	Four high identity matches were found to GAPDHs of *P. Americana*, with an E-score ranging from 7E-160 to 0.0 and identities from 66% to 83%.
NCBI BLASTP (No keyword)	Many high scoring identity matches with homologous proteins from bacteria, yeasts, and animals, including those from *A. oryzae* (E-score 6E-165, identity 67% over 329 AA) and from *P. roqueforti* (E-score 5E-170, identity 69% over 328 AA).
Conclusion: There is a minimal risk of allergy based on the alignment with GAPDH from *T. aestivum*. Moreover, GAPDH is an ancestrally conserved, housekeeping protein, found ubiquitously in nature, with highly conserved sequences that possess similarity with GAPDH homologues present in *Aspergillus oryzae*, *Penicillium roqueforti*, and humans. Thus, the risk of allergy is likely to be very low.	

Table 29. Assessment of Potential Toxigenicity of significant match LegH Prep 2.0 proteins (Reyes *et al.*, 2021)

Search method	Assessment of potential toxigenicity
The 333 AA GAPDH, isozyme 3 (XP_002491345.1)	
NCBI BLASTP (Toxin or Toxic)	Many high scoring alignments were identified using toxin and toxic keywords to alcohol dehydrogenase enzymes of bacteria that are associated with toxicity (*V. cholerae*, cyanobacterium *Planktothrix*, *E. cloacae*, *E. coli* and *Streptococcus* sp.). However, there is no published evidence to support that these are proteins associated with toxicity.
Conclusion: There were alignments to proteins identified as being from toxic organisms, but only as common homologues. There is no published evidence to substantiate that these proteins identified are associated with toxicity. This is a widely spread conserved protein whose homolog can be also found in *A. oryzae* (E-score 6E-165, identity 67% over 329 AA), from *P. roqueforti*, and from human with E-score of 2E-165 and 68% sequence identity. Thus, the likely risk of toxicity is low.	

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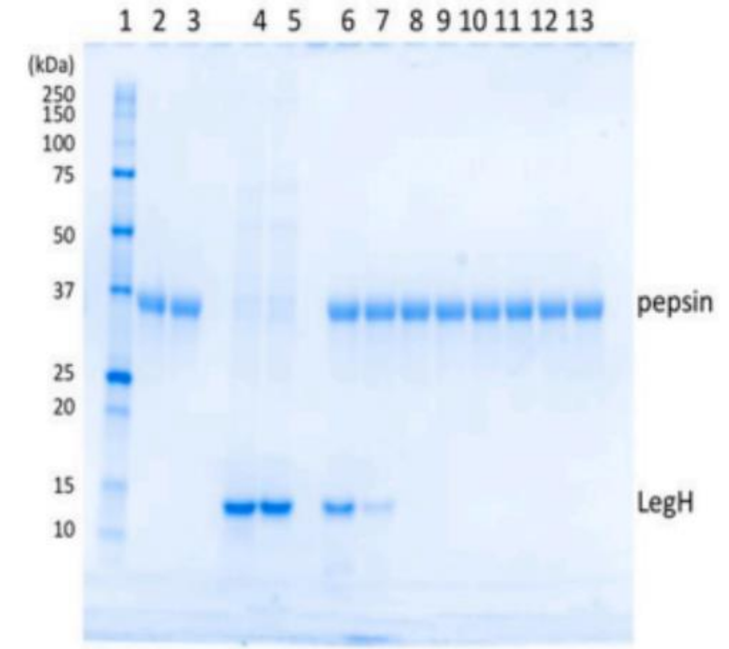
To ascertain whether any of the proteins within LegH Prep contained digestion-resistant polypeptides, an *in vitro* digestion assay was conducted using simulated gastric fluid (SGF). The *in vitro* pepsin digestion protocol used is comparable in procedure as to the one used in a multi-laboratory study demonstrating the power, versatility, and reproducibility of this procedure in a total of nine laboratories (Thomas *et al.*, 2004). This assay is utilized as there is a strong predictive value of the digestibility of a protein when comparing the stability of allergenic and non-allergenic dietary proteins (Bannon *et al.*, 2003). To simulate gastric fluid, assays were conducted at pH 2.0 with pepsin-protein ratios of 10:1 (U/µg) or 1:1 (U/µg), and the proteins in LegH Prep were compared to three control proteins [hemoglobin (Hb), bovine serum albumin (BSA), and ovalbumin (OVA)]. Hb and BSA were digested within 30 seconds, whereas OVA possessed more stability, with 10% visually stainable full-length protein band remaining after 30 minutes of digestion using 10 U of pepsin. The original LegH protein was rapidly digested by pepsin, to below 10% of the band intensity detected in a control digestion, in two minutes. In addition, the 17 *P. pastoris* proteins found in the original LegH Prep product were digested and no longer visible at 0.5 minutes under the conditions of the *in vitro* digestion assay. The authors concluded that:

> [T]he results demonstrated that the LegHb [LegH] protein and the associated *Pichia pastoris* proteins were more than 90% digested within 2 min in SGF plus pepsin at 37°C, at ratios of either 10 units or 1 unit of pepsin activity *per* 1 µg of total protein, based on Coomassie Blue staining detection (Jin *et al*., 2018).

In summary, based on the work to assess the potential allergenicity of the original LegH Prep product, the authors concluded "…our analysis found no evidence to suggest that food products containing the LegH protein and associated minor components from *Pichia pastoris* pose any significant risk of allergy or toxicity to consumers" (Jin *et al.*, 2018).

In addition, Reyes *et al.* (2021) conducted an *in vitro* digestion assay to determine if any of the protein of interest might include digestion-resistant polypeptides (Thomas *et al.*, 2004). The acidic (pH = 2) gastric fluid environment was simulated using pepsin-protein ratios of 10:1 (U/µg) or 1:1 (U/µg), and compared the proteins found in the LegH Prep to pepsin only or LegH Prep alone. The results show the protein was digested to non-detectable levels within two minutes (Figure 2). The authors concluded,

> Collectively, the literature search, the bioinformatics data (sequence homology) and the pepsin digest assay show that consumption of GAPDH from *P. pastoris* is unlikely to result in a food allergenic reaction. Overall, the sequence-related putative allergens identified in this search were not potent, nor were any of them known to be allergenic when orally consumed. As the LegH protein only makes up to 0.8% in the final plant-based meat products the abundance of each *Pichia* protein is in trace amounts. Furthermore, no published evidence was found to indicate that these proteins are toxins. Based on results from the proteomic analysis none of these proteins pose a significant risk of allergenicity (Reyes *et al.*, 2021).

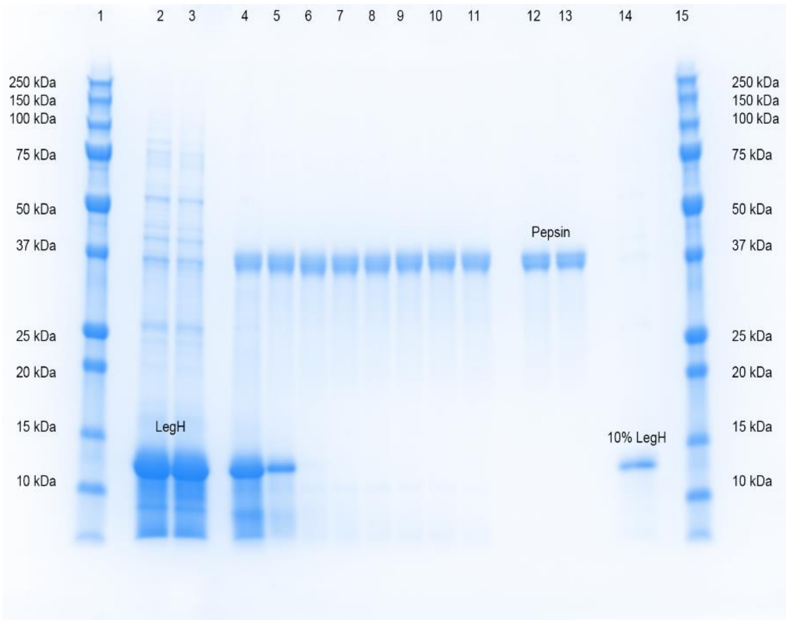


1. Ladder
2. Pepsin, 0 min
3. Pepsin, 60 min
4. LegH Prep 2.0, 0 min
5. LegH Prep 2.0, 60 min
6. Pepsin + LegH Prep, 0 min
7. Pepsin + LegH Prep, 0.5 min
8. Pepsin + LegH Prep, 2 min
9. Pepsin + LegH Prep, 5 min
10. Pepsin + LegH Prep, 10 min
11. Pepsin + LegH Prep, 20 min
12. Pepsin + LegH Prep, 30 min
13. Pepsin + LegH Prep, 60 min

Figure 2. Pepsin digestion LegH Prep

6.6.2 FDA requested pepsin digestibility assay for LegH Prep

In response to a request made by the FDA, a pepsin digestibility assay was conducted by IFI to evaluate the digestibility of the current LegH Prep (IFI, 2022). To this end, IFI subjected the LegH Prep to SGF using an *in vitro* digestibility assay as previously described by Ofori-Anti *et al.* (2008), which was modified by increasing the ratio between LegH Prep and pepsin by 100-fold (1:10 to 10:1). The digestion reaction contained 4.0 mg of LegH Prep (2.5 mg/ml) and 400 units of pepsin (20 units/ml), which was incubated at pH = 2.0 at 37.0°C for 0-60 minutes. The subsequent resulting digestion products after the different incubation periods were analyzed by SDS-PAGE (Lanes 4-11) (Figure 3). Controls that were utilized included LegH protein-only (Lanes 2 and 3) and pepsin only (Lanes 12 and 13) samples, which were incubated in SGF under the same temperature and incubation conditions to ensure they were stable. A diluted (10% of original concentration before digestion) sample of LegH Prep (Lane 14) was also loaded onto the gel for comparison purposes. The results of SDS-PAGE analysis indicated that all proteins, including LegH protein and other *P. pastoris* proteins disappeared within five minutes after digestion initiation (Figure 3). However, there was no difference in gel band intensity before or after the 60-minute SGF incubation period in lanes containing exclusively LegH or pepsin, as evidenced by SDS-PAGE analysis, which indicated both LegH protein and pepsin were stable in SGF during the incubation period (Figure 3). The authors concluded, "no protease resistant protein was present in the LegH preparation sample, and more than 98% of the LegH was digested within 5 min. The result is similar to the previous study on the LegH preparation" (IFI, 2022).



1. Ladder
2. LegH Prep + SGF, 0 min
3. LegH Prep + SGF, 0 min
4. Pepsin + LegH Prep, 0 min
5. Pepsin + LegH Prep, 0.5 min
6. Pepsin + LegH Prep, 2 min
7. Pepsin + LegH Prep, 5 min
8. Pepsin + LegH Prep, 10 min
9. Pepsin + LegH Prep, 20 min
10. Pepsin + LegH Prep, 30 min
11. Pepsin + LegH Prep, 60 min
12. Pepsin + SGF, 0 min
13. Pepsin + SGF, 60 min
14. 10% Diluted LegH Prep
15. Ladder

Figure 3. Pepsin digestion of the LegH preparation

6.7 Summary and conclusions

Based on the literature and database evaluations, bioinformatics and comparisons to known allergens and toxins, and the *in vitro* digestibility assays, there is minimal likelihood of allergic reactions when consuming LegH Prep in the intended food categories of use.

6.8 Regulatory status

The GRAS notices for LegH Prep and the production organism, *Pichia pastoris*, are listed in Table 30. LegH Prep has been the subject of two GRAS conclusions, one for LegH derived from the *P. pastoris* strain, Bg10, (GRN No. 540), which was submitted by IFI (FDA, 2015), and the other for the original LegH Prep derived from a strain of *P. pastoris*, MXY0291 (GRN No. 737), which was again submitted by IFI (FDA, 2017b). While the first GRAS ceased to be evaluated by the FDA at the request of the notifier, GRN No. 737 received a "no questions" response from the FDA. The regulation of LegH and *Pichia pastoris* under the 21 Code of Federal Regulations (CFR) are listed in Table 31. LegH Protein is approved as a color additive (21 CFR 73.520) and dried P. pastoris is approved up to 10% weight for use in feed formulations of broilers (21 CFR 573.750).

The *P. pastoris* parent strain obtained from the ATCC (NRRL Y-11430) is classified as a "Biosafety Level 1" organism, a classification indicating that the organism is well-characterized and "not known to consistently cause disease in immunocompetent adult humans, and present minimal potential hazard to laboratory personnel and the environment" (CDC, 2009). *P. pastoris* is approved as a production organism for a phospholipase C enzyme preparation (GRN No. 204) as a processing aid for degumming edible vegetable oils (Ciofalo *et al.*, 2006), as notified to FDA under the FDA GRAS notification process (FDA, 2006; Tarantino, 2006). *P. pastoris* has also been concluded as GRAS (GRN No. 1001) for a myoglobin preparation that is expressed from the *Bos taurus* myoglobulin gene (FDA, 2021b) and for a pepsin A enzyme preparation (GRN No. 1025), in response to GRAS notification the FDA had no questions while FDA's response to GRN No. 1025 is still pending (FDA, 2021c). Two nontoxigenic/nonpathogenic strains of *P. pastoris* are used in the manufacture of rebaudioside M, a general purpose and tabletop sweetener, for use in foods (FDA, 2016; Adams and Keefe, 2017; EFSA, 2019a). The FDA also provided a "no questions" response to GRN No. 667, for the use of rebaudioside M as a tabletop sweetener and as a general-purpose non-nutritive sweetener for incorporation into foods in 2016 (FDA, 2017a). The European Union's Scientific Committee on Food has ruled that 6-phytase and 3-phytase, both produced by genetically modified strains of *P. pastoris*, are approved for use in poultry, such as laying and fattening hens (EFSA, 2015; EFSA, 2016). The European Food Safety Authority (EFSA) has stated that *K. pastoris* (*syn*., *P. pastoris*) has a qualified presumption of safety (QPS) when the species is used for enzyme production; however, the final enzyme product must not contain residual *P. pastoris* (EFSA, 2017). EFSA also determined that phospholipase C derived from a genetically modified *P. pastoris* did not pose a safety concern and the likelihood for an allergic reaction would be low (EFSA, 2019a). EFSA also made the same determination for rebaudioside M produced by and purified from *P. pastoris* for use as a food additive (EFSA, 2019b).

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Table 30. GRAS notices for *Pichia pastoris*

GRAS Notice No.	Substance	Intended Use	FDA response
204	Phospholipase C enzyme preparation from *Pichia pastoris* expressing a heterologous phospholipase C gene	As an enzyme in degumming vegetable oils for food use.	FDA has no questions.
540	Soybean leghemoglobin from *Pichia pastoris*	As a flavor in meat analogue products at a level not to exceed 1.4 *percent* in the finished product.	At the notifier's request, FDA ceased to evaluate this notice.
667	Rebaudioside M	Intended to be used as a tabletop sweetener and as a general purpose non-nutritive sweetener for incorporation into foods in general, other than infant formulas and meat and poultry products, at *per* serving levels reflecting good manufacturing practices and principles, in that the quantity added to foods should not exceed the amount reasonably required to accomplish its intended technical effect.	FDA has no questions.
737	Soy leghemoglobin preparation from a strain of *Pichia pastoris*	Resubmission of GRN No. 540. For use at levels up to 0.8% soybean leghemoglobin protein to optimize flavor in ground beef analogue products intended to be cooked.	FDA has no questions.
1001	Myoglobin preparation from a strain of *Pichia pastoris* expressing the myoglobin gene from *Bos taurus*	Intended to impart flavor and aroma at levels up to 2% myoglobin in ground meat and poultry analogue products.	FDA has no questions.
1025	Pepsin A enzyme preparation produced by *Pichia pastoris* to overexpress the gene encoding pepsin A	Intended for use as an enzyme at a maximum level of 34 mg total organic solids/kg raw material, in the production of hydrolyzed vegetable and animal protein ingredients.	Pending

Table 31. Regulation of *Pichia pastoris* under the Code of Federal Regulations (CFR)

21 CFR	Substance	Intended Use
73.520	Soy leghemoglobin	(a) **Identity**. (1) The color additive soy leghemoglobin is a stabilized product of controlled fermentation of a non-pathogenic and non-toxicogenic strain of the yeast, *Pichia pastoris*, genetically engineered to express soy leghemoglobin protein. Soy leghemoglobin protein is the principal coloring component of the color additive and imparts a reddish-brown color. (2) Color additive mixtures made with soy leghemoglobin may contain only those diluents that are suitable and are listed in this subpart as safe for use in color additive mixtures for coloring foods. (b) **Specifications**. Soy leghemoglobin shall conform to the following specifications and shall be free from impurities, other than those named, to the extent that such impurities may be avoided by good manufacturing practice: (1) Soy leghemoglobin protein purity on protein basis (weight/weight), not less than 65 *percent*, as determined by sodium dodecyl sulfate-polyacrylamide gel electrophoresis. (2) Lead, not more than 0.4 milligrams *per* kilogram (mg/kg) (0.4 parts *per* million (ppm)). (3) Arsenic, not more than 0.05 mg/kg (0.05 ppm). (4) Mercury, not more than 0.05 mg/kg (0.05 ppm). (5) Cadmium, not more than 0.2 mg/kg (0.2 ppm). (c) **Uses and restrictions.** Soy leghemoglobin may be safely used in ground beef analogue products such that the amount of soy leghemoglobin protein does not exceed 0.8 *percent* by weight of the uncooked ground beef analogue product. (d) **Labeling**. The label of the color additive and of any mixture prepared therefrom intended solely or in part for coloring purposes must conform to §70.25 of this chapter. (e) **Exemption from certification**. Certification of this color additive is not necessary for the protection of the public health, and therefore batches thereof are exempt from the certification requirements of section 721(c) of the Federal Food, Drug, and Cosmetic Act.
573.750	*Pichia pastoris* dried yeast	a) **Identity**. The food additive *Pichia pastoris* dried yeast may be used in feed formulations of broiler chickens as a source of protein not to exceed 10 *percent* by weight of the total formulation. (b) **Specifications**. The additive shall conform to the following *percent*-by-weight specifications: (1) Crude protein, not less than 60 *percent*. (2) Crude fat, not less than 2 *percent*. (3) Crude fiber, not more than 2 *percent*. (4) Ash, not more than 13 *percent*. (5) Moisture, not more than 6 *percent*. (c) **Use**. To ensure safe use, the labeling of the additive and any feed additive supplement, concentrate, or premix prepared therefrom shall bear, in addition to other required information, the name of the additive, directions for use to provide not more than 10 *percent* by weight of the total ration, and the statement "Caution: Not to be used in layers or other poultry intended for breeding."

6.9 Safety evaluation summary

LegH Prep is an ingredient derived from the methanotrophic yeast, *Pichia pastoris* (*Komagataella pastoris*), which is to be used as a flavor and nutrient component (providing iron) in plant-derived meat analogue products up to 2.0%. The LegH protein contained within LegH Prep exhibits at least 65% purity and comprises up to 15% of the LegH Prep. The LegH Prep described in this GRAS notification has not changed in terms of how it is manufactured and has comparable composition, specifications, and stability as the LegH Prep described in GRN No. 737. The only thing that has changed with this LegH Prep, with respect to LegH Prep described in GRN No. 737, is the number and types of meat analogue products into which LegH Prep is to be added and the maximum amount that is to be added to these products (≤2.0% as opposed to ≤0.8% in GRN No. 737). LegH Prep passed all microbial and heavy metal specifications, and all of the compositional specifications set by IFI, which are the same as those provided for LegH Prep in GRN No. 737. LegH Prep exhibited an average LegH protein content of 11.68%, a LegH protein purity of 85.2%, a salt concentration of 75 mM, a total protein content of 12.96%, a fat content of 0.10%, a carbohydrates content of 2.41%, an ash content of 1.25%, consisting of 16.92% solids, and a moisture content of 83.10%. These compositional parameters were comparable to those in the LegH Prep that was the subject of GRN No. 737. Stability studies also indicated the neat substance is stable for up to 91 days at -20°C, and comparable to the LegH Prep in GRN No. 737. The potential allergenicity and toxicity of *Pichia pastoris* proteins in IFI's LegH Prep 2.0 was also included in this dossier to support the lack of toxicity and allergenicity of LegH Prep. Although LegH Prep 2.0 is a less purified version of the original LegH Prep and therefore contains more *P. pastoris* components, it is comparable in composition, macronutrient LegH protein (≥ 65%) and overall protein content and utilizes the same production organism as described in GRN No. 737 (and as described for the LegH Prep that is the subject of this GRAS notification). Therefore, studies for LegH Prep 2.0 were also utilized to establish the lack of allergenicity and toxigenicity of this LegH Prep.

The calculated EDI for LegH Prep at the 90th *percentile* of consumption is 40.43 mg/kg bw/day, which is based on the amount of dried LegH Prep (from dried solids) that would be required to be added to the selected food categories, described in Table 5, to get the desired LegH protein concentration in the meat analogues present in these foods. The LegH protein represents approximately 11.7% of the total hydrated version of LegH Prep; however, the NOAEL derived from the 90-day study utilized the dried LegH Prep (see Section 5.2), with dried solids representing approximately 17% of the LegH Prep. Therefore, to achieve the desired amounts of LegH protein within the selected foods provided in Table 5, and to calculate an accurate estimation of the EDI for the LegH Prep, the *percentage* of LegH protein within the dried solids (which is approximately 17% of the LegH Prep) was utilized, which provided the 90th *percentile* EDI of 40.43 mg/kg bw/day (Table 7).

The NOAEL for LegH Prep in rats is 4798.3 mg/kg bw/day in male rats and 5761.5 mg/kg bw/day in female rats (equivalent to 287,892 and 357,213 mg/day, respectively, in a 60 kg human). Normally, it is desirable to have a 100-fold safety factor between the Acceptable Daily Intake (ADI) of the substance and the experimentally achieved NOAEL from a 90-day safety study. However, LegH Prep and LegH protein have a documented history of use, and LegH Prep has been declared GRAS in GRN No. 737 (original LegH Prep); therefore, a 75-fold safety factor was deemed acceptable when calculating the ADI. Therefore, the ADI is 63.97 mg/kg bw/day based on a 75-fold safety factor and the NOAEL of the 90-day study conducted by IFI (4798 mg/kg bw/day). Using this 75-fold safety factor, the 90th *percentile* EDI is less than the ADI. Even when the 15% loss of the LegH protein observed over a four-day period in the 90-day study is taken into consideration, which would result in an ADI of 54.37 mg/kg bw/day from the rat study NOAEL, the 90th *percentile* of consumption of 40.43 mg/kg bw/day would still be within the new ADI.

LegH Prep was comprehensively evaluated for general toxicity in rats, palatability, genotoxicity, and potential allergenicity. The additional proteins in *Pichia pastoris* were also evaluated for allergenicity using bioinformatics and an *in vitro* pepsin digestibility assay. The strain of *P. pastoris* utilized, is derived with a lineage with an established history of use. Moreover, *P. pastoris* is a non-pathogenic and non-toxigenic organism used as a protein expression system for use in human food), potable water treatment, animal feed, and FDA approved therapeutics. LegH Prep also has a well-established history of use and has been GRAS since 2017 (GRN No. 737), which demonstrated that LegH Prep is safe, as well as the production organism. Similarly, the macronutrient content of this LegH Prep is consistent with the LegH Prep contained within GRN No. 737, which does not affect the above stated parameters. LegH Prep will be added to plant-derived meats such as beef-, pork-, pork sausage-, chicken tender/nugget- and chicken luncheon meat-type analogue products.

LegH Prep, like the original LegH Prep (GRN No. 737), did not exhibit any adverse effects, clinical signs of toxicity, or palatability issues. When IFI conducted a 14-day palatability study, LegH Prep was palatable, well tolerated, and did not induce any clinically significant effects up to 12,500 mg/kg day. The 90-day toxicity study also corroborated the lack of toxicity for the LegH Prep. All observations noted were considered incidental and of no toxicological significance, as there were no trends in observations that increased with dietary level. There were no changes in body weight or body weight gain or food consumption in male and female rats attributed to the dietary administration of LegH Prep. There were significant increases noted in some hematological, urinary, and thyroid hormone parameters; however, all statistically significant changes were within historical control ranges, and without histopathological correlate, were not considered adverse. Histopathological analysis revealed that there were no LegH Prep attributable macroscopic observations, organ weight changes, or microscopic findings. There were significant changes in the absolute and relative thymus weight, epididymides-to-body weight, and significant increases in kidney-to-brain weights; however, these changes did not exhibit a dose-dependent relationship and were not associated with any clinically adverse effects, and therefore deemed incidental.

Utilizing studies not conducted by IFI and provided by publicly available literature, the pathogenicity of *P. pastoris* was assessed in mice by oral gavage and IV administration, which resulted in no adverse effects, behavioral changes, or indications of deleterious effects as evidenced by histopathological analysis. Studies also indicated that *P. pastoris* does not induce any adverse effects in the GI tract or other organs, when consumed as viable, probiotic cultures. *P. pastoris* was also well tolerated and did not induce any adverse effects in broiler chicken chicks up to 49 days. Genotoxicity of LegH Prep was assessed using the bacterial reverse mutation assay and the chromosomal aberration assay in human lymphocytes. LegH Prep was found to be non-mutagenic and non-clastogenic in each assay.

The original LegH Prep in GRN No. 737 has been comprehensively evaluated for allergenicity, using literature and sequence database searches, bioinformatics. Sequence comparison to known allergens or toxins revealed that LegH and all proteins that had greater than 1% expression of the total proteome did not pose an allergenic or toxicological concern. The most abundant *Pichia* proteins within LegH Prep are ubiquitous in nature and contain high sequence identity to homologues in yeast and molds that are commonly used in making cheese, wine, bread, and beer. AOL search results and bioinformatics showed no significant matches, except for the highly evolutionarily conserved GAPDH protein from wheat (*Triticum aestivum*). There was a high degree of alignment to homologous proteins to *Saccharomyces* sp. and *Aspergillus oryzae*, and proteins present in toxic organisms, which suggests the allergenicity/toxigenicity risk from exposure to these proteins would be minimal. GAPDH is a common, ancestrally conserved, house-keeping protein, which also has homologues present in *Aspergillus oryzae*, *Penicillium roqueforti*, and humans. GAPDH is not a known toxic protein. The most abundant *Pichia* proteins within LegH Prep are ubiquitous in nature and contain high sequence identity to homologues in yeast and molds that are commonly used in making cheese, wine, bread, and beer. The results of *in vitro* pepsin digestibility assays also confirmed *P. pastoris* does not contain any digestion-resistant peptides. Moreover, the *in vitro* digestibility study conducted by IFI, in which a 100-fold increase in the ratio of LegH Prep and pepsin, indicated that there are no protease resistant proteins within this LegH Prep.

Thus, based on the experimental data establishing the LegH Prep NOAEL at 4798.3 mg/kg/day, the Expert Panel agrees to a safety factor of 75 and established a conservative acceptable daily intake (ADI) of LegH Prep 54.37 mg/kg bw/day or 3262.2 mg/day for a 60 kg individual, and that LegH Prep is safe to consume at the 90th *percentile* of 40.43 mg/kg bw/day or 2,425.8 mg/day for a 60 kg individual, but greater levels have not been evaluated.

6.10 GRAS Panel statement

At the request of IFI, a Panel of Experts (the "GRAS Panel") of independent scientists, qualified by their scientific training and relevant national and international experience in the safety evaluation of food ingredients, conducted a critical and comprehensive assessment of data and information pertinent to the safety of soy leghemoglobin preparation (LegH Prep) produced from the methylotrophic yeast *P. pastoris* when used as used as an ingredient to be used as a flavor and nutrient component (i.e., iron) in meat analogue products, replacing meat, as described in the GRAS dossier, to evaluate the LegH Prep status as Generally Recognized As Safe (GRAS) based on scientific procedures. The GRAS Panel consisted of the below-signed scientific experts: Eric A. Johnson (Eric A. Johnson Consulting, LLC); Professor Emeritus (Department of Food Science) and Director Emeritus (Food Research Institute) Michael W. Pariza (University of Wisconsin-Madison); and Professor Emeritus Stephen L. Taylor (University of Nebraska).

The GRAS Panel critically evaluated a comprehensive package of publicly available scientific data and information compiled from a search of the scientific literature through March 2023 by the Burdock Group and from IFI documents, and summarized by the Burdock Group in a dossier entitled “Dossier in support of the generally recognized as safe (GRAS) status of a *Pichia pastoris* based food ingredient containing soy LegH protein for human consumption under the conditions of use cited herein“ that included an evaluation of the available scientific data and information (favorable and unfavorable) relevant to the safety of intended use of LegH Prep manufactured from the methylotrophic yeast *Pichia pastoris*. The dossier included information on the characterization, identity and purity of the LegH Prep ingredient, manufacturing information, ingredient specifications, analytical data, stability information, conditions of use, consumption information and safety data. IFI assures Burdock Group that all relevant, unpublished information in its possession related to the safety of LegH Prep has been supplied to Burdock Group and has been summarized in this dossier. This dossier and supporting documentation were made available to the GRAS Panel. The GRAS Panel also independently evaluated other materials deemed appropriate and necessary and unanimously concluded LegH Prep GRAS under the stated intended conditions of use. The following page contains the conclusion statement and signatures of the GRAS Expert Panel.

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8. CONCLUSION

Following a critical evaluation of the information available, the Expert Panel has determined that, based on common knowledge throughout the scientific community knowledgeable about the safety of substances directly or indirectly added to food, there is reasonable certainty that LegH Prep produced using *Pichia pastoris*, and produced in accordance with current Good Manufacturing Practice (cGMP), is safe under the intended conditions of use, and is therefore Generally Recognized As Safe (GRAS), by scientific procedures, when used as an ingredient to be used as a flavor and nutrient component (*i.e.*, iron) in meat analog products, replacing meat so that total daily consumption of LegH Prep from all sources up to 2,425.8 mg LegH Prep/day, but higher levels of use have not been evaluated. In particular, the Expert Panel has evaluated the proposed use of LegH Prep at specified levels in the foods listed in Table 8 of this document and has concluded that such use results in an estimated intake that is below the ADI of 3262.2 mg/day LegH Prep and is therefore Generally Recognized As Safe (GRAS).

It is our opinion that other experts qualified by scientific training and experience to evaluate the safety of food and food ingredients would concur with these conclusions.

9. SIGNATURES

[redacted]

5 Dec 2023

Eric A. Johnson, Sc.D.
Eric A. Johnson Consulting, LLC

Date

[redacted]

29 Nov 2023

Michael W. Pariza, Ph.D.
Michael W. Pariza Consulting, LLC
Professor Emeritus, Department of Food Science
Director Emeritus, Food Research Institute
University of Wisconsin-Madison

Date

[redacted]

24 November 2023

Steve L. Taylor, Ph.D.
Taylor Consulting, LLC
Professor Emeritus, Department of Food Science & Technology
University of Nebraska-Lincoln

Date

Part 7: Supporting Data and Information

A comprehensive search[34] of the scientific literature was conducted through November, 2023 for safety and toxicity information on LegH Prep, which contains soy leghemoglobin isolated from the root nodules of the soy plant (*Glycine max*) derived from the methylotrophic yeast, *P. pastoris*, and related substances and was described in Part 6 of this GRAS notice.

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[34] Relevant literature cited in the electronic database search was reviewed. Literature not cited in the search or literature published subsequent to the search, may not have been included in the review.

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7.2 Appendices

7.2.1. Appendix I. Specifications for individual LegH Prep Lots

			Batch Analysis Results				
Analysis	**Method**	**Specification**	**LH-20-151-152-190FG**	**LH-20-163-164-190FG**	**LH-21-227-228-190FG**	**LH-22-313-314-190HF**	**LH-22-377-378-190HG**
Soy LegH Protein (*w/w*)[1]	HPLC/UPLC*	6 – 15%	8.5%	8.0%	9.9%	10.8	10.6
Soy LegH Protein Purity (*w/w*)[2]	SDS-PAGE*	≥65%	79%	84%	91%	92	80%
Calculated salt concentration (mM)	AOAC 983.14	≤250	125	113	60	41	36
Total Protein (*w/w*)[3]	AOAC 991.20.I	≤16%	11.40%	12.13%	12.00%	14.5	14.78%
Fat (*w/w*)	AOAC 989.05	≤2%	0.18%	0.1%	0.1	0.10	0.02
Carbohydrates (*w/w*)	Calculation (Ref AOAC 986.25)	≤6%	1.54%	3.2%	1.6	1.8	3.9
Ash (*w/w*)	AOAC 945.46	≤4%	1.17%	1.7%	0.9	1.1	1.4
Solids (*w/w*)[4]	Calculated as 100 - Moisture	≤26%	15.25%	17.13%	14.6	17.5	20.1
Moisture (*w/w*)	AOAC 948.12	≥74%	84.75%	82.87%	85.4	82.6	79.9
pH	Internal method	6.0 – 9.7	6.5	6.8	9.4	9.2	6.5
Lead (ppm)	AOAC2015.01Mod<2232>	<0.4	<0.01	<0.01	<0.01	<0.01	<0.01
Arsenic (ppm)	AOAC2015.01Mod<2232>	<0.05	0.01	0.010	<0.01	0.02	0.02
Mercury (ppm)	AOAC2015.01Mod<2232>	<0.05	<0.005	<0.005	<0.005	<0.005	<0.005
Cadmium (ppm)	AOAC2015.01Mod<2232>	<0.2	<0.001	<0.001	0.001	<0.001	<0.001
Aerobic Plate Count (CFU/g)	MB073.06 (AOAC-RI PTM 051702)	<10,000	<10	<10	<10	<10	90
E. coli 0157:H7[5] (neg *per* 25 g)	MB217.01 (AOAC-RI # 100701)	Absent by test	Absent	Absent	Absent	Absent	Absent
Salmonella spp.[6] (neg *per* 25 g)	MB316.01 (AOAC-RI # 021201)	Absent by test	Absent	Absent	Absent	Absent	Absent
Listeria monocytogenes[7] (neg *per* 25 g)	MB217.01 (AOAC-RI # 100701)	Absent by test	Absent	Absent	Absent	Absent	Absent

* Internal Method

[1]Soy leghemoglobin (LegH) protein may exceed 15% if additional water (moisture) is removed during the concentration step of the manufacturing process. Additional concentration (*i.e.*, less water) does not change the composition of the dry solids.

[2]The balance of the proteins in the preparation is residual *Pichia* proteins.

[3]Estimated *via* Kjeldahl analysis for total nitrogen content.

[4]*Percent* solids specification is based on the sum of the maximum concentrations of total protein, fat, carbohydrates, and ash. Maximum total protein was calculated as the maximum soy leghemoglobin protein concentration divided by the minimum soy leghemoglobin protein purity.

AOAC = Association of Official Agricultural Chemists; AVG = average; CFU = colony forming units; HPLC = high performance liquid chromatography; mM = millimolar; neg = negative; ppm = parts *per* million; SDS-PAGE = sodium dodecyl sulfate polyacrylamide gel electrophoresis; w/w = weight/weight; UPLC = ultra-performance liquid chromatography.

7.2.2 Appendix II. Food Groups selected for the addition of LegH Prep

Food code	Description	Amount (mg/g)
20000000	Meat, NFS	10
21000100	Beef, NS as to cut, cooked, NS as to fat eaten	10
21000110	Beef, NS as to cut, cooked, lean and fat eaten	10
21000120	Beef, NS as to cut, cooked, lean only eaten	10
21001000	Steak, NS as to type of meat, cooked, NS as to fat eaten	10
21001010	Steak, NS as to type of meat, cooked, lean and fat eaten	10
21001020	Steak, NS as to type of meat, cooked, lean only eaten	10
21003000	Beef, NS as to cut, fried, NS to fat eaten	10
21101000	Beef steak, NS as to cooking method, NS as to fat eaten	10
21101010	Beef steak, NS as to cooking method, lean and fat eaten	10
21101020	Beef steak, NS as to cooking method, lean only eaten	10
21101110	Beef steak, broiled or baked, NS as to fat eaten	10
21101120	Beef steak, broiled or baked, lean and fat eaten	10
21101130	Beef steak, broiled or baked, lean only eaten	10
21102110	Beef steak, fried, NS as to fat eaten	10
21102120	Beef steak, fried, lean and fat eaten	10
21102130	Beef steak, fried, lean only eaten	10
21103110	Beef steak, breaded or floured, baked or fried, NS as to fat eaten	6.8
21103120	Beef steak, breaded or floured, baked or fried, lean and fat eaten	6.8
21103130	Beef steak, breaded or floured, baked or fried, lean only eaten	6.8
21104110	Beef steak, battered, fried, NS as to fat eaten	6.8
21104120	Beef steak, battered, fried, lean and fat eaten	6.8
21104130	Beef steak, battered, fried, lean only eaten	6.8
21105110	Beef steak, braised, NS as to fat eaten	10
21105120	Beef steak, braised, lean and fat eaten	10
21105130	Beef steak, braised, lean only eaten	10
21401000	Beef, roast, roasted, NS as to fat eaten	10
21401110	Beef, roast, roasted, lean and fat eaten	10
21401120	Beef, roast, roasted, lean only eaten	10
21401400	Beef, roast, canned	10
21407000	Beef, pot roast, braised or boiled, NS as to fat eaten	10
21407110	Beef, pot roast, braised or boiled, lean and fat eaten	10
21407120	Beef, pot roast, braised or boiled, lean only eaten	10
21410000	Beef, stew meat, cooked, NS as to fat eaten	10
21410110	Beef, stew meat, cooked, lean and fat eaten	10
21410120	Beef, stew meat, cooked, lean only eaten	10
21417100	Beef brisket, cooked, NS as to fat eaten	10
21417110	Beef brisket, cooked, lean and fat eaten	10

Food code	Description	Amount (mg/g)
21417120	Beef brisket, cooked, lean only eaten	10
21420100	Beef, sandwich steak, flaked, formed, thinly sliced	10
23200100	Veal, NS as to cut, cooked, NS as to fat eaten	1
23200110	Veal, NS as to cut, cooked, lean and fat eaten	1
23200120	Veal, NS as to cut, cooked, lean only eaten	1
23204010	Veal cutlet or steak, NS as to cooking method, NS as to fat eaten	1
23204020	Veal cutlet or steak, NS as to cooking method, lean and fat eaten	1
23204030	Veal cutlet or steak, NS as to cooking method, lean only eaten	1
23204200	Veal cutlet or steak, broiled, NS as to fat eaten	1
23204210	Veal cutlet or steak, broiled, lean and fat eaten	1
23204220	Veal cutlet or steak, broiled, lean only eaten	1
23205010	Veal cutlet or steak, fried, NS as to fat eaten	1
23205020	Veal cutlet or steak, fried, lean and fat eaten	1
23205030	Veal cutlet or steak, fried, lean only eaten	1
23210010	Veal, roasted, NS as to fat eaten	1
23210020	Veal, roasted, lean and fat eaten	1
23210030	Veal, roasted, lean only eaten	1
23220030	Veal patty, breaded, cooked	0.68
20000200	Ground meat, NFS	10
21500000	Ground beef, raw	10
21500100	Ground beef, cooked	10
21500310	Ground beef patty, cooked	10
23220010	Veal, ground or patty, cooked	0.9
22000100	Pork, NS as to cut, cooked, NS as to fat eaten	4
22000110	Pork, NS as to cut, cooked, lean and fat eaten	4
22000120	Pork, NS as to cut, cooked, lean only eaten	4
22000200	Pork, NS as to cut, fried, NS as to fat eaten	4
22000210	Pork, NS as to cut, fried, lean and fat eaten	4
22000220	Pork, NS as to cut, fried, lean only eaten	4
22000300	Pork, NS as to cut, breaded or floured, fried, NS as to fat eaten	2.72
22000310	Pork, NS as to cut, breaded or floured, fried, lean and fat eaten	2.72
22000320	Pork, NS as to cut, breaded or floured, fried, lean only eaten	2.72
22001000	Pork, pickled, NS as to cut	4
22002000	Pork, ground or patty, cooked	4
22002100	Pork, ground or patty, breaded, cooked	2.72
22201000	Pork steak or cutlet, NS as to cooking method, NS as to fat eaten	4
22201010	Pork steak or cutlet, NS as to cooking method, lean and fat eaten	4
22201020	Pork steak or cutlet, NS as to cooking method, lean only eaten	4
22201050	Pork steak or cutlet, battered, fried, NS as to fat eaten	2.72

Food code	Description	Amount (mg/g)
22201060	Pork steak or cutlet, battered, fried, lean and fat eaten	2.72
22201070	Pork steak or cutlet, battered, fried, lean only eaten	2.72
22201100	Pork steak or cutlet, broiled or baked, NS as to fat eaten	4
22201110	Pork steak or cutlet, broiled or baked, lean and fat eaten	4
22201120	Pork steak or cutlet, broiled or baked, lean only eaten	4
22201200	Pork steak or cutlet, fried, NS as to fat eaten	4
22201210	Pork steak or cutlet, fried, lean and fat eaten	4
22201220	Pork steak or cutlet, fried, lean only eaten	4
22201300	Pork steak or cutlet, breaded or floured, broiled or baked, NS as to fat eaten	2.72
22201310	Pork steak or cutlet, breaded or floured, broiled or baked, lean and fat eaten	2.72
22201320	Pork steak or cutlet, breaded or floured, broiled or baked, lean only eaten	2.72
22201400	Pork steak or cutlet, breaded or floured, fried, NS as to fat eaten	2.72
22201410	Pork steak or cutlet, breaded or floured, fried, lean and fat eaten	2.72
22201420	Pork steak or cutlet, breaded or floured, fried, lean only eaten	2.72
22210300	Pork, tenderloin, cooked, NS as to cooking method	4
22210310	Pork, tenderloin, breaded, fried	2.72
22210350	Pork, tenderloin, braised	4
22210400	Pork, tenderloin, baked	4
22210450	Pork, tenderloin, battered, fried	2.72
22301000	Ham, fresh, cooked, NS as to fat eaten	4
22301110	Ham, fresh, cooked, lean and fat eaten	4
22301120	Ham, fresh, cooked, lean only eaten	4
22400100	Pork roast, NS as to cut, cooked, NS as to fat eaten	4
22400110	Pork roast, NS as to cut, cooked, lean and fat eaten	4
22400120	Pork roast, NS as to cut, cooked, lean only eaten	4
22401000	Pork roast, loin, cooked, NS as to fat eaten	4
22401010	Pork roast, loin, cooked, lean and fat eaten	4
22401020	Pork roast, loin, cooked, lean only eaten	4
22402510	Fried pork chunks, Puerto Rican style	4
22411000	Pork roast, shoulder, cooked, NS as to fat eaten	4
22411010	Pork roast, shoulder, cooked, lean and fat eaten	4
22411020	Pork roast, shoulder, cooked, lean only eaten	4
23000100	Lamb, NS as to cut, cooked	20
23120100	Lamb, roast, cooked, NS as to fat eaten	20
23120110	Lamb, roast, cooked, lean and fat eaten	20
23120120	Lamb, roast, cooked, lean only eaten	20
23132000	Lamb, ground or patty, cooked	20
23150100	Goat, boiled	10

Food code	Description	Amount (mg/g)
23150200	Goat, fried	10
23150250	Goat, baked	10
23310000	Rabbit, NS as to domestic or wild, cooked	4
23311100	Rabbit, domestic, NS as to cooking method	4
23311120	Rabbit, NS as to domestic or wild, breaded, fried	4
23311200	Rabbit, wild, cooked	4
23321000	Venison/deer, NFS	20
23321100	Venison/deer, roasted	20
23321200	Venison/deer steak, cooked, NS as to cooking method	20
23321250	Venison/deer steak, breaded or floured, cooked, NS as to cooking method	20
23322400	Venison/deer, stewed	20
23326100	Bison, cooked	20
24100000	Chicken, NS as to part and cooking method, NS as to skin eaten	1
24100010	Chicken, NS as to part and cooking method, skin eaten	1
24100020	Chicken, NS as to part and cooking method, skin not eaten	1
24102000	Chicken, NS as to part, baked, broiled, or roasted, NS as to skin eaten	1
24102010	Chicken, NS as to part, baked, broiled, or roasted, skin eaten	1
24102020	Chicken, NS as to part, baked, broiled, or roasted, skin not eaten	1
24102050	Chicken, NS as to part, rotisserie, NS as to skin eaten	1
24102060	Chicken, NS as to part, rotisserie, skin eaten	1
24102070	Chicken, NS as to part, rotisserie, skin not eaten	1
24103000	Chicken, NS as to part, stewed, NS as to skin eaten	1
24103010	Chicken, NS as to part, stewed, skin eaten	1
24103020	Chicken, NS as to part, stewed, skin not eaten	1
24103050	Chicken, NS as to part, grilled without sauce, NS as to skin eaten	1
24103055	Chicken, NS as to part, grilled without sauce, skin eaten	1
24103060	Chicken, NS as to part, grilled without sauce, skin not eaten	1
24103070	Chicken, NS as to part, grilled with sauce, NS as to skin eaten	1
24103075	Chicken, NS as to part, grilled with sauce, skin eaten	1
24103080	Chicken, NS as to part, grilled with sauce, skin not eaten	1
24104049	Chicken, NS as to part, sauteed, skin eaten	1
24104051	Chicken, NS as to part, sauteed, skin not eaten	1
24107070	Chicken, NS as to part, fried, coated, skin / coating eaten	1
24107071	Chicken, NS as to part, fried, coated, skin / coating not eaten	1
24107080	Chicken, NS as to part, baked, coated, skin / coating eaten	1
24107081	Chicken, NS as to part, baked, coated, skin / coating not eaten	1
24120110	Chicken breast, NS as to cooking method, skin eaten	0.5
24120120	Chicken breast, NS as to cooking method, skin not eaten	0.5
24122130	Chicken breast, baked, broiled, or roasted, skin eaten, from raw	0.5

Food code	Description	Amount (mg/g)
24122131	Chicken breast, baked, broiled, or roasted, skin not eaten, from raw	0.5
24122140	Chicken breast, baked or broiled, skin eaten, from pre-cooked	0.5
24122141	Chicken breast, baked or broiled, skin not eaten, from pre-cooked	0.5
24122150	Chicken breast, baked or broiled, skin eaten, from fast food/ restaurant	0.5
24122151	Chicken breast, baked or broiled, skin not eaten, from fast food/ restaurant	0.5
24122160	Chicken breast, baked, broiled, or roasted with marinade, skin eaten, from raw	0.5
24122161	Chicken breast, baked, broiled, or roasted with marinade, skin not eaten from raw	0.5
24150210	Chicken thigh, NS as to cooking method, skin eaten	1
24150220	Chicken thigh, NS as to cooking method, skin not eaten	1
24152230	Chicken thigh, baked, broiled, or roasted, skin eaten, from raw	1
24152231	Chicken thigh, baked, broiled, or roasted, skin not eaten, from raw	1
24152240	Chicken thigh, baked or broiled, skin eaten, from pre-cooked	1
24152241	Chicken thigh, baked or broiled, skin not eaten, from pre-cooked	1
24152250	Chicken thigh, baked or broiled, skin eaten, from fast food/ restaurant	1
24152251	Chicken thigh, baked or broiled, skin not eaten, from fast food restaurant	1
24152300	Chicken thigh, rotisserie, skin eaten	1
24152301	Chicken thigh, rotisserie, skin not eaten	1
24153210	Chicken thigh, stewed, skin eaten	1
24153220	Chicken thigh, stewed, skin not eaten	1
24154010	Chicken thigh, grilled without sauce, skin eaten	1
24154011	Chicken thigh, grilled without sauce, skin not eaten	1
24154020	Chicken thigh, grilled with sauce, skin eaten	0.8
24154021	Chicken thigh, grilled with sauce, skin not eaten	0.8
24154300	Chicken thigh, sauteed, skin eaten	1
24154301	Chicken thigh, sauteed, skin not eaten	1
24157300	Chicken thigh, fried, coated, skin / coating eaten, from raw	0.68
24157301	Chicken thigh, fried, coated, skin / coating not eaten, from raw	0.68
24157302	Chicken thigh, fried, coated, prepared skinless, coating eaten from raw	0.68
24157310	Chicken thigh, fried, coated, skin / coating eaten, from pre-cooked	0.68
24157311	Chicken thigh, fried, coated, skin / coating not eaten, from pre-cooked	0.68
24157320	Chicken thigh, fried, coated, skin / coating eaten, from fast food	0.68
24157321	Chicken thigh, fried, coated, skin / coating not eaten, from fast food	0.68
24157330	Chicken thigh, fried, coated, skin / coating eaten, from restaurant	0.68
24157331	Chicken thigh, fried, coated, skin / coating not eaten, from restaurant	0.68
24157400	Chicken thigh, baked, coated, skin / coating eaten	0.68
24157401	Chicken thigh, baked, coated, skin / coating not eaten	0.68
24168030	Chicken "wings", boneless, with hot sauce, from fast food / restaurant	0.5

Food code	Description	Amount (mg/g)
24168031	Chicken "wings", boneless, with hot sauce, from other sources	0.5
24198570	Chicken, canned, meat only	0.5
24198720	Chicken, ground	0.5
24198840	Fried chicken chunks, Puerto Rican style	0.5
24201000	Turkey, NFS	0.5
24201020	Turkey, light meat, skin not eaten	0.5
24201030	Turkey, light meat, skin eaten	0.5
24201060	Turkey, light meat, breaded, baked or fried, skin not eaten	0.5
24201070	Turkey, light meat, breaded, baked or fried, skin eaten	0.5
24201120	Turkey, light meat, roasted, skin not eaten	0.5
24201130	Turkey, light meat, roasted, skin eaten	0.5
24201220	Turkey, dark meat, roasted, skin not eaten	1
24201230	Turkey, dark meat, roasted, skin eaten	1
24201320	Turkey, light and dark meat, roasted, skin not eaten	0.5
24201330	Turkey, light and dark meat, roasted, skin eaten	0.5
24201360	Turkey, light or dark meat, fried, coated, skin not eaten	0.68
24201370	Turkey, light or dark meat, fried, coated, skin eaten	0.68
24201410	Turkey, light or dark meat, stewed, skin not eaten	1
24201420	Turkey light or dark meat, stewed, skin eaten	1
24201510	Turkey, light or dark meat, smoked, skin eaten	1
24201520	Turkey, light or dark meat, smoked, skin not eaten	1
24202460	Turkey, thigh, cooked, skin eaten	1
24202500	Turkey, thigh, cooked, skin not eaten	1
24206000	Turkey, canned	1
24207000	Turkey, ground	1
24208000	Turkey, nuggets	1
24300110	Duck, cooked, skin eaten	5
24300120	Duck, cooked, skin not eaten	5
24301010	Duck, roasted, skin eaten	5
24301020	Duck, roasted, skin not eaten	5
24302010	Duck, pressed, Chinese	5
24311010	Goose, wild, roasted	5
24400010	Cornish game hen, cooked, skin eaten	5
24400020	Cornish game hen, cooked, skin not eaten	5
24401010	Cornish game hen, roasted, skin eaten	5
24401020	Cornish game hen, roasted, skin not eaten	5
26100100	Fish, NS as to type, raw	4
26100110	Fish, NS as to type, cooked, NS as to cooking method	4
26100120	Fish, NS as to type, baked or broiled, made with oil	4

Food code	Description	Amount (mg/g)
26100121	Fish, NS as to type, baked or broiled, made with butter	4
26100122	Fish, NS as to type, baked or broiled, made with margarine	4
26100123	Fish, NS as to type, baked or broiled, no added fat	4
26100124	Fish, NS as to type, baked or broiled, made with cooking spray	4
26100130	Fish, NS as to type, coated, baked or broiled, made with oil	4
26100131	Fish, NS as to type, coated, baked or broiled, made with butter	4
26100132	Fish, NS as to type, coated, baked or broiled, made with margarine	4
26100133	Fish, NS as to type, coated, baked or broiled, no added fat	4
26100134	Fish, NS as to type, coated, baked or broiled, made with cooking spray	4
26100140	Fish, NS as to type, coated, fried, made with oil	4
26100141	Fish, NS as to type, coated, fried, made with butter	4
26100142	Fish, NS as to type, coated, fried, made with margarine	4
26100143	Fish, NS as to type, coated, fried, no added fat	4
26100144	Fish, NS as to type, coated, fried, made with cooking spray	4
26100160	Fish, NS as to type, steamed	4
26100170	Fish, NS as to type, dried	4
26100180	Fish, NS as to type, canned	4
26100190	Fish, NS as to type, smoked	4
26100200	Fish, NS as to type, from fast food	4
26149110	Swordfish, cooked, NS as to cooking method	4
26149120	Swordfish, baked or broiled, fat added	4
26149121	Swordfish, baked or broiled, no added fat	4
26149130	Swordfish, coated, baked or broiled, fat added	2.72
26149131	Swordfish, coated, baked or broiled, no added fat	2.72
26149140	Swordfish, coated, fried	2.72
26149160	Swordfish, steamed or poached	4
26153100	Tuna, fresh, raw	4
26153110	Tuna, fresh, cooked, NS as to cooking method	4
26153120	Tuna, fresh, baked or broiled, fat added	4
26153122	Tuna, fresh, baked or broiled, no added fat	4
26153130	Tuna, fresh, coated, baked or broiled, fat added	2.72
26153131	Tuna, fresh, coated, baked or broiled, no added fat	2.72
26153140	Tuna, fresh, coated, fried	2.72
26153160	Tuna, fresh, steamed or poached	4
26153170	Tuna, fresh, dried	4
26153190	Tuna, fresh, smoked	4
26155110	Tuna, canned, NS as to oil or water pack	1
26155180	Tuna, canned, oil pack	1
26155190	Tuna, canned, water pack	1

Food code	Description	Amount (mg/g)
32130190	Egg omelet or scrambled egg, with meat, NS as to fat	1.2
32130200	Egg omelet or scrambled egg, with meat, made with margarine	1.2
32130210	Egg omelet or scrambled egg, with meat, made with oil	1.2
32130220	Egg omelet or scrambled egg, with meat, made with butter	1.2
32130240	Egg omelet or scrambled egg, with meat, made with animal fat or meat drippings	1.2
32130260	Egg omelet or scrambled egg, with meat, made with cooking spray	1.2
32130265	Egg omelet or scrambled egg, with meat, NS as to fat type	1.2
32130270	Egg omelet or scrambled egg, with meat, no added fat	1.2
32130290	Egg omelet or scrambled egg, with cheese and meat, NS as to fat	1.2
32130300	Egg omelet or scrambled egg, with cheese and meat, made with margarine	1.2
32130310	Egg omelet or scrambled egg, with cheese and meat, made with oil	1.2
32130320	Egg omelet or scrambled egg, with cheese and meat, made with butter	1.2
32130340	Egg omelet or scrambled egg, with cheese and meat, made with animal fat or meat drippings	1.2
32130360	Egg omelet or scrambled egg, with cheese and meat, made with cooking spray	1.2
32130365	Egg omelet or scrambled egg, with cheese and meat, NS as to fat type	1.2
32130370	Egg omelet or scrambled egg, with cheese and meat, no added fat	1.2
32130800	Egg omelet or scrambled egg, with meat and tomatoes, fat added	1.2
32130810	Egg omelet or scrambled egg, with meat and tomatoes, no added fat	1.2
32130820	Egg omelet or scrambled egg, with meat and tomatoes, NS as to fat	1.2
32130830	Egg omelet or scrambled egg, with meat and dark-green vegetables, fat added	1.2
32130840	Egg omelet or scrambled egg, with meat and dark-green vegetables, no added fat	1.2
32130850	Egg omelet or scrambled egg, with meat and dark-green vegetables, NS as to fat	1.2
32130860	Egg omelet or scrambled egg, with meat, tomatoes, and dark-green vegetables, fat added	0.7
32130870	Egg omelet or scrambled egg, with meat, tomatoes, and dark-green vegetables, no added fat	0.7
32130880	Egg omelet or scrambled egg, with meat, tomatoes, and dark-green vegetables, NS as to fat	0.7
32130890	Egg omelet or scrambled egg, with meat and vegetables other than dark-green and/or tomatoes, fat added	0.7
32130900	Egg omelet or scrambled egg, with meat and vegetables other than dark-green and/or tomatoes, no added fat	0.7
32130910	Egg omelet or scrambled egg, with meat and vegetables other than dark-green and/or tomatoes, NS as to fat	0.7
32131000	Egg omelet or scrambled egg, with cheese, meat, and tomatoes, fat added	0.7
32131010	Egg omelet or scrambled egg, with cheese, meat, and tomatoes, no added fat	0.7
32131020	Egg omelet or scrambled egg, with cheese, meat, and tomatoes, NS as to fat	0.7

Food code	Description	Amount (mg/g)
32131030	Egg omelet or scrambled egg, with cheese, meat, and dark-green vegetables, fat added	0.7
32131040	Egg omelet or scrambled egg, with cheese, meat, and dark-green vegetables, no added fat	0.7
32131050	Egg omelet or scrambled egg, with cheese, meat, and dark-green vegetables, NS as to fat	0.7
32131060	Egg omelet or scrambled egg, with cheese, meat, tomatoes, and dark-green vegetables, fat added	0.7
32131070	Egg omelet or scrambled egg, with cheese, meat, tomatoes, and dark-green vegetables, no added fat	0.7
32131080	Egg omelet or scrambled egg, with cheese, meat, tomatoes, and dark-green vegetables, NS as to fat	0.7
32131090	Egg omelet or scrambled egg, with cheese, meat, and vegetables other than dark-green and/or tomatoes, fat added	0.7
32131100	Egg omelet or scrambled egg, with cheese, meat, and vegetables other than dark-green and/or tomatoes, no added fat	0.7
32131110	Egg omelet or scrambled egg, with cheese, meat, and vegetables other than dark-green and/or tomatoes, NS as to fat	0.7
32400200	Egg white, omelet, scrambled, or fried, with meat	1.2
32400400	Egg white, omelet, scrambled, or fried, with cheese and meat	1.2
32400600	Egg white, omelet, scrambled, or fried, with meat and vegetables	1.2
32400700	Egg white, omelet, scrambled, or fried, with cheese, meat, and vegetables	0.7
33401100	Egg substitute, omelet, scrambled, or fried, with meat	1.2
33401300	Egg substitute, omelet, scrambled, or fried, with cheese and meat	1.2
33401500	Egg substitute, omelet, scrambled, or fried, with meat and vegetables	1.2
33401600	Egg substitute, omelet, scrambled, or fried, with cheese, meat, and vegetables	0.7
21416000	Corned beef, cooked, NS as to fat eaten	5
21416110	Corned beef, cooked, lean and fat eaten	5
21416120	Corned beef, cooked, lean only eaten	5
21416150	Corned beef, canned, ready-to-eat	5
21602000	Beef, dried, chipped, uncooked	5
21602010	Beef, dried, chipped, cooked in fat	5
21602100	Beef jerky	10
22002800	Pork jerky	8
22300130	Ham, fried, lean and fat eaten	4
22300140	Ham, fried, lean only eaten	4
22300150	Ham, breaded or floured, fried, NS as to fat eaten	2.72
22300160	Ham, breaded or floured, fried, lean and fat eaten	2.72
22300170	Ham, breaded or floured, fried, lean only eaten	2.72
22311000	Ham, smoked or cured, cooked, NS as to fat eaten	4
22311010	Ham, smoked or cured, cooked, lean and fat eaten	4
22311020	Ham, smoked or cured, cooked, lean only eaten	4

Food code	Description	Amount (mg/g)
22311450	Ham, prosciutto	2.4
22311500	Ham, smoked or cured, canned, NS as to fat eaten	4
22311510	Ham, smoked or cured, canned, lean and fat eaten	4
22311520	Ham, smoked or cured, canned, lean only eaten	4
22321110	Ham, smoked or cured, ground patty	4
22421000	Pork roast, smoked or cured, cooked, NS as to fat eaten	4
22421010	Pork roast, smoked or cured, cooked, lean and fat eaten	4
22421020	Pork roast, smoked or cured, cooked, lean only eaten	4
22431000	Pork roll, cured, fried	4
22501010	Canadian bacon, cooked	4
23321050	Venison/deer, cured	5
23321900	Venison/deer jerky	5
25220410	Bologna, NFS	4
25220425	Bologna, made from any kind of meat, reduced fat	4
25220435	Bologna, made from any kind of meat, reduced sodium	4
25220445	Bologna, made from any kind of meat, reduced fat and reduced sodium	4
25221215	Pastrami, NFS	5
25221220	Pastrami, made from any kind of meat, reduced fat	5
25221250	Pepperoni, NFS	4
25221255	Pepperoni, reduced fat	4
25221260	Pepperoni, reduced sodium	4
25221500	Salami, NFS	4
25221505	Salami, made from any type of meat, reduced fat	4
25221515	Salami, made from any type of meat, reduced sodium	4
25230110	Luncheon meat, NFS	4
25230210	Ham, prepackaged or deli, luncheon meat	4
25230220	Ham, prepackaged or deli, luncheon meat, reduced sodium	4
25230320	Chicken, prepackaged or deli, luncheon meat	1
25230340	Chicken, prepackaged or deli, luncheon meat, reduced sodium	1
25230420	Ham luncheon meat, loaf type	4
25230530	Ham and pork, canned luncheon meat, chopped, minced, pressed, spiced	4
25230540	Ham, pork and chicken, canned luncheon meat, chopped, minced, pressed, spiced, reduced fat and reduced sodium	4
25230550	Ham, pork, and chicken, canned luncheon meat, chopped, minced, pressed, spiced, reduced sodium	4
25230610	Luncheon meat, loaf type	5
25230780	Turkey, prepackaged or deli, luncheon meat	1
25230785	Turkey, prepackaged or deli, luncheon meat, reduced sodium	1
25230800	Turkey ham, prepackaged or deli, luncheon meat	1
25231110	Beef, prepackaged or deli, luncheon meat	5

Food code	Description	Amount (mg/g)
25231120	Beef, prepackaged or deli, luncheon meat, reduced sodium	5
21601000	Beef, bacon, cooked	2.5
21601010	Beef, bacon, reduced sodium, cooked	2.5
22600100	Bacon, NS as to type of meat, cooked	2.5
22600110	Bacon, NS as to type of meat, reduced sodium, cooked	2.5
22600200	Pork bacon, NS as to fresh, smoked or cured, cooked	2
22600210	Pork bacon, NS as to fresh, smoked or cured, reduced sodium, cooked	2
22601000	Pork bacon, smoked or cured, cooked	2
22601040	Bacon or side pork, fresh, cooked	2
22602010	Pork bacon, smoked or cured, reduced sodium, cooked	2
22621000	Salt pork, cooked	2
24208500	Turkey bacon, cooked	0.5
24208510	Turkey bacon, reduced sodium, cooked	0.5
25210110	Frankfurter or hot dog, NFS	4
25210150	Frankfurter or hot dog, cheese-filled	2.4
25210210	Frankfurter or hot dog, beef	4
25210220	Frankfurter or hot dog, beef and pork	4
25210240	Frankfurter or hot dog, beef and pork, reduced fat or light	4
25210250	Frankfurter or hot dog, meat and poultry, fat free	2
25210280	Frankfurter or hot dog, meat and poultry	2
25210290	Frankfurter or hot dog, meat and poultry, reduced fat or light	2
25210410	Frankfurter or hot dog, turkey	1
25210620	Frankfurter or hot dog, beef, reduced fat or light	4
25210750	Frankfurter or hot dog, reduced fat or light, NFS	4
25220105	Beef sausage	5
25220106	Beef sausage, reduced fat	5
25220108	Beef sausage, reduced sodium	5
25220150	Beef sausage with cheese	5
25220350	Bratwurst	2
25220360	Bratwurst, with cheese	2
25220650	Turkey or chicken and beef sausage	2
25220710	Chorizo	4
25221310	Polish sausage	1
25221350	Italian sausage	2
25221400	Sausage, NFS	6
25221405	Pork sausage	2
25221406	Pork sausage, reduced fat	2
25221408	Pork sausage, reduced sodium	2
25221460	Pork and beef sausage	4

Food code	Description	Amount (mg/g)
25221830	Turkey or chicken sausage	1
25221855	Turkey or chicken sausage, reduced sodium	1
25221860	Turkey or chicken sausage, reduced fat	1
25221870	Turkey or chicken and pork sausage	1.5
25221875	Turkey or chicken, pork, and beef sausage, reduced sodium	2
25221910	Vienna sausage, canned	4
25221950	Pickled sausage	4
27111000	Beef with tomato-based sauce	7
27111100	Beef goulash	7
27111200	Beef burgundy	7
27111300	Beef stew, no potatoes, tomato-based sauce, Mexican style	2
27111310	Beef stew, no potatoes, tomato-based sauce, with chili peppers, Mexican style	2
27111400	Chili con carne, NS as to beans	2
27111405	Chili con carne with beans, from restaurant	2
27111406	Chili con carne with beans, home recipe	2
27111407	Chili con carne with beans, canned	2
27111410	Chili con carne with beans	2
27111420	Chili con carne without beans	2
27111430	Chili con carne, NS as to beans, with cheese	2
27111440	Chili con carne with beans and cheese	2
27111500	Beef sloppy joe, no bun	7
27112000	Beef with gravy	7
27112010	Salisbury steak with gravy	7
27113000	Beef with cream or white sauce	7
27113100	Beef stroganoff	2
27113200	Creamed chipped or dried beef	7
27113300	Swedish meatballs with cream or white sauce	4
27114000	Beef with mushroom sauce	7
27116100	Beef curry	7
27116110	Beef curry with rice	2
27116200	Beef with barbecue sauce	7
27116350	Stewed seasoned ground beef, Mexican style	2
27116400	Steak tartare	7
27118110	Meatballs, Puerto Rican style	4
27118120	Stewed seasoned ground beef, Puerto Rican style	4
27118130	Stewed dried beef, Puerto Rican style	7
27118140	Stuffed pot roast, Puerto Rican style, NFS	2
27118180	Beef stew, meat with gravy, no potatoes, Puerto Rican style	2
27120020	Ham or pork with gravy	2.8

Food code	Description	Amount (mg/g)
27120030	Ham or pork with barbecue sauce	2.8
27120080	Ham stroganoff	0.8
27120090	Ham or pork with mushroom sauce	2.8
27120100	Ham or pork with tomato-based sauce	2.8
27120110	Sausage with tomato-based sauce	3.2
27120120	Sausage gravy	3.2
27120130	Pork stew, no potatoes, tomato-based sauce, Mexican style	2.8
27120210	Frankfurter or hot dog, with chili, no bun	2
27120250	Frankfurters or hot dogs with tomato-based sauce	2
27121000	Pork with chili and tomatoes	0.8
27121010	Stewed pork, Puerto Rican style	0.8
27121410	Chili con carne with beans, made with pork	0.8
27130010	Lamb or mutton with gravy	14
27130050	Lamb or mutton goulash	4
27133010	Stewed goat, Puerto Rican style	2.8
27160010	Meat with barbecue sauce, NS as to type of meat	7
27160100	Meatballs, NS as to type of meat, with sauce	4
27161010	Meat loaf, Puerto Rican style	5.5
27162010	Meat with tomato-based sauce	7
27162500	Stewed, seasoned, ground beef and pork, Mexican style	2
27163010	Meat with gravy, NS as to type of meat,	7
27211000	Beef and potatoes, no sauce	5
27211100	Beef stew with potatoes, tomato-based sauce	2
27211110	Beef stew with potatoes, tomato-based sauce, Mexican style	2
27211150	Beef goulash with potatoes	2
27211190	Beef and potatoes with cream sauce, white sauce or mushroom sauce	2
27211200	Beef stew with potatoes, gravy	2
27211300	Beef, roast, hash	2
27211500	Beef and potatoes with cheese sauce	2
27211550	Stewed, seasoned, ground beef with potatoes, Mexican style	2
27212000	Beef and noodles, no sauce	2
27212050	Beef and macaroni with cheese sauce	2
27212100	Beef and noodles with tomato-based sauce	2
27212120	Chili con carne with beans and macaroni	2
27212150	Beef goulash with noodles	2
27212200	Beef and noodles with gravy	2
27212300	Beef and noodles with cream or white sauce	2
27212350	Beef stroganoff with noodles	2
27212400	Beef and noodles with mushroom sauce	2

Food code	Description	Amount (mg/g)
27213000	Beef and rice, no sauce	2
27213100	Beef and rice with tomato-based sauce	2
27213120	Porcupine balls with tomato-based sauce	2
27213150	Chili con carne with beans and rice	2
27213200	Beef and rice with gravy	2
27213300	Beef and rice with cream sauce	2
27213400	Beef and rice with mushroom sauce	2
27213420	Porcupine balls with mushroom sauce	2
27213600	Beef and rice with cheese sauce	2
27214100	Meat loaf made with beef	2
27214110	Meat loaf made with beef, with tomato-based sauce	2
27214300	Beef wellington	1.8
27218110	Stuffed pot roast, with potatoes, Puerto Rican style	2
27218210	Beef stew with potatoes, Puerto Rican style	2
27218310	Stewed corned beef, Puerto Rican style	2
27220010	Meat loaf made with ham	0.8
27220020	Ham and noodles with cream or white sauce	0.8
27220030	Ham and rice with mushroom sauce	0.8
27220050	Ham or pork with stuffing	0.8
27220080	Ham croquette	0.8
27220110	Pork and rice with tomato-based sauce	0.8
27220120	Sausage and rice with tomato-based sauce	0.4
27220150	Sausage and rice with mushroom sauce	0.4
27220170	Sausage and rice with cheese sauce	0.4
27220190	Sausage and noodles with cream or white sauce	0.4
27220210	Ham and noodles, no sauce	0.8
27220310	Ham or pork and rice, no sauce	0.8
27220510	Ham or pork and potatoes with gravy	0.8
27220520	Ham or pork and potatoes with cheese sauce	0.8
27221150	Pork stew, with potatoes, tomato-based sauce, Mexican style	0.8
27230010	Lamb or mutton loaf	14
27231000	Lamb or mutton and potatoes with gravy	4
27232000	Lamb or mutton and potatoes with tomato-based sauce	4
27233000	Lamb or mutton and noodles with gravy	4
27235000	Meat loaf made with venison/deer	4
27260010	Meat loaf, NS as to type of meat	5.5
27260050	Meatballs, with breading, NS as to type of meat, with gravy	5.6
27260080	Meat loaf made with beef and pork	3.85
27260090	Meat loaf made with beef, veal and pork	2.8

Food code	Description	Amount (mg/g)
27260100	Meat loaf made with beef and pork, with tomato-based sauce	2.8
27260110	Hash, NS as to type of meat	2.8
27260500	Vienna sausages stewed with potatoes, Puerto Rican style	0.8
27261500	Stewed, seasoned, ground beef and pork with potatoes, Mexican style	1.8
27311110	Beef, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	1.8
27311120	Beef, potatoes, and vegetables, excluding carrots, broccoli, and dark-green leafy; no sauce	1.8
27311210	Corned beef, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	1.8
27311220	Corned beef, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	1.8
27311310	Beef stew with potatoes and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	1.8
27311320	Beef stew with potatoes and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	1.8
27311410	Beef stew with potatoes and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	1.8
27311420	Beef stew with potatoes and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	1.8
27311510	Shepherd's pie with beef	1.8
27311600	Beef, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	1.8
27311605	Beef, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	1.8
27311610	Beef, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; cream sauce, white sauce, or mushroom sauce	1.8
27311620	Beef, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; cream sauce, white sauce, or mushroom sauce	1.8
27311625	Beef, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	1.8
27311630	Beef, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	1.8
27311635	Beef, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; cheese sauce	1.8
27311640	Beef, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; cheese sauce	1.8
27313010	Beef, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	1.8
27313020	Beef, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	1.8
27313210	Beef, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	1.8
27313220	Beef, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	1.8
27313310	Beef, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; mushroom sauce	1.8
27313320	Beef, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; mushroom sauce	1.8

Food code	Description	Amount (mg/g)
27313410	Beef, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	1.8
27313420	Beef, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	1.8
27315010	Beef, rice, and vegetables including carrots, broccoli, and/ or dark-green leafy; no sauce	1.8
27315020	Beef, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	1.8
27315210	Beef, rice, and vegetables including carrots, broccoli, and/ or dark-green leafy; tomato-based sauce	1.8
27315220	Beef, rice, and vegetables excluding carrots, broccoli, and/ or dark-green leafy; tomato-based sauce	1.8
27315250	Stuffed cabbage rolls with beef and rice	1.8
27315270	Stuffed grape leaves with beef and rice	1.8
27315310	Beef, rice, and vegetables including carrots, broccoli, and/ or dark-green leafy; mushroom sauce	1.8
27315320	Beef, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; mushroom sauce	1.8
27315330	Beef, rice, and vegetables including carrots, broccoli, and/ or dark-green leafy; cheese sauce	1.8
27315340	Beef, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; cheese sauce	1.8
27315410	Beef, rice, and vegetables including carrots, broccoli, and/ or dark-green leafy; gravy	1.8
27315420	Beef, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	1.8
27317010	Beef pot pie	1.8
27317100	Beef, dumplings, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	1.8
27317110	Beef, dumplings, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	1.8
27319010	Stuffed green pepper, Puerto Rican style	1.8
27320025	Ham or pork, noodles and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	0.8
27320027	Ham or pork, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	0.8
27320030	Ham or pork, noodles and vegetables excluding carrots, broccoli, and dark-green leafy; cheese sauce	0.8
27320040	Pork, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	0.8
27320070	Ham or pork, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	0.8
27320080	Sausage, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	1.2
27320090	Sausage, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	1.2
27320100	Pork, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	0.8
27320110	Pork, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	0.8

Food code	Description	Amount (mg/g)
27320120	Sausage, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	1.2
27320130	Sausage, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	1.2
27320140	Pork, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	0.8
27320150	Pork, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	0.8
27320210	Pork, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	0.8
27320340	Pork, rice, and vegetables including carrots, broccoli, and/ or dark-green leafy; tomato-based sauce	0.8
27320350	Pork, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	0.8
27320410	Ham, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	0.8
27320450	Ham, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	0.8
27330010	Shepherd's pie with lamb	4
27330030	Lamb or mutton stew with potatoes and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	4
27330050	Lamb or mutton, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	4
27330060	Lamb or mutton, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	4
27330080	Lamb or mutton, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	4
27330110	Lamb or mutton stew with potatoes and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	4
27330170	Stuffed grape leaves with lamb and rice	4
27330210	Lamb or mutton stew with potatoes and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	4
27330220	Lamb or mutton stew with potatoes and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	4
27360000	Stew, NFS	2
27360010	Goulash, NFS	2
27360050	Meat pie, NFS	2
27410210	Beef and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, no sauce	2
27410220	Beef and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, no sauce	2
27410250	Beef shish kabob with vegetables, excluding potatoes	2
27411100	Beef with vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, tomato-based sauce	2
27411120	Swiss steak	2
27411150	Beef rolls, stuffed with vegetables or meat mixture, tomato-based sauce	2
27411200	Beef with vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, tomato-based sauce	2

Food code	Description	Amount (mg/g)
27414100	Beef with vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, mushroom sauce	2
27414200	Beef with vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, mushroom sauce	2
27416100	Beef and vegetables, Hawaiian style	2
27416150	Pepper steak	2
27416200	Beef, ground, with egg and onion	2
27416250	Beef salad	2
27416450	Beef and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, gravy	2
27416500	Beef and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, gravy	2
27418110	Seasoned shredded soup meat	2
27418210	Beef stew with vegetables excluding potatoes, Puerto Rican style	2
27418310	Corned beef with tomato sauce and onion, Puerto Rican style	2
27418410	Beef steak with onions, Puerto Rican style	2
27420010	Cabbage with ham hocks	0.8
27420020	Ham or pork salad	0.8
27420040	Frankfurters or hot dogs and sauerkraut	0.8
27420060	Pork and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, no sauce	0.8
27420200	Pork hash	0.8
27420250	Ham and vegetables including carrots broccoli, and/or dark-green leafy; no potatoes, no sauce	0.8
27420270	Ham and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, no sauce	0.8
27420350	Pork and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, no sauce	0.8
27420400	Pork and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, tomato-based sauce	0.8
27420410	Pork and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, tomato-based sauce	0.8
27420450	Sausage and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, tomato-based sauce	0.8
27420460	Sausage and vegetables, excluding carrots, broccoli, and dark-green leafy; no potatoes, tomato-based sauce	0.8
27420470	Sausage and peppers, no sauce	0.8
27420520	Pork shish kabob with vegetables, excluding potatoes	0.8
27430400	Lamb or mutton stew with vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, gravy	4
27430410	Lamb or mutton stew with vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, gravy	4
27460490	Julienne salad, meat, cheese, eggs, vegetables, no dressing	2
27460510	Antipasto with ham, fish, cheese, vegetables	0.8
27560705	Sausage balls, made with biscuit mix and cheese	0.8
28101000	Frozen dinner, NFS	2

Food code	Description	Amount (mg/g)
28110000	Beef dinner, NFS, frozen meal	2
28110150	Beef with vegetable, diet frozen meal	2
28110220	Sirloin, chopped, with gravy, mashed potatoes, vegetable, frozen meal	2
28110250	Sirloin tips, with gravy, potatoes, vegetable, frozen meal	2
28110270	Sirloin beef, with gravy, potatoes, vegetable, frozen meal	2
28110300	Salisbury steak dinner, NFS, frozen meal	2
28110310	Salisbury steak with gravy, potatoes, vegetable, frozen meal	2
28110330	Salisbury steak with gravy, whipped potatoes, vegetable, dessert, frozen meal	2
28110350	Salisbury steak with gravy, potatoes, vegetable, dessert, frozen meal	2
28110370	Salisbury steak with gravy, macaroni and cheese, vegetable, frozen meal	2
28110380	Salisbury steak with gravy, macaroni and cheese, frozen meal	2
28110390	Salisbury steak, potatoes, vegetable, dessert, diet frozen meal	2
28110510	Beef, sliced, with gravy, potatoes, vegetable, frozen meal	2
28110660	Meatballs, Swedish, in gravy, with noodles, diet frozen meal	1.3
28113110	Salisbury steak, baked, with tomato sauce, vegetable, diet frozen meal	2
28113140	Beef with spaetzle or rice, vegetable, frozen meal	2
28160300	Meat loaf dinner, NFS, frozen meal	0.52
28160310	Meat loaf with potatoes, vegetable, frozen meal	0.52
58117310	Kibby, Puerto Rican style	1
58117510	Hayacas, Puerto Rican style	1
58302050	Beef and noodles with meat sauce and cheese, diet frozen meal	2
58310210	Sausage and french toast, frozen meal	0.8
58310310	Pancakes and sausage, frozen meal	0.8
77205610	Ripe plantain meat pie, Puerto Rican style	2
77230210	Cassava Pasteles, Puerto Rican style	2
77250110	Stuffed tannier fritters, Puerto Rican style	2
77272010	Puerto Rican pasteles	2
77316010	Stuffed cabbage, with meat, Puerto Rican style	2
77316510	Stuffed cabbage, with meat and rice, Syrian dish, Puerto Rican style	2
77316600	Eggplant and meat casserole	2
77513010	Spanish stew	2
77563010	Puerto Rican stew	2
24209000	Turkey with barbecue sauce, skin eaten	0.7
24209001	Turkey with barbecue sauce, skin not eaten	0.7
27141000	Chicken or turkey cacciatore	0.2
27141050	Stewed chicken with tomato-based sauce, Mexican style	0.2
27141500	Chili con carne with chicken or turkey and beans	0.2
27142000	Chicken with gravy	0.7

Food code	Description	Amount (mg/g)
27142100	Chicken or turkey fricassee	0.2
27142200	Turkey with gravy	0.7
27143000	Chicken or turkey with cream sauce	0.7
27144000	Chicken or turkey with mushroom sauce	0.7
27146011	Chicken, shredded or pulled, with barbecue sauce	0.7
27146150	Chicken curry	0.2
27146155	Chicken curry with rice	0.2
27146160	Chicken with mole sauce	0.2
27146200	Chicken or turkey with cheese sauce	0.2
27146250	Chicken or turkey cordon bleu	0.2
27146300	Chicken or turkey parmigiana	0.2
27146400	Chicken kiev	0.2
27148010	Stuffed chicken, drumstick or breast, Puerto Rican style	0.2
27241000	Chicken or turkey hash	0.2
27241010	Chicken or turkey and potatoes with gravy	0.2
27242000	Chicken or turkey and noodles, no sauce	0.2
27242200	Chicken or turkey and noodles with gravy	0.2
27242250	Chicken or turkey and noodles with mushroom sauce	0.2
27242300	Chicken or turkey and noodles with cream or white sauce	0.2
27242310	Chicken or turkey and noodles with cheese sauce	0.2
27242350	Chicken or turkey tetrazzini	0.2
27242400	Chicken or turkey and noodles with tomato-based sauce	0.2
27243000	Chicken or turkey and rice, no sauce	0.2
27243300	Chicken or turkey and rice with cream sauce	0.2
27243400	Chicken or turkey and rice with mushroom sauce	0.2
27243500	Chicken or turkey and rice with tomato-based sauce	0.2
27246100	Chicken or turkey with dumplings	0.2
27246200	Chicken or turkey with stuffing	0.2
27246300	Chicken or turkey cake, patty, or croquette	0.2
27246400	Chicken or turkey souffle	0.2
27246500	Meat loaf made with chicken or turkey	0.2
27246505	Meat loaf made with chicken or turkey, with tomato-based sauce	0.2
27341000	Chicken or turkey, potatoes, corn, and cheese, with gravy	0.2
27341010	Chicken or turkey, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	0.2
27341020	Chicken or turkey, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	0.2
27341025	Chicken or turkey, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	0.2
27341030	Chicken or turkey, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	0.2

Food code	Description	Amount (mg/g)
27341035	Chicken or turkey, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; cream sauce, white sauce, or mushroom sauce	0.2
27341040	Chicken or turkey, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; cream sauce, white sauce, or mushroom sauce	0.2
27341045	Chicken or turkey, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; cheese sauce	0.2
27341050	Chicken or turkey, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; cheese sauce	0.2
27341055	Chicken or turkey, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	0.2
27341060	Chicken or turkey, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	0.2
27341310	Chicken or turkey stew with potatoes and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	0.2
27341320	Chicken or turkey stew with potatoes and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	0.2
27341510	Chicken or turkey stew with potatoes and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	0.2
27341520	Chicken or turkey stew with potatoes and vegetables excluding carrots, broccoli, and dark-green leafy; tomato- based sauce	0.2
27343010	Chicken or turkey, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	0.2
27343020	Chicken or turkey, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	0.2
27343410	Chicken or turkey, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	0.2
27343420	Chicken or turkey, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	0.2
27343470	Chicken or turkey, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; cream sauce, white sauce, or mushroom sauce	0.2
27343480	Chicken or turkey, noodles, and vegetables excluding carrots, broccoli, and/or dark-green leafy; cream sauce, white sauce, or mushroom sauce	0.2
27343510	Chicken or turkey, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	0.2
27343520	Chicken or turkey, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	0.2
27343950	Chicken or turkey, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; cheese sauce	0.2
27343960	Chicken or turkey, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; cheese sauce	0.2
27345010	Chicken or turkey, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	0.2
27345020	Chicken or turkey, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; no sauce	0.2
27345210	Chicken or turkey, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; gravy	0.2
27345220	Chicken or turkey, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	0.2
27345230	Chicken or turkey, rice, corn, and cheese, with gravy	0.2

Food code	Description	Amount (mg/g)
27345410	Chicken or turkey, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; cream sauce, white sauce, or mushroom sauce	0.2
27345420	Chicken or turkey, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; cream sauce, white sauce, or mushroom sauce	0.2
27345440	Chicken or turkey, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; cheese sauce	0.2
27345450	Chicken or turkey, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; cheese sauce	0.2
27345510	Chicken or turkey, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	0.2
27345520	Chicken or turkey, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; tomato-based sauce	0.2
27347100	Chicken or turkey pot pie	0.2
27347200	Chicken or turkey, stuffing, and vegetables including carrots, broccoli, and/or dark-green leafy; no sauce	0.2
27347210	Chicken or turkey, stuffing, and vegetables excluding carrots, broccoli, and dark green leafy; no sauce	0.2
27347220	Chicken or turkey, stuffing, and vegetables including carrot s, broccoli, and/or dark-green leafy; gravy	0.2
27347230	Chicken or turkey, stuffing, and vegetables excluding carrots, broccoli, and dark-green leafy; gravy	0.2
27347240	Chicken or turkey, dumplings, and vegetables including carrots, broccoli, and/or dark green leafy; gravy	0.2
27347250	Chicken or turkey, dumplings, and vegetables excluding carrots, broccoli, and dark green leafy; gravy	0.2
27348100	Chicken fricassee, Puerto Rican style	0.2
27440110	Chicken or turkey and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, no sauce	0.2
27440120	Chicken or turkey and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, no sauce	0.2
27440130	Chicken or turkey shish kabob with vegetables, excluding potatoes	0.2
27441120	Chicken or turkey creole, without rice	0.2
27442110	Chicken or turkey and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, gravy	0.2
27442120	Chicken or turkey and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, gravy	0.2
27443110	Chicken or turkey a la king with vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, cream, white, or soup-based sauce	0.2
27443120	Chicken or turkey a la king with vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, cream, white, or soup-based sauce	0.2
27443150	Chicken or turkey divan	0.2
27445125	Chicken or turkey and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, tomato-based sauce	0.2
27445130	Chicken or turkey and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, tomato-based sauce	0.2
27446200	Chicken or turkey salad, made with mayonnaise	0.2
27446205	Chicken or turkey salad with nuts and/or fruits	0.2
27446220	Chicken or turkey salad with egg	0.2

Food code	Description	Amount (mg/g)
27446225	Chicken or turkey salad, made with light mayonnaise	0.2
27446230	Chicken or turkey salad, made with mayonnaise-type salad dressing	0.2
27446235	Chicken or turkey salad, made with light mayonnaise-type salad dressing	0.2
27446240	Chicken or turkey salad, made with creamy dressing	0.2
27446245	Chicken or turkey salad, made with light creamy dressing	0.2
27446250	Chicken or turkey salad, made with Italian dressing	0.2
27446255	Chicken or turkey salad, made with light Italian dressing	0.2
27446260	Chicken or turkey salad, made with any type of fat free dressing	0.2
27446300	Chicken or turkey garden salad, chicken and/or turkey, tomato and/or carrots, other vegetables, no dressing	0.2
27446310	Chicken or turkey garden salad, chicken and/or turkey, other vegetables excluding tomato and carrots, no dressing	0.2
27446315	Chicken or turkey garden salad with bacon and cheese, chicken and/or turkey, bacon, cheese, lettuce and/or greens, tomato and/or carrots, other vegetables, no dressing	0.2
27446320	Chicken or turkey, breaded, fried, garden salad with bacon and cheese, chicken and/or turkey, bacon, cheese, lettuce and/or greens, tomato and/or carrots, other vegetables, no dressing	0.2
27446330	Chicken or turkey garden salad with cheese, chicken and/or turkey, cheese, lettuce and/or greens, tomato and/or carrots, other vegetables, no dressing	0.2
27446332	Chicken or turkey, breaded, fried, garden salad with cheese, chicken and/or turkey, cheese, lettuce and/or greens, tomato and/or carrots, other vegetables, no dressing	0.2
27446350	Asian chicken or turkey garden salad, chicken and/or turkey, lettuce, fruit, nuts, no dressing	0.2
27446355	Asian chicken or turkey garden salad with crispy noodles, chicken and/or turkey, lettuce, fruit, nuts, crispy noodles, no dressing	0.2
27446360	Chicken or turkey Caesar Garden salad, chicken and/or turkey, lettuce, tomato, cheese, no dressing	0.2
27446362	Chicken or turkey, breaded, fried, Caesar Garden salad, chicken and/or turkey, lettuce, tomatoes, cheese, no dressing	0.2
27446400	Chicken or turkey and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, cheese sauce	0.2
27446410	Chicken or turkey and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, cheese sauce	0.2
27448020	Chicken or turkey fricassee, with sauce, no potatoes, potatoes reported separately, Puerto Rican style	0.2
27448030	Chicken or turkey fricassee, no sauce, no potatoes, Puerto Rican style	0.2
27460710	Livers, chicken, chopped, with eggs and onion	0.2
27463000	Stewed gizzards, Puerto Rican style	0.2
28140100	Chicken dinner, NFS, frozen meal	0.2
28140320	Chicken and noodles with vegetable, dessert, frozen meal	0.2
28140710	Chicken, fried, with potatoes, vegetable, frozen meal	0.2
28140720	Chicken patty, or nuggets, boneless, breaded, potatoes, vegetable, frozen meal	0.2

Food code	Description	Amount (mg/g)
28140740	Chicken patty or nuggets, boneless, breaded, with pasta and tomato sauce, fruit, dessert, frozen meal	0.2
28140810	Chicken, fried, with potatoes, vegetable, dessert, frozen meal	0.2
28141010	Chicken, fried, with potatoes, vegetable, dessert, frozen meal, large meat portion	0.2
28141250	Chicken with rice and vegetable, diet frozen meal	0.2
28141650	Chicken and vegetables au gratin with rice, diet frozen entree	0.2
28143020	Chicken and vegetable entree with rice, diet frozen meal	0.2
28143080	Chicken with noodles and cheese sauce, diet frozen meal	0.2
28143130	Chicken and vegetable entree with noodles, frozen meal	0.2
28143150	Chicken and vegetable entree with noodles, diet frozen meal	0.2
28143170	Chicken in cream sauce with noodles and vegetable, frozen meal	0.2
28143180	Chicken in butter sauce with potatoes and vegetable, diet frozen meal	0.2
28143190	Chicken in mushroom sauce, white and wild rice, vegetable, frozen meal	0.2
28144100	Chicken and vegetable entree with noodles and cream sauce, frozen meal	0.2
28145000	Turkey dinner, NFS, frozen meal	0.2
28145100	Turkey with gravy, dressing, vegetable and fruit, diet frozen meal	0.2
28145110	Turkey with vegetable, stuffing, diet frozen meal	0.2
28145210	Turkey with gravy, dressing, potatoes, vegetable, frozen meal	0.2
58128110	Chicken cornbread	0.2
58137300	Adobo, with noodles	0.2
58150530	Adobo, with rice	0.2
77141010	Potato chicken pie, Puerto Rican style	0.2
27150050	Fish timbale or mousse	0.8
27150120	Tuna with cream or white sauce	2.8
27150250	Fish moochim	0.8
27150310	Fish with tomato-based sauce	0.8
27150320	Fish curry	0.8
27150325	Fish curry with rice	0.8
27151030	Ceviche	0.8
27250050	Fish cake or patty, NS as to fish	0.8
27250150	Tuna loaf	2.8
27250160	Tuna cake or patty	0.8
27250610	Tuna noodle casserole with cream or white sauce	0.8
27250630	Tuna noodle casserole with mushroom sauce	0.8
27250710	Tuna and rice with mushroom sauce	0.8
27350070	Tuna pot pie	0.8
27350080	Tuna noodle casserole with vegetables, cream or white sauce	0.8
27350090	Fish, noodles, and vegetables including carrots, broccoli, and/or dark green leafy; cheese sauce	0.8

Food code	Description	Amount (mg/g)
27350100	Fish, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; cheese sauce	0.8
27350110	Bouillabaisse	0.8
27350310	Seafood stew with potatoes and vegetables including carrots, broccoli, and/or dark-green leafy; tomato-based sauce	0.8
27350410	Tuna noodle casserole with vegetables and mushroom sauce	0.8
27450060	Tuna salad, made with mayonnaise	0.8
27450061	Tuna salad, made with light mayonnaise	0.8
27450062	Tuna salad, made with mayonnaise-type salad dressing	0.8
27450063	Tuna salad, made with light mayonnaise-type salad dressing	0.8
27450064	Tuna salad, made with creamy dressing	0.8
27450065	Tuna salad, made with light creamy dressing	0.8
27450066	Tuna salad, made with Italian dressing	0.8
27450067	Tuna salad, made with light Italian dressing	0.8
27450068	Tuna salad, made with any type of fat free dressing	0.8
27450090	Tuna salad with cheese	0.8
27450100	Tuna salad with egg	0.8
27450150	Fish, tofu, and vegetables, tempura	0.8
27450510	Tuna casserole with vegetables and mushroom sauce, no noodles	0.8
27450700	Fish and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, tomato-based sauce	0.8
27450710	Fish and vegetables excluding carrots, broccoli, and dark- green leafy; no potatoes, tomato-based sauce	0.8
28150510	Fish in lemon-butter sauce with starch item, vegetable, frozen meal	0.8
58162090	Stuffed pepper, with meat	2
58162110	Stuffed pepper, with rice and meat	2
58162120	Stuffed pepper, with rice, meatless	2
58162130	Stuffed tomato, with rice and meat	2
72125500	Channa Saag	2
72202010	Broccoli casserole with noodles	0.2
73305020	Squash, winter, souffle	2
74203010	Tomatoes, scalloped	0.2
75144100	Lettuce, wilted, with bacon dressing	0.8
75400500	Artichokes, stuffed	0.8
75403020	Green bean casserole	0.8
75412070	Eggplant with cheese and tomato sauce	1.8
75414020	Mushrooms, stuffed	1.8
75418040	Squash, summer, casserole, with cheese sauce	1.8
75418060	Squash, summer, souffle	1.8
75418220	Creamed christophine, Puerto Rican style	1.8
27213010	Biryani with meat	2

Food code	Description	Amount (mg/g)
27243100	Biryani with chicken	0.2
27363100	Jambalaya with meat and rice	2
55701000	Cake made with glutinous rice	0.2
55702000	Cake or pancake made with rice flour and/or dried beans	0.8
58155110	Rice with chicken, Puerto Rican style	0.2
58155310	Paella with meat, Valenciana style	0.8
58156210	Rice with vienna sausage, Puerto Rican style	0.8
58156310	Rice with Spanish sausage, Puerto Rican style	0.8
58157300	Congee, with meat, poultry, and/or seafood	2
58157310	Congee, with meat, poultry, and/or seafood, and vegetables	2
58160400	Rice, white, with corn, NS as to fat	1.8
58160410	Rice, white, with corn, no added fat	1.8
58160420	Rice, white, with corn, fat added	1.8
58160430	Rice, white, with peas, NS as to fat	1.8
58160440	Rice, white, with peas, no added fat	1.8
58160450	Rice, white, with peas, fat added	1.8
58160460	Rice, white, with carrots, NS as to fat	0.2
58160470	Rice, white, with carrots, no added fat	0.2
58160530	Rice, white, with tomatoes and/or tomato-based sauce, no added fat	1.8
58160540	Rice, white, with tomatoes and/or tomato-based sauce, fat added	1.8
58160550	Rice, white, with dark green vegetables, NS as to fat	1.8
58160560	Rice, white, with dark green vegetables, no added fat	1.8
58160810	Rice, white, with lentils, no added fat	1.8
58161200	Rice, cooked with coconut milk	1.8
58161430	Rice, brown, with peas, NS as to fat	1.8
58161440	Rice, brown, with peas and carrots, NS as to fat	1.8
58161442	Rice, brown, with peas and carrots, no added fat	1.8
58161444	Rice, brown, with peas and carrots, fat added	1.8
58161492	Rice, brown, with dark green vegetables and tomatoes and/or tomato-based sauce, no added fat	1.8
58161494	Rice, brown, with dark green vegetables and tomatoes and/or tomato-based sauce, fat added	1.8
58161502	Rice, brown, with carrots and dark green vegetables, no added fat	1.8
58161504	Rice, brown, with carrots and dark green vegetables, fat added	1.8
58161520	Rice, brown, with carrots, dark green vegetables, and tomatoes and/or tomato-based sauce, NS as to fat	1.8
58161522	Rice, brown, with carrots, dark green vegetables, and tomatoes and/or tomato-based sauce, no added fat	0.2
58161530	Rice, brown, with other vegetables, NS as to fat	0.2
58161532	Rice, brown, with other vegetables, no added fat	0.2
58161710	Rice croquette	0.2

Food code	Description	Amount (mg/g)
58163510	Rice dressing	1.8
58164210	Rice dessert or salad with fruit	1.8
58164500	Rice, white, with cheese and/or cream-based sauce, NS as to fat	1.8
58164520	Rice, white, with cheese and/or cream-based sauce, fat added	1.8
58164530	Rice, white, with gravy, NS as to fat	0.2
58164550	Rice, white, with gravy, fat added	0.2
58164800	Rice, brown, with cheese and/or cream-based sauce, NS as to fat	0.2
58164820	Rice, brown, with cheese and/or cream-based sauce, fat added	0.2
58165430	Rice, brown, with vegetables and gravy, NS as to fat	1.8
58165450	Rice, brown, with vegetables and gravy, fat added	1.8
58122210	Gnocchi, cheese	1.8
58130011	Lasagna with meat	2
58130013	Lasagna with meat, canned	2
58130014	Lasagna with meat, from restaurant	2
58130015	Lasagna with meat, home recipe	2
58130016	Lasagna with meat, frozen	2
58130020	Lasagna with meat and spinach	2
58130140	Lasagna with chicken or turkey	0.2
58130150	Lasagna, with chicken or turkey, and spinach	0.2
58131310	Ravioli, meat-filled, no sauce	2
58131320	Ravioli, meat-filled, with tomato sauce or meat sauce	2
58131323	Ravioli, meat-filled, with tomato sauce or meat sauce, canned	2
58131330	Ravioli, meat-filled, with cream sauce	2
58134610	Tortellini, meat-filled, with tomato sauce	2
58134613	Tortellini, meat-filled, with tomato sauce, canned	2
58134650	Tortellini, meat-filled, no sauce	2
58140110	Spaghetti with corned beef, Puerto Rican style	2
58140310	Macaroni with tuna, Puerto Rican style	2
58146120	Pasta with tomato-based sauce, cheese and meat	2
58146315	Pasta with sauce and meat, from school lunch	2
58146321	Pasta with tomato-based sauce and meat, restaurant	2
58146323	Pasta with tomato-based sauce and meat, ready-to-heat	2
58146331	Pasta with tomato-based sauce, meat, and added vegetables, restaurant	2
58146333	Pasta with tomato-based sauce, meat, and added vegetables, ready-to-heat	2
58146341	Pasta with tomato-based sauce and poultry, restaurant	0.2
58146343	Pasta with tomato-based sauce and poultry, ready-to-heat	0.2
58146351	Pasta with tomato-based sauce, poultry, and added vegetables, restaurant	0.2
58146353	Pasta with tomato-based sauce, poultry, and added vegetables, ready-to-heat	0.2

Food code	Description	Amount (mg/g)
58146401	Pasta with cream sauce and meat, restaurant	2
58146403	Pasta with cream sauce and meat, ready-to-heat	2
58146411	Pasta with cream sauce, meat, and added vegetables, restaurant	2
58146413	Pasta with cream sauce, meat, and added vegetables, ready-to-heat	2
58146421	Pasta with cream sauce and poultry, restaurant	0.2
58146423	Pasta with cream sauce and poultry, ready-to-heat	0.2
58146431	Pasta with cream sauce, poultry, and added vegetables, restaurant	0.2
58146433	Pasta with cream sauce, poultry, and added vegetables, ready-to-heat	0.2
58146621	Pasta, whole grain, with tomato-based sauce and meat, restaurant	2
58146623	Pasta, whole grain, with tomato-based sauce and meat, ready- to-heat	2
58146631	Pasta, whole grain, with tomato-based sauce, meat, and added vegetables, restaurant	2
58146633	Pasta, whole grain, with tomato-based sauce, meat, and added vegetables, ready-to-heat	2
58146641	Pasta, whole grain, with tomato-based sauce and poultry, restaurant	0.2
58146643	Pasta, whole grain, with tomato-based sauce and poultry, ready-to-heat	0.2
58146651	Pasta, whole grain, with tomato-based sauce, poultry, and added vegetables, restaurant	0.2
58146653	Pasta, whole grain, with tomato-based sauce, poultry, and added vegetables, ready-to-heat	0.2
58146701	Pasta, whole grain, with cream sauce and meat, restaurant	2
58146703	Pasta, whole grain, with cream sauce and meat, ready-to-heat	2
58146711	Pasta, whole grain, with cream sauce, meat, and added vegetables, restaurant	2
58146712	Pasta, whole grain, with cream sauce, meat, and added vegetables, home recipe	2
58146713	Pasta, whole grain, with cream sauce, meat, and added vegetables, ready-to-heat	2
58146721	Pasta, whole grain, with cream sauce and poultry, restaurant	0.2
58146723	Pasta, whole grain, with cream sauce and poultry, ready-to-heat	0.2
58146731	Pasta, whole grain, with cream sauce, poultry, and added vegetables, restaurant	0.2
58146732	Pasta, whole grain, with cream sauce, poultry, and added vegetables, home recipe	0.2
58146733	Pasta, whole grain, with cream sauce, poultry, and added vegetables ready-to-heat	0.2
58148130	Macaroni or pasta salad with tuna	0.8
58148160	Macaroni or pasta salad with tuna and egg	0.8
58148170	Macaroni or pasta salad with chicken	0.2
58148550	Macaroni or pasta salad with meat	2
58301050	Lasagna with cheese and meat sauce, diet frozen meal	2
58304010	Spaghetti and meatballs dinner, NFS, frozen meal	2
58304050	Spaghetti with meat and mushroom sauce, diet frozen meal	2
58304060	Spaghetti with meat sauce, diet frozen meal	2

Food code	Description	Amount (mg/g)
58145120	Macaroni or noodles with cheese and tuna	2
58145135	Macaroni or noodles with cheese and meat	2
58145136	Macaroni or noodles with cheese and meat, prepared from Hamburger Helper mix	2
58145160	Macaroni or noodles with cheese and frankfurters or hot dogs	0.8
58145170	Macaroni or noodles with cheese and egg	0.2
58145190	Macaroni or noodles with cheese and chicken or turkey	0.2
58147340	Macaroni or noodles, creamed, with cheese and tuna	0.8
58116110	Meat turnover, Puerto Rican style	2
58116120	Empanada, Mexican turnover, filled with meat and vegetables	2
58116130	Empanada, Mexican turnover, filled with chicken and vegetables	0.2
58116210	Meat pie, Puerto Rican style	2
58120110	Crepe, filled with meat, poultry, or seafood, with sauce	2
58120120	Crepe, filled with meat, poultry, or seafood, no sauce	2
58121510	Dumpling, meat-filled	2
58122330	Knish, meat	2
58124230	Pastry, meat / poultry-filled	2
58125110	Quiche with meat, poultry or fish	2
58126000	Turnover filled with ground beef and cabbage	2
58126110	Turnover, meat-filled, no gravy	2
58126120	Turnover, meat-filled, with gravy	2
58126130	Turnover, meat- and cheese-filled, no gravy	2
58126140	Turnover, meat- and bean-filled, no gravy	2
58126150	Turnover, meat- and cheese-filled, tomato-based sauce	2
58126170	Turnover filled with meat and vegetable, no potatoes, no gravy	2
58126180	Turnover, meat-, potato-, and vegetable-filled, no gravy	2
58126270	Turnover, chicken- or turkey-, and cheese-filled, no gravy	0.2
58126280	Turnover, chicken- or turkey-, and vegetable-filled, lower in fat	0.2
58126290	Turnover, meat- and cheese-filled, lower in fat	2
58126300	Turnover, meat- and cheese-filled, tomato-based sauce, lower in fat	2
58126400	Turnover, filled with egg, meat and cheese	2
58126410	Turnover, filled with egg, meat, and cheese, lower in fat	2
58128120	Cornmeal dressing with chicken or turkey and vegetables	0.2
58128220	Dressing with chicken or turkey and vegetables	0.2
58128250	Dressing with meat and vegetables	2
58174100	Dosa (Indian), with filling	2
75410550	Stuffed jalapeno pepper	1
27313110	Beef chow mein or chop suey with noodles	1
27320310	Pork chow mein or chop suey with noodles	0.8
27343910	Chicken or turkey chow mein or chop suey with noodles	0.2

Food code	Description	Amount (mg/g)
27360080	Chow mein or chop suey, NS as to type of meat, with noodles	2
27360120	Chow mein or chop suey, various types of meat, with noodles	2
27415150	Beef chow mein or chop suey, no noodles	2
27420390	Pork chow mein or chop suey, no noodles	0.8
27446100	Chicken or turkey chow mein or chop suey, no noodles	0.2
27460010	Chow mein or chop suey, NS as to type of meat, no noodles	1
28143040	Chicken chow mein with rice, diet frozen meal	0.5
58135110	Chow fun noodles with meat and vegetables	1
58136140	Lo mein, with pork	0.8
58136150	Lo mein, with beef	2
58136160	Lo mein, with chicken	0.2
58137230	Pad Thai with chicken	0.2
58137250	Pad Thai with meat	2
58147520	Yat Ga Mein with meat, fish, or poultry	2
58150320	Rice, fried, with chicken	0.2
58150330	Rice, fried, with pork	0.8
58150340	Rice, fried, with beef	2
27115000	Beef with soy-based sauce	7
27115100	Steak teriyaki	7
27116300	Beef with sweet and sour sauce	7
27120060	Sweet and sour pork	2.8
27120150	Pork or ham with soy-based sauce	2.8
27145000	Chicken or turkey with teriyaki	0.7
27146100	Sweet and sour chicken or turkey	0.7
27146110	Sweet and sour chicken or turkey, without vegetables	0.7
27146350	Orange chicken	0.7
27146360	Sesame chicken	0.7
27212500	Beef and noodles with soy-based sauce	2
27213500	Beef and rice with soy-based sauce	2
27242500	Chicken or turkey and noodles with soy-based sauce	0.2
27243600	Chicken or turkey and rice with soy-based sauce	0.2
27311645	Beef, potatoes, and vegetables including carrots, broccoli, and/or dark-green leafy; soy-based sauce	2
27311650	Beef, potatoes, and vegetables excluding carrots, broccoli, and dark-green leafy; soy-based sauce	2
27313150	Beef, noodles, and vegetables including carrots, broccoli, and/or dark-green leafy; soy-based sauce	2
27313160	Beef, noodles, and vegetables excluding carrots, broccoli, and dark-green leafy; soy-based sauce	2
27315510	Beef, rice, and vegetables including carrots, broccoli, and/ or dark-green leafy; soy-based sauce	2

Food code	Description	Amount (mg/g)
27315520	Beef, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; soy-based sauce	2
27320320	Pork, rice, and vegetables including carrots, broccoli, and/ or dark-green leafy; soy-based sauce	0.8
27320330	Pork, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; soy-based sauce	0.8
27320500	Sweet and sour pork with rice	0.8
27345310	Chicken or turkey, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; soy-based sauce	0.2
27345320	Chicken or turkey, rice, and vegetables excluding carrots, broccoli, and dark-green leafy; soy-based sauce	0.2
27415100	Beef and vegetables including carrots, broccoli, and/or dark -green leafy; no potatoes, soy-based sauce	2
27415110	Beef and broccoli	2
27415120	Beef, tofu, and vegetables including carrots, broccoli, and/ or dark-green leafy; no potatoes, soy-based sauce	2
27415130	Szechuan beef	2
27415140	Hunan beef	2
27415170	Kung Pao beef	2
27415200	Beef and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, soy-based sauce	2
27415220	Beef, tofu, and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, soy-based sauce	2
27416400	Stir fried beef and vegetables in soy sauce	2
27420100	Pork, tofu, and vegetables including carrots, broccoli, and/ or dark-green leafy; no potatoes, soy-based sauce	0.8
27420110	Pork and vegetables, Hawaiian style	0.8
27420120	Pork and watercress with soy-based sauce	0.8
27420150	Kung Pao pork	0.8
27420160	Moo Shu pork, without Chinese pancake	0.8
27420170	Pork and onions with soy-based sauce	0.8
27420370	Pork, tofu, and vegetables, excluding carrots, broccoli, and dark-green leafy; no potatoes, soy-based sauce	0.8
27420500	Pork and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, soy-based sauce	0.8
27420510	Pork and vegetables excluding carrots, broccoli, and dark- green leafy; no potatoes, soy-based sauce	0.8
27445110	Chicken or turkey and vegetables including carrots, broccoli, and/or dark-green leafy; no potatoes, soy-based sauce	0.2
27445120	Chicken or turkey and vegetables excluding carrots, broccoli, and dark-green leafy; no potatoes, soy-based sauce	0.2
27445150	General Tso chicken	0.2
27445180	Moo Goo Gai Pan	0.2
27445220	Kung pao chicken	0.2
27445250	Almond chicken	0.2
28141201	Teriyaki chicken with rice and vegetable, diet frozen meal	0.2

Food code	Description	Amount (mg/g)
28143200	Chicken in soy-based sauce, rice and vegetables, frozen meal	0.2
32105210	Chicken egg foo yung	0.2
32105220	Pork egg foo yung	0.8
32105240	Beef egg foo yung	2
58110130	Egg roll, with beef and/or pork	2
58110170	Egg roll, with chicken or turkey	0.2
58110200	Roll with meat and/or shrimp, vegetables and rice paper, not fried	2
58111110	Wonton, fried, filled with meat, poultry, or seafood,	2
58111130	Wonton, fried, filled with meat, poultry, or seafood, and vegetable	2
58112510	Dumpling, steamed, filled with meat, poultry, or seafood	2
58123110	Sweet bread dough, filled with meat, steamed	2
58151220	Sushi roll tuna	0.8
58151440	Sushi, topped with tuna	0.8
27416300	Beef taco filling: beef, cheese, tomato, taco sauce	2
58100100	Burrito with meat	2
58100120	Burrito with meat and beans	2
58100125	Burrito with meat and beans, from fast food	2
58100135	Burrito with meat and sour cream	2
58100140	Burrito with meat, beans, and sour cream	2
58100145	Burrito with meat, beans, and sour cream, from fast food	2
58100160	Burrito with meat, beans, and rice	2
58100165	Burrito with meat, beans, rice, and sour cream	2
58100200	Burrito with chicken	0.2
58100220	Burrito with chicken and beans	0.2
58100235	Burrito with chicken and sour cream	0.2
58100245	Burrito with chicken, beans, and sour cream	0.2
58100255	Burrito with chicken, beans, and rice	0.2
58100260	Burrito with chicken, beans, rice, and sour cream	0.2
58101320	Taco or tostada with meat	2
58101323	Taco or tostada with meat, from fast food	2
58101325	Taco or tostada with meat and sour cream	2
58101345	Soft taco with meat	2
58101347	Soft taco with meat, from fast food	2
58101350	Soft taco with meat and sour cream	2
58101357	Soft taco with meat and sour cream, from fast food	2
58101450	Soft taco with chicken	0.2
58101457	Soft taco with chicken, from fast food	0.2
58101460	Soft taco with chicken and sour cream	0.2
58101520	Taco or tostada with chicken	0.2

Food code	Description	Amount (mg/g)
58101525	Taco or tostada with chicken and sour cream	0.2
58101620	Soft taco with meat and beans	2
58101625	Soft taco with chicken and beans	0.2
58101630	Soft taco with meat, beans, and sour cream	2
58101635	Soft taco with chicken, beans, and sour cream	0.2
58101730	Taco or tostada with meat and beans	2
58101733	Taco or tostada with meat and beans, from fast food	2
58101735	Taco or tostada with chicken and beans	0.2
58101745	Taco or tostada with meat, beans, and sour cream	2
58101750	Taco or tostada with chicken, beans, and sour cream	0.2
58101930	Taco or tostada salad with meat	2
58101935	Taco or tostada salad with chicken	0.2
58101945	Taco or tostada salad with meat and sour cream	2
58101950	Taco or tostada salad with chicken and sour cream	0.2
58104130	Nachos with meat and cheese	2
58104160	Nachos with chili	0.2
58104180	Nachos with meat, cheese, and sour cream	2
58104190	Nachos with chicken, cheese, and sour cream	0.2
58100520	Enchilada with meat and beans, red-chile or enchilada sauce	2
58100525	Enchilada with meat and beans, green-chile or enchilada sauce	2
58100530	Enchilada with meat, red-chile or enchilada sauce	2
58100535	Enchilada with meat, green-chile or enchilada sauce	2
58100620	Enchilada with chicken and beans, red-chile or enchilada sauce	0.2
58100625	Enchilada with chicken and beans, green-chile or enchilada sauce	0.2
58100630	Enchilada with chicken, red-chile or enchilada sauce	0.2
58100635	Enchilada with chicken, green-chile or enchilada sauce	0.2
58101820	Mexican casserole made with ground beef, beans, tomato sauce, cheese, taco, seasonings, and corn chips	2
58101830	Mexican casserole made with ground beef, tomato sauce, cheese, taco seasonings, and corn chips	2
58103120	Tamale with meat	2
58103130	Tamale with chicken	0.2
58103310	Tamale casserole with meat	2
58104290	Gordita, sope, or chalupa with meat	2
58104320	Gordita, sope, or chalupa with chicken and sour cream	0.2
58104340	Gordita, sope, or chalupa with chicken	0.2
58104500	Chimichanga with meat	2
58104530	Chimichanga with chicken	0.2
58104535	Chimichanga with meat and sour cream	2
58104550	Chimichanga with chicken and sour cream	0.2

Food code	Description	Amount (mg/g)
58104730	Quesadilla with meat	2
58104740	Quesadilla with chicken	0.2
58104745	Quesadilla with chicken, from fast food	0.2
58104760	Quesadilla with vegetables and meat	2
58104770	Quesadilla with vegetables and chicken	0.2
58104820	Taquito or flauta with meat	2
58104825	Taquito or flauta with meat and cheese	2
58104830	Taquito or flauta with chicken	0.2
58104835	Taquito or flauta with chicken and cheese	0.2
58105000	Fajita with chicken and vegetables	0.2
58105050	Fajita with meat and vegetables	2
58105110	Pupusa, meat-filled	2
58306010	Beef enchilada dinner, NFS, frozen meal	2
58306020	Beef enchilada, chili gravy, rice, refried beans, frozen meal	2
75410530	Chiles rellenos, filled with meat and cheese	2
14620320	Topping from meat pizza	1.666
14620330	Topping from meat and vegetable pizza	1.666
58106512	Pizza with pepperoni, from frozen, thin crust	0.67
58106514	Pizza with pepperoni, from frozen, medium crust	0.67
58106516	Pizza with pepperoni, from frozen, thick crust	0.67
58106540	Pizza with pepperoni, from restaurant or fast food, NS as to type of crust	0.67
58106550	Pizza with pepperoni, from restaurant or fast food, thin crust	0.67
58106555	Pizza with pepperoni, from restaurant or fast food, medium crust	0.67
58106560	Pizza with pepperoni, from restaurant or fast food, thick crust	0.67
58106565	Pizza with pepperoni, stuffed crust	0.67
58106570	Pizza with pepperoni, from school lunch, thin crust	0.67
58106578	Pizza, with pepperoni, from school lunch, medium crust	0.67
58106580	Pizza with pepperoni, from school lunch, thick crust	0.67
58106602	Pizza with meat other than pepperoni, from frozen, thin crust	1.666
58106604	Pizza with meat other than pepperoni, from frozen, medium crust	1.666
58106606	Pizza with meat other than pepperoni, from frozen, thick crust	1.666
58106610	Pizza with meat other than pepperoni, from restaurant or fast food, NS as to type of crust	1.666
58106620	Pizza with meat other than pepperoni, from restaurant or fast food, thin crust	1.666
58106625	Pizza with meat other than pepperoni, from restaurant or fast food, medium crust	1.666
58106630	Pizza with meat other than pepperoni, from restaurant or fast food, thick crust	1.666
58106633	Pizza, with meat other than pepperoni, stuffed crust	1.666

Food code	Description	Amount (mg/g)
58106634	Pizza, with meat other than pepperoni, from school lunch, medium crust	1.666
58106635	Pizza, with meat other than pepperoni, from school lunch, thin crust	1.666
58106636	Pizza, with meat other than pepperoni, from school lunch, thick crust	1.666
58106650	Pizza with extra meat, thin crust	1.666
58106655	Pizza with extra meat, medium crust	1.666
58106660	Pizza with extra meat, thick crust	1.666
58106700	Pizza with meat and vegetables, from frozen, thin crust	1.666
58106702	Pizza with meat and vegetables, from frozen, medium crust	1.666
58106705	Pizza with meat and vegetables, from frozen, thick crust	1.666
58106720	Pizza with meat and vegetables, from restaurant or fast food, thin crust	1.666
58106725	Pizza with meat and vegetables, from restaurant or fast food, medium crust	1.666
58106730	Pizza with meat and vegetables, from restaurant or fast food, thick crust	1.666
58106736	Pizza with extra meat and extra vegetables, thin crust	1.666
58106737	Pizza with extra meat and extra vegetables, thick crust	1.666
58106738	Pizza with extra meat and extra vegetables, medium crust	1.666
58106750	Pizza with meat and fruit, thin crust	1.666
58106755	Pizza with meat and fruit, medium crust	1.666
58106760	Pizza with meat and fruit, thick crust	1.666
58107222	White pizza, cheese, with meat, thin crust	1.666
58107224	White pizza, cheese, with meat, thick crust	1.666
58107232	White pizza, cheese, with meat and vegetables, thin crust	1.666
58107234	White pizza, cheese, with meat and vegetables, thick crust	1.666
58108010	Calzone, with meat and cheese	1.666
58109030	Pizza, with meat, whole wheat thin crust	1.666
58109040	Pizza, with meat, whole wheat thick crust	1.666
58109120	Pizza, with meat, gluten-free thin crust	1.666
58109130	Pizza, with meat, gluten-free thick crust	1.666
27510140	Cheeseburger slider, from fast food	6
27510155	Cheeseburger, NFS	6
27510160	Cheeseburger, from fast food, 1 small patty	6
27510170	Cheeseburger (Burger King)	6
27510171	Whopper Jr with cheese (Burger King)	6
27510172	Cheeseburger (McDonalds)	6
27510190	Cheeseburger, from school cafeteria	6
27510191	Cheeseburger slider	6
27510195	Cheeseburger, on white bun, 1 small patty	6
27510196	Cheeseburger, on wheat bun, 1 small patty	6
27510215	Cheeseburger, from fast food, 1 medium patty	6

Food code	Description	Amount (mg/g)
27510229	Quarter Pounder (McDonalds)	6
27510231	Whopper with cheese (Burger King)	6
27510232	Quarter Pounder with cheese (McDonalds)	6
27510235	Cheeseburger submarine sandwich with lettuce, tomato and spread	6
27510241	Cheeseburger, on white bun, 1 medium patty	6
27510242	Cheeseburger, on wheat bun, 1 medium patty	6
27510245	Cheeseburger, on white bun, 1 large patty	6
27510246	Cheeseburger, on wheat bun, 1 large patty	6
27510254	Double cheeseburger, on white bun, 2 small patties	6
27510255	Double cheeseburger, on wheat bun, 2 small patties	6
27510257	Double cheeseburger, on white bun, 2 medium patties	6
27510258	Double cheeseburger, on wheat bun, 2 medium patties	6
27510261	Cheeseburger, from fast food, 1 large patty	6
27510262	Double cheeseburger, on white bun, 2 large patties	6
27510263	Double cheeseburger, on wheat bun, 2 large patties	6
27510371	Double cheeseburger, from fast food, 2 small patties	6
27510386	Double cheeseburger (Burger King)	6
27510387	Double cheeseburger (McDonalds)	6
27510388	McDouble (McDonalds)	6
27510389	Big Mac (McDonalds)	6
27510401	Double cheeseburger, from fast food, 2 medium patties	6
27510405	Double cheeseburger, from fast food, 2 large patties	6
27510501	Hamburger slider, from fast food	6
27510521	Hamburger, NFS	6
27510531	Hamburger, from fast food, 1 small patty	6
27510551	Hamburger (Burger King)	6
27510552	Whopper Jr (Burger King)	6
27510553	Hamburger (McDonalds)	6
27510565	Hamburger, from school cafeteria	6
27510573	Hamburger slider	6
27510575	Hamburger, on white bun, 1 small patty	6
27510576	Hamburger, on wheat bun, 1 small patty	6
27510601	Hamburger, from fast food, 1 medium patty	6
27510605	Hamburger, from fast food, 1 large patty	6
27510615	Whopper (Burger King)	6
27510631	Hamburger, on white bun, 1 medium patty	6
27510632	Hamburger, on wheat bun, 1 medium patty	6
27510635	Hamburger, on white bun, 1 large patty	6
27510636	Hamburger, on wheat bun, 1 large patty	6

Food code	Description	Amount (mg/g)
27510649	Double hamburger, on white bun, 2 small patties	6
27510652	Double hamburger, on wheat bun, 2 small patties	6
27510655	Double hamburger, on white bun, 2 medium patties	6
27510657	Double hamburger, on wheat bun, 2 medium patties	6
27510658	Double hamburger, on white bun, 2 large patties	6
27510659	Double hamburger, on wheat bun, 2 large patties	6
27510661	Double hamburger, from fast food, 2 small patties	6
27510671	Double hamburger, from fast food, 2 medium patties	6
27510675	Double hamburger, from fast food, 2 large patties	6
27510705	Chiliburger, with or without cheese, on bun	6
27517000	Hamburger wrap sandwich, from fast food	6
27545100	Turkey or chicken burger, on white bun	0.6
27545110	Turkey or chicken burger, on wheat bun	0.6
27560300	Corn dog, frankfurter or hot dog with cornbread coating	2.4
27560350	Pig in a blanket, frankfurter or hot dog wrapped in dough	2.4
27564000	Frankfurter or hot dog sandwich, NFS, plain, on white bun	2.4
27564001	Frankfurter or hot dog sandwich, NFS, plain, on wheat bun	2.4
27564002	Frankfurter or hot dog sandwich, NFS, plain, on whole wheat bun	2.4
27564003	Frankfurter or hot dog sandwich, NFS, plain, on whole grain white bun	2.4
27564004	Frankfurter or hot dog sandwich, NFS, plain, on multigrain bun	2.4
27564010	Frankfurter or hot dog sandwich, NFS, plain, on white bread	2.4
27564020	Frankfurter or hot dog sandwich, NFS, plain, on wheat bread	2.4
27564030	Frankfurter or hot dog sandwich, NFS, plain, on whole wheat bread	2.4
27564040	Frankfurter or hot dog sandwich, NFS, plain, on whole grain white bread	2.4
27564050	Frankfurter or hot dog sandwich, NFS, plain, on multigrain bread	2.4
27564060	Frankfurter or hot dog sandwich, beef, plain, on white bun	2.4
27564061	Frankfurter or hot dog sandwich, beef, plain, on wheat bun	2.4
27564062	Frankfurter or hot dog sandwich, beef, plain, on whole wheat bun	2.4
27564063	Frankfurter or hot dog sandwich, beef, plain, on whole grain white bun	2.4
27564064	Frankfurter or hot dog sandwich, beef, plain, on multigrain	2.4
27564070	Frankfurter or hot dog sandwich, beef, plain, on white bread	2.4
27564080	Frankfurter or hot dog sandwich, beef, plain, on wheat bread	2.4
27564090	Frankfurter or hot dog sandwich, beef, plain, on whole wheat bread	2.4
27564100	Frankfurter or hot dog sandwich, beef, plain, on whole grain white bread	2.4
27564110	Frankfurter or hot dog sandwich, beef, plain, on multigrain bread	2.4
27564120	Frankfurter or hot dog sandwich, beef and pork, plain, on white bun	2.4
27564121	Frankfurter or hot dog sandwich, beef and pork, plain, on wheat bun	2.4
27564122	Frankfurter or hot dog sandwich, beef and pork, plain, on whole wheat bun	2.4

Food code	Description	Amount (mg/g)
27564123	Frankfurter or hot dog sandwich, beef and pork, plain, on whole grain white bun	2.4
27564124	Frankfurter or hot dog sandwich, beef and pork, plain, on multigrain bun	2.4
27564130	Frankfurter or hot dog sandwich, beef and pork, plain, on white bread	2.4
27564140	Frankfurter or hot dog sandwich, beef and pork, plain, on wheat bread	2.4
27564150	Frankfurter or hot dog sandwich, beef and pork, plain, on whole wheat bread	2.4
27564160	Frankfurter or hot dog sandwich, beef and pork, plain, on whole grain white bread	2.4
27564170	Frankfurter or hot dog sandwich, beef and pork, plain, on multigrain bread	2.4
27564180	Frankfurter or hot dog sandwich, meat and poultry, plain, on white bun	1.5
27564181	Frankfurter or hot dog sandwich, meat and poultry, plain, on wheat bun	1.5
27564182	Frankfurter or hot dog sandwich, meat and poultry, plain, on whole wheat bun	1.5
27564183	Frankfurter or hot dog sandwich, meat and poultry, plain, on whole grain white bun	1.5
27564184	Frankfurter or hot dog sandwich, meat and poultry, plain, on multigrain bun	1.5
27564190	Frankfurter or hot dog sandwich, meat and poultry, plain, on white bread	1.5
27564200	Frankfurter or hot dog sandwich, meat and poultry, plain, on wheat bread	1.5
27564210	Frankfurter or hot dog sandwich, meat and poultry, plain, on whole wheat bread	1.5
27564220	Frankfurter or hot dog sandwich, meat and poultry, plain, on whole grain white bread	1.5
27564230	Frankfurter or hot dog sandwich, meat and poultry, plain, on multigrain bread	1.5
27564240	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on white bun	0.6
27564241	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on wheat bun	0.6
27564242	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on whole wheat bun	0.6
27564243	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on whole grain white bun	0.6
27564244	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on multigrain bun	0.6
27564250	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on white bread	0.6
27564260	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on white bread	0.6
27564270	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on whole wheat bread	0.6
27564280	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on whole grain white bread	0.6
27564290	Frankfurter or hot dog sandwich, chicken and/or turkey, plain, on multigrain bread	0.6

Food code	Description	Amount (mg/g)
27564300	Frankfurter or hot dog sandwich, reduced fat or light, plain, on white bun	2.4
27564301	Frankfurter or hot dog sandwich, reduced fat or light, plain, on wheat bun	2.4
27564302	Frankfurter or hot dog sandwich, reduced fat or light, plain, on whole wheat bun	2.4
27564303	Frankfurter or hot dog sandwich, reduced fat or light, plain, on whole grain white bun	2.4
27564304	Frankfurter or hot dog sandwich, reduced fat or light, plain, on multigrain bun	2.4
27564310	Frankfurter or hot dog sandwich, reduced fat or light, plain, on white bread	2.4
27564320	Frankfurter or hot dog sandwich, reduced fat or light, plain, on wheat bread	2.4
27564330	Frankfurter or hot dog sandwich, reduced fat or light, plain, on whole wheat bread	2.4
27564340	Frankfurter or hot dog sandwich, reduced fat or light, plain, on whole grain white bread	2.4
27564350	Frankfurter or hot dog sandwich, reduced fat or light, plain, on multigrain bread	2.4
27564360	Frankfurter or hot dog sandwich, fat free, plain, on white bun	2.4
27564361	Frankfurter or hot dog sandwich, fat free, plain, on wheat bun	2.4
27564362	Frankfurter or hot dog sandwich, fat free, plain, on whole wheat bun	2.4
27564363	Frankfurter or hot dog sandwich, fat free, plain, on whole grain white bun	2.4
27564364	Frankfurter or hot dog sandwich, fat free, plain, on multigrain bun	2.4
27564370	Frankfurter or hot dog sandwich, fat free, plain, on white bread	2.4
27564380	Frankfurter or hot dog sandwich, fat free, plain, on wheat bread	2.4
27564390	Frankfurter or hot dog sandwich, fat free, plain, on whole wheat bread	2.4
27564400	Frankfurter or hot dog sandwich, fat free, plain, on whole grain bread	2.4
27564410	Frankfurter or hot dog sandwich, fat free, plain, on multigrain bread	2.4
27564418	Frankfurter or hot dog sandwich, reduced sodium	2.4
27564440	Frankfurter or hot dog sandwich, with chili, on white bun	1.2
27564441	Frankfurter or hot dog sandwich, with chili, on wheat bun	1.2
27564442	Frankfurter or hot dog sandwich, with chili, on whole wheat bun	1.2
27564443	Frankfurter or hot dog sandwich, with chili, on whole grain white bun	1.2
27564444	Frankfurter or hot dog sandwich, with chili, on multigrain bun	1.2
27564450	Frankfurter or hot dog sandwich, with chili, on white bread	1.2
27564460	Frankfurter or hot dog sandwich, with chili, on wheat bread	1.2
27564470	Frankfurter or hot dog sandwich, with chili, on whole wheat bread	1.2
27564480	Frankfurter or hot dog sandwich, with chili, on whole grain white bread	1.2
27564490	Frankfurter or hot dog sandwich, with chili, on multigrain bread	1.2
27540110	Sliced chicken sandwich, with spread	0.6
27540111	Sliced chicken sandwich, with cheese and spread	0.6

Food code	Description	Amount (mg/g)
27540120	Chicken salad or chicken spread sandwich	0.6
27540130	Chicken barbecue sandwich	0.6
27540132	Chicken fillet sandwich, NFS	0.6
27540139	Chicken fillet sandwich, from school cafeteria	0.6
27540145	Chicken fillet biscuit, from fast food	0.6
27540146	Chicken fillet sandwich, fried, from fast food	0.6
27540147	Chicken fillet sandwich, fried, from fast food, with cheese	0.5
27540152	Chicken fillet sandwich, grilled, from fast food	0.6
27540153	Chicken fillet sandwich, grilled, from fast food, with cheese	0.5
27540160	Chicken fillet sandwich, NS as to fried or grilled, from fast food	0.6
27540175	Chicken fillet sandwich, fried, on white bun	0.6
27540176	Chicken fillet sandwich, fried, on white bun; with cheese	0.5
27540185	Chicken fillet sandwich, fried, on wheat bun	0.6
27540186	Chicken fillet sandwich, fried, on wheat bun, with cheese	0.5
27540195	Chicken fillet sandwich, grilled, on white bun	0.6
27540196	Chicken fillet sandwich, grilled, on white bun, with cheese	0.5
27540205	Chicken fillet sandwich, grilled, on wheat bun	0.6
27540206	Chicken fillet sandwich, grilled, on wheat bun, with cheese	0.5
27540210	Chicken fillet wrap sandwich, fried, from fast food	0.6
27540285	Chicken, bacon, and tomato club sandwich, with lettuce and spread	0.5
27540290	Chicken submarine sandwich, with lettuce, tomato and spread	0.5
27540291	Chicken submarine sandwich, with cheese, lettuce, tomato and spread	0.5
27540295	Buffalo chicken submarine sandwich	0.6
27540296	Buffalo chicken submarine sandwich with cheese	0.5
27540300	Chicken fillet wrap sandwich, grilled, from fast food	0.6
27540310	Turkey sandwich, with spread	3
27540320	Turkey salad or turkey spread sandwich	3
27540350	Turkey submarine sandwich, with cheese, lettuce, tomato and spread	3
27540360	Turkey and bacon submarine sandwich, with lettuce, tomato and spread	3
27540361	Turkey and bacon submarine sandwich, with cheese, lettuce, tomato and spread	3
27520140	Bacon and egg sandwich	0.375
27520170	Bacon on biscuit	0.375
27520250	Ham on biscuit	0.6
27520380	Ham and cheese on English muffin	0.6
27560650	Sausage on biscuit	0.9
27560660	Sausage griddle cake sandwich	0.9
27560670	Sausage and cheese on English muffin	0.9
27560710	Sausage sandwich	0.9

Food code	Description	Amount (mg/g)
32202000	Egg, cheese, ham, and bacon on bun	0.6
32202010	Egg, cheese, and ham on English muffin	0.6
32202020	Egg, cheese, and ham on biscuit	0.6
32202025	Egg, cheese and ham on bagel	0.75
32202030	Egg, cheese, and sausage on English muffin	0.9
32202034	Egg, cheese, and sausage on bun	0.9
32202045	Egg, cheese, and steak on bagel	1.5
32202050	Egg, cheese, and sausage on biscuit	0.9
32202055	Egg, cheese, and sausage griddle cake sandwich	0.9
32202060	Egg and sausage on biscuit	0.9
32202070	Egg, cheese, and bacon on biscuit	0.375
32202075	Egg, cheese, and bacon griddle cake sandwich	0.375
32202080	Egg, cheese, and bacon on English muffin	0.375
32202085	Egg, cheese and bacon on bagel	0.375
32202090	Egg and bacon on biscuit	0.375
32202110	Egg and ham on biscuit	0.6
32202120	Egg, cheese and sausage on bagel	0.9
32202130	Egg and steak on biscuit	1.5
58100010	Burrito, taco, or quesadilla with egg and breakfast meat	1.5
58100013	Burrito, taco, or quesadilla with egg and breakfast meat, from fast food	1.5
58100015	Burrito, taco, or quesadilla with egg, potato, and breakfast meat	1.5
58100017	Burrito, taco, or quesadilla with egg, potato, and breakfast meat, from fast food	1.5
58100020	Burrito, taco, or quesadilla with egg, beans, and breakfast meat	1.5
58104905	Taquito or flauta with egg and breakfast meat	1.5
58127210	Croissant sandwich, filled with ham and cheese	0.6
58127270	Croissant sandwich with sausage and egg	0.9
58127290	Croissant sandwich with bacon and egg	0.375
58127310	Croissant sandwich with ham, egg, and cheese	0.6
58127330	Croissant sandwich with sausage, egg, and cheese	0.9
58127350	Croissant sandwich with bacon, egg, and cheese	0.375
27500100	Meat sandwich, NFS	4.5
27510000	Beef sandwich, NFS	4.5
27510130	Beef barbecue submarine sandwich, on bun	4.5
27510700	Meatball and spaghetti sauce submarine sandwich	4.5
27510910	Corned beef sandwich	4.5
27510950	Reuben sandwich, corned beef sandwich with sauerkraut and cheese with spread	4.5
27511010	Pastrami sandwich	4.5
27513010	Roast beef sandwich	4.5

Food code	Description	Amount (mg/g)
27513020	Roast beef sandwich, with gravy	4.5
27513040	Roast beef submarine sandwich, with lettuce, tomato and spread	4.5
27513041	Roast beef submarine sandwich, with cheese, lettuce, tomato and spread	4.5
27513050	Roast beef sandwich with cheese	4.5
27513060	Roast beef sandwich with bacon and cheese sauce	4.5
27513070	Roast beef submarine sandwich, on roll, au jus	4.5
27515000	Steak submarine sandwich with lettuce and tomato	4.5
27515010	Steak sandwich, plain, on roll	4.5
27515020	Steak and cheese submarine sandwich, with lettuce and tomato	4.5
27515030	Steak and cheese sandwich, plain, on roll	4.5
27515040	Steak and cheese submarine sandwich, plain, on roll	4.5
27515050	Fajita-style beef sandwich with cheese, on pita bread, with lettuce and tomato	4.5
27515070	Steak and cheese submarine sandwich, with fried peppers and onions, on roll	4.5
27515080	Steak sandwich, plain, on biscuit	4.5
27516010	Gyro sandwich (pita bread, beef, lamb, onion, condiments), with tomato and spread	4.5
27520110	Bacon sandwich, with spread	0.375
27520120	Bacon and cheese sandwich, with spread	0.375
27520150	Bacon, lettuce, and tomato sandwich with spread	0.375
27520155	Bacon, lettuce, and tomato submarine sandwich, with spread	0.375
27520156	Bacon, lettuce, tomato, and cheese submarine sandwich, with spread	0.375
27520300	Ham sandwich, with spread	1.8
27520310	Ham sandwich with lettuce and spread	1.8
27520320	Ham and cheese sandwich, with lettuce and spread	1.8
27520340	Ham salad sandwich	1.8
27520360	Ham and cheese sandwich, on bun, with lettuce and spread	1.8
27520370	Hot ham and cheese sandwich, on bun	1.8
27520390	Ham and cheese submarine sandwich, with lettuce, tomato and spread	1.8
27520410	Cuban sandwich, with spread	1.8
27520420	Midnight sandwich, with spread	1.8
27520500	Pork sandwich, on white roll, with onions, dill pickles and barbecue sauce	2.25
27520510	Pork barbecue sandwich or Sloppy Joe, on bun	2.25
27520520	Pork sandwich	2.25
27541000	Turkey, ham, and roast beef club sandwich, with lettuce, tomato and spread	1.125
27541001	Turkey, ham, and roast beef club sandwich with cheese, lettuce, tomato, and spread	1.125
27560000	Luncheon meat sandwich, NFS, with spread	1.8

Food code	Description	Amount (mg/g)
27560110	Bologna sandwich, with spread	1.8
27560120	Bologna and cheese sandwich, with spread	1.8
27560410	Puerto Rican sandwich	1.8
27560500	Pepperoni and salami submarine sandwich, with lettuce, tomato and spread	1.8
27560510	Salami sandwich, with spread	1.8
27560720	Sausage and spaghetti sauce sandwich	1.8
27560910	Cold cut submarine sandwich, with cheese, lettuce, tomato and spread	1.8
27563010	Meat spread or potted meat sandwich	1.8
27550720	Tuna salad sandwich, on bread	2.4
27550730	Tuna salad sandwich, on bread, with cheese	2.4
27550740	Tuna salad sandwich, on bun	2.4
27550745	Tuna salad sandwich, on bun, with cheese	2.4
27550755	Tuna salad wrap sandwich	2.4
28310110	Beef, broth, bouillon, or consomme	10
28310150	Oxtail soup	2
28310230	Meatball soup, home recipe, Mexican style	4.4
28310320	Beef noodle soup, Puerto Rican style	5
28310330	Pho	6
28310420	Beef and rice soup, Puerto Rican style	5.5
28311010	Pepperpot soup	5.5
28311020	Menudo soup, home recipe	5.5
28311030	Menudo soup, canned, prepared with water or ready-to-serve	5.5
28315050	Beef vegetable soup with potato, pasta, or rice, chunky style, canned, or ready-to-serve	5.5
28315140	Beef vegetable soup, home recipe, Mexican style	5.5
28315150	Meat and corn hominy soup, home recipe, Mexican style	5.5
28315160	Italian Wedding Soup	5.5
28317010	Beef stroganoff soup, chunky style, home recipe, canned or ready-to-serve	5.5
28320140	Ham, noodle, and vegetable soup, Puerto Rican style	1.1
28320160	Pork vegetable soup with potato, pasta, or rice, stew type, chunky style	1.1
28320300	Pork with vegetable excluding carrots, broccoli and/or dark-green leafy; soup, Asian Style	1.1
28321130	Bacon soup, cream of, prepared with water	1.375
28331110	Lamb, pasta, and vegetable soup, Puerto Rican style	5.5
28340110	Chicken or turkey broth, bouillon, or consomme	1
28340150	Mexican style chicken broth soup stock	0.7
28340179	Beef broth, less or reduced sodium, canned or ready-to-serve	10
28340180	Chicken or turkey broth, less or reduced sodium, canned or ready to serve	1
28340210	Chicken rice soup, Puerto Rican style	0.55

Food code	Description	Amount (mg/g)
28340220	Chicken soup with noodles and potatoes, Puerto Rican style	0.55
28340310	Chicken or turkey gumbo soup, home recipe, canned or ready-to serve	0.55
28340510	Chicken or turkey noodle soup, chunky style, canned or ready to serve	0.55
28340550	Sweet and sour soup	0.55
28340580	Chicken or turkey soup with vegetables, broccoli, carrots, celery, potatoes and onions, Asian style	0.55
28340590	Chicken or turkey corn soup with noodles, home recipe	0.55
28340600	Chicken or turkey vegetable soup, canned, prepared with water or ready-to-serve	0.55
28340610	Chicken or turkey vegetable soup, stew type	0.55
28340630	Chicken or turkey vegetable soup with rice, stew type, chunky style	0.55
28340640	Chicken or turkey vegetable soup with noodles, stew type, chunky style, canned or ready-to-serve	0.55
28340690	Chicken or turkey vegetable soup with potato and cheese, chunky style, canned or ready-to-serve	0.55
28340700	Bird's nest soup	0.55
28340750	Hot and sour soup	0.55
28340800	Chicken or turkey soup with vegetables and fruit, Asian Style	0.55
28345010	Chicken or turkey soup, cream of, canned, reduced sodium, NS as to made with milk or water	0.55
28345020	Chicken or turkey soup, cream of, canned, reduced sodium, made with milk	0.55
28345030	Chicken or turkey soup, cream of, canned, reduced sodium, made with water	0.55
28345110	Chicken or turkey soup, cream of, NS as to prepared with milk or water	0.55
28345120	Chicken or turkey soup, cream of, prepared with milk	0.55
28345130	Chicken or turkey soup, cream of, prepared with water	0.55
28345160	Chicken or turkey mushroom soup, cream of, prepared with milk	0.55
28350050	Fish chowder	1.375
28351110	Fish and vegetable soup, no potatoes, Mexican style	1.375
28351120	Fish soup with potatoes, Mexican style	1.375
28360100	Meat broth, Puerto Rican style	5.5
41601020	Bean with bacon or ham soup, canned or ready-to-serve	1.1
41601110	Bean and ham soup, chunky style, canned or ready-to-serve	1.1
41601160	Bean and ham soup, canned, reduced sodium, prepared with water or ready-to-serve	1.1
41602010	Pea and ham soup, chunky style, canned or ready-to-serve	1.1
41602030	Split pea and ham soup	1.1
58402010	Beef noodle soup, canned or ready-to-serve	5.5
58402020	Beef dumpling soup, home recipe, canned or ready-to-serve	5.5
58402030	Beef rice soup, home recipe, canned or ready-to-serve	5.5
58402100	Beef noodle soup, home recipe	5.5

Food code	Description	Amount (mg/g)
58403010	Chicken or turkey noodle soup, canned or ready-to-serve	0.55
58403040	Chicken or turkey noodle soup, home recipe	0.55
58403050	Chicken or turkey noodle soup, cream of, home recipe, canned, or ready-to-serve	0.55
58403060	Chicken or turkey noodle soup, reduced sodium, canned or ready-to-serve	0.55
58404010	Chicken or turkey rice soup, canned, or ready-to-serve	0.55
58404030	Chicken or turkey rice soup, home recipe	0.55
58404040	Chicken or turkey rice soup, reduced sodium, canned, prepared with water or ready-to-serve	0.55
58404050	Chicken or turkey rice soup, reduced sodium, canned, prepared with milk	0.55
58404510	Chicken or turkey soup with dumplings and potatoes, home rec ipe, canned, or ready-to-serve	0.55
58404520	Chicken or turkey soup with dumplings, home recipe, canned or ready-to-serve	0.55
58407050	Instant soup, noodle with egg, shrimp or chicken	0.55
58408010	Wonton soup	0.99
72308000	Dark-green leafy vegetable soup with meat, Asian style	5.5
74603010	Tomato beef soup, prepared with water	5.5
74604010	Tomato beef noodle soup, prepared with water	5.5
74604100	Tomato beef rice soup, prepared with water	5.5
75601210	Cabbage with meat soup, home recipe, canned or ready-to-serve	5.5
75651020	Vegetable beef soup, canned, prepared with water, or ready-to-serve	5.5
75651030	Vegetable beef noodle soup, prepared with water	5.5
75651080	Vegetable beef soup with rice, canned, prepared with water or ready-to-serve	5.5
75652010	Vegetable beef soup, home recipe	5.5
75652030	Vegetable beef soup, canned, prepared with milk	5.5
75652040	Vegetable beef soup with noodles or pasta, home recipe	5.5
75652050	Vegetable beef soup with rice, home recipe	5.5
75656060	Vegetable beef soup, chunky style	5.5
53729000	Nutrition bar or meal replacement bar, NFS	20
27141030	Spaghetti sauce with poultry	0.2
27141035	Spaghetti sauce with poultry and added vegetables	0.2
27162040	Spaghetti sauce with meat	2
27162060	Spaghetti sauce with meat and added vegetables	2
14650170	Alfredo sauce with meat	2
14650175	Alfredo sauce with meat and added vegetables	2
14650180	Alfredo sauce with poultry	0.2
14650185	Alfredo sauce with poultry and added vegetables	0.2
95220000	Nutritional powder mix, NFS	20

Food code	Description	Amount (mg/g)
95220010	Nutritional powder mix, high protein, NFS	20
95230010	Nutritional powder mix, protein, soy based, NFS	20
95230020	Nutritional powder mix, protein, light, NFS	20
95230030	Nutritional powder mix, protein, NFS	20
89901000	Bacon, for use with vegetables	2.8
89901002	Ham, for use with vegetables	2.8
89901004	Beef, for use with vegetables	7
89901006	Chicken, for use with vegetables	0.7
89902100	Bacon, for use on a sandwich	0.6
99992230	Breakfast meat as ingredient in omelet	2

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Form Approved: OMB No. 0910-0342; Expiration Date: 08/31/2025
(See last page for OMB Statement)

DEPARTMENT OF HEALTH AND HUMAN SERVICES
Food and Drug Administration

GENERALLY RECOGNIZED AS SAFE *(GRAS)* NOTICE (Subpart E of Part 170)

FDA USE ONLY

GRN NUMBER	DATE OF RECEIPT
001202	Jun 20, 2024
ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
NAME FOR INTERNET	
KEYWORDS	

Transmit completed form and attachments electronically via the Electronic Submission Gateway *(see Instructions)*; OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety *(HFS-200)*, Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission *(Check one)*

☒ New ☐ Amendment to GRN No.______ ☐ Supplement to GRN No. ______

2. ☒ All electronic files included in this submission have been checked and found to be virus free. *(Check box to verify)*

3 Most recent presubmission meeting *(if any)* with FDA on the subject substance *(yyyy/mm/dd)*: ______

4 For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? *(Check one)* ☐ Yes ☐ No If yes, enter the date of communication *(yyyy/mm/dd)*: ______

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier

Name of Contact Person	Position or Title
Teresa Chan	Director of Regulatory Affairs

Organization *(if applicable)*: Impossible Foods Inc

Mailing Address *(number and street)*: 400 Saginaw Dr.

City	State or Province	Zip Code/Postal Code	Country
Redwood City	California	94063	United States of America

Telephone Number	Fax Number	E-Mail Address
4158550347		teresa.chan@impossiblefoods.com

1b. Agent or Attorney *(if applicable)*

Name of Contact Person	Position or Title
Erik Duane Hedrick	Associate Scientist

Organization *(if applicable)*: Burdock Group Consultants

Mailing Address *(number and street)*: 59 Outer Road

City	State or Province	Zip Code/Postal Code	Country
ORLANDO	Florida	32814	United States of America

Telephone Number	Fax Number	E-Mail Address
407-802-1400		ehedrick@burdockgroup.com

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Pichia Pastoris Based Food Ingredient Containing Soy Leghemoglobin (LegH) Protein (LegH Prep)

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway

☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes

Total number of pages

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)* ☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN

☐ b) GRAS Affirmation Petition No. GRP

☐ c) Food Additive Petition No. FAP

☐ d) Food Master File No. FMF

☐ e) Other or Additional *(describe or enter information as above)*

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))* ☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8))

☐ Yes *(Proceed to Item 8*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

LegH Prep is an ingredient to be used as a flavor and nutrient component, which provides iron in plant-derived meat analogue products. This GRAS notification will discuss LegH Prep, which has comparable LegH protein and overall protein content, other macronutrient content, and is manufactured by the same methodology as described for LegH Prep in GRN No. 737 (FDA, 2017b), with improved process modifications. LegH Prep is a nutritive, plant-sourced flavor and nutrient component, intended to be used as an ingredient in meat analogue and meat analogue-containing products, including but not limited to beef (ground and unground), pork, lamb, goat, game, liver and organ meats, chicken, turkey, duck and other poultry, fish, cold cuts and cured meats, bacon,

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☐ No , you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).

☒ PART 3 of a GRAS notice: Dietary exposure (170.235).

☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).

☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).

☒ PART 6 of a GRAS notice: Narrative (170.250).

☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Impossible Foods Inc
(name of notifier)

has concluded that the intended use*(s)* of Pichia Pastoris Based Food Ingredient Containing Soy Leghemoglobin (LegH) Protein (LegH Pr
(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Impossible Foods *(name of notifier)* agrees to make the data and information that are the basis for the conclusion of GRAS status available to FDA if FDA asks to see them; agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

Burdock Group, 859 Outer Rd, Orlando, FL 32814
(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance.The notifying party certifies that the information provided herein is accurate and complete to the best or his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official, Agent, or Attorney	Printed Name and Title	Date (mm/dd/yyyy)
Erik Hedrick Digitally signed by Erik Hedrick DN: cn=Erik Hedrick, o=Burdock Group, ou=Associate Scientist, email=ehedrick@burdockgroup.com, c=US Date: 2024.06.20 10:50:01 -04'00'	Erik Hedrick Ph.D., Associate Scientist	06/20/2024

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	COSM_3667_20281_ImpossibleFoodsInc.pdf	Administrative
	22.IMPO002.02-GRAS_notification_FINAL_06072024.pdf	Administrative
	Abdel-Hafezetal.,1977.pdf	Administrative
	Ahmadetal.,2014.pdf	Administrative
	AlimentariusCodex,2009.pdf	Administrative
	AOACOfficialMethod2015.01heavymetalsinfood.pdf	Administrative
	AOAC-Method-2015.01.pdf	Administrative
	ATCC(2022)KomagataellaphaffiiKurtzman-76273.pdf	Administrative

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Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Aunstrupetal.,1979.pdf	Administrative
	Balamuruganetal.,2007.pdf	Administrative
	BanerjeeandVerma,2000.pdf	Administrative
	Bannonetal.,2003.pdf	Administrative
	Becanaetal.,1995.pdf	Administrative
	Becerril-Garciaetal.,2022.pdf	Administrative
	BretthauerandCastellino,2000.pdf	Administrative
	Cardonaetal.,2018.pdf	Administrative
	CarverandWalker,1995.pdf	Administrative

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Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	CDC-Biosafetycriteria-BMBL5,2009.pdf	Administrative
	Ciofaloetal.,2006.pdf	Administrative
	Commissionregulation(EU)2017-735B.49.pdf	Administrative
	EFSAJournal-2019a--SafetyfoodenzymephospholipaseC.pdf	Administrative
	EFSAJournal-2019b-SafetyoftheproposedamendmentforsteviolglycosidesE960.pdf	Administrative
	EFSAJournal2015-OPTIPHOS6-phytase.pdf	Administrative
	EFSAJournal2016-Ppastorisproductionorganism-3-phytase.pdf	Administrative
	EFSA,2017-updatedQPSlist.pdf	Administrative
	FDA-(1992)FoodsDerivedfromNewPlantVarieties.pdf	Administrative

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Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Francaetal.,2015.pdf	Administrative
	Fraseretal.,2018.pdf	Administrative
	Fraseretal.,2018-supplementaltables.pdf	Administrative
	González-Manceboetal.,2014.pdf	Administrative
	GRAS-No.1025.pdf	Administrative
	GRNNo.204.pdf	Administrative
	GRN204-Ppastorissafetystatements.pdf	Administrative
	GRNNo.355.pdf	Administrative
	GRNNo.540.pdf	Administrative

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Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	GRNNo.632.pdf	Administrative
	GRNNo.667.pdf	Administrative
	GRNNo.940.pdf	Administrative
	ICHguidelinesS2A(1996).pdf	Administrative
	ICHguidelinesS2B(1997).pdf	Administrative
	IFI-2022.pdf	Administrative
	ImafidonandSosulski,1990.pdf	Administrative
	Invitrogen-Ppastoristechnicalinformationsheet-2006.pdf	Administrative
	Jinetal.,2018.pdf	Administrative

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	JointFAO-WHO,2021.pdf	Administrative
	Karbalaeietal.,2020.pdf	Administrative
	Keefe(2018)-GRAS-Notice-GRN-737-Agency-Response-Letter.pdf	Administrative
	Khanaruksombatetal.,2014.pdf	Administrative
	Kosmachevskayaetal.,2021.pdf	Administrative
	Kurtzman2009.pdf	Administrative
	Liuetal.,2015.pdf	Administrative
	OECD407-RepDose28-dayOralToxicityrodents.pdf	Administrative
	OECD408-Repdose90-dayoraltoxstudyinrodents.pdf	Administrative

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	OECD471bacterialreversemutassay.pdf	Administrative
	OECD473-InVitroMammChromAberTest.pdf	Administrative
	OECD487-micronucleus.pdf	Administrative
	Ofori-Antietal.,2008.pdf	Administrative
	OliveiraandDominiques,2001.pdf	Administrative
	ParizaandJohnson,2001.pdf	Administrative
	Prattipatietal.,2020.pdf	Administrative
	Prielhoferetal.,2015.pdf	Administrative
	Regulation(EC)2017-735-wholedocument.pdf	Administrative

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Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Reyesetal.,2021.pdf	Administrative
	Reyesetal-2023.pdf	Administrative
	Ruethersetal.,2018.pdf	Administrative
	Ruethersetal.,2020.pdf	Administrative
	Sanderetal.,2011.pdf	Administrative
	Santosetal.,2011.pdf	Administrative
	Sturmbergeretal.,2016.pdf	Administrative
	Tarantino,2006-GRASNoticeInventory_AgencyResponseLetterGRASNoticeNo.GRN000204.pdf	Administrative
	TaylorandBaumert,2013.pdf	Administrative

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Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Thomasetal-2004.pdf	Administrative
	USDA,2021WWEIA.pdf	Administrative
	Vogletal.,2013.pdf	Administrative
	safderetal.,2018.pdf	Administrative
	FDAcommunication7JUNE2024.pdf	Administrative

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November 18, 2024

Ellen T. Anderson
Regulatory Review Scientist
Division of Food Ingredients
Office of Food Additive Safety
Center for Food Safety and Applied Nutrition
U.S. Food and Drug Administration
Tel: 240-402-1309
Ellen.Anderson@fda.hhs.gov

In re: FDA QUESTIONS ON GRN 001202 AND IMPOSSIBLE FOODS INC. RESPONSES

Dear Dr. Anderson:

Following this salutary statement, please find our responses to FDA's questions concerning GRN001202.

If you have any questions or find our responses to not be as complete as they need to be, please do not hesitate to contact me.

Best regards,

Erik D. Hedrick, Ph.D.
Director of Toxicology, Burdock Group

Copy: Teresa Chan

Administrative

FDA Question 1

1. *Per* 21 CFR 170.225, Part 1 of a GRAS notice must be dated and signed. Dr. Hedrick is listed as the agent to the notifier under the Certification section on page 6 of the notice, but the signature and date fields are blank. Please provide a Certification page that is signed and dated.

   Impossible Foods Inc. Response:

   Please find attached the signed and dated Certification Page that is now included in GRN 001202.

Intended Use

FDA Question 2

2. The notifier states that the intended use of soy leghemoglobin preparation is "substitutive for existing meat analog products, rather than additive in use" and that soy leghemoglobin preparation would be added to "meat analogue and meat analogue-containing products identified herein."
   a. The notifier acknowledges that the intended use of soy leghemoglobin preparation presented in the notice includes an increase in the number and types of meat analogue products that would contain the ingredient and that the maximum use level represents an increase compared to the previously notified uses. Please clarify the statement that the intended use is substitutive for existing uses and would not be expected to be additive.

      Impossible Foods Inc. Response:

      We clarify that the statement that the intended use is substitutive and not additive. The statement above "the maximum use level represents an increase compared to the previously notified uses" refers to the fact that the LegH Prep in this notification (GRN 001202) has a higher inclusion rate (≤2.0%) and a greater number of foods to which LegH Prep will be added as compared to the original LegH Prep that was the subject of GRN No. 737 (≤0.8%).

   b. On pages 5 and 20 of the notice, the notifier lists the types of meat analogue products that would contain soy leghemoglobin and notes that the use is not limited to those listed. Please clarify this statement.

      Impossible Foods Inc. Response:

      On pages 5 and 20, the food categories that are listed are representative but not exhaustive of all the food codes represented in Appendix II. Appendix II is a very large list of all the food codes to which LegH Prep would be added, and spans over 40 pages. Accordingly, listing all the different meat analogue products and food codes present in Appendix II on pages 5 and 20 would not be practical; therefore, we state on pages 5 and 20 "not limited to" as opposed to writing out the whole list of meat analogue products presented in Appendix II.

c. We note that the intended food categories listed in Table 5 (pages 20 and 21) include various mixed dishes, i.e., foods that are mixtures of component foods, as well as foods that would not be expected to contain a meat analogue component. Please clarify whether the intended use levels described in Table 5 are on the basis of the meat analogue component of the food or for the whole mixture.

Impossible Foods Inc. Response:

We clarify that the intended use levels described in Table 5 are on the basis of the meat analogue component of the food and not the whole mixture.

FDA Question 3a

3. Table 5 on page 20 of the notice includes the intended food categories and maximum use levels in which the notifier intends to use soy leghemoglobin preparation. Appendix II of the notice includes the food codes, descriptions, and use levels utilized in the notifier's dietary exposure assessment.

    a. We note that the use levels listed in Appendix II do not always match the maximum use levels listed for the corresponding food categories in Table 5. Please clarify whether the maximum use level considered in the dietary exposure assessment (i.e., the levels listed in Appendix II) are adjusted in some way. If the use levels are adjusted, please discuss the rationale used and provide a citation to any data and information used to determine the resulting use level.

        i. For example, the use levels listed for turkey in Appendix II (food codes starting with 2420*) are in the range of 0.5 to 1 mg/g; however, the maximum use level for 'turkey, duck, and other poultry' in Table 5 is 5 mg/g. Similarly, turkey mixed dishes (food codes starting with 2724*) are 0.2 mg/g whereas, the level in Table 5 for 'poultry mixed dishes' is 1 mg/g. On the other hand, the level for turkey sandwiches (food codes starting with 2754*) is listed as 3 mg/g, which matches the level in Table 5 for 'chicken/turkey sandwiches' (3 mg/g).

Impossible Foods Inc. Response:

The maximum use of 5 mg/g for "turkey, duck, and other poultry" reflects the collective use of turkey, duck, chicken, and other species of poultry that can be used in a myriad of food products, represented by a myriad of food codes, that would consist of turkey, duck, or chicken. On the other hand, 1 mg/g for "poultry mixed dishes" and 3 mg/g for "chicken/turkey sandwiches" reflects the collective use in turkey, duck, and other poultry in all "mixed dishes" and the collective use of turkey and chicken in all "sandwiches", respectively. However, the food code "turkey mixed dishes", which would be included under the food category "poultry mixed dishes", and the food code "turkey sandwiches", which would be included under the food category "chicken/turkey sandwiches" are exclusively composed of turkey. Therefore, the concentrations of turkey in these food codes would be less or within 1 and 3 mg/g, respectively. Both of these uses would also be within the maximum use of 5 mg/g for the food category "turkey, duck, and other poultry", as again this category encompasses the maximum use of turkey, duck, chicken, and other species of poultry in several food code products contained within Appendix II.

It should also be noted that the concentrations of LegH Prep for the respective food codes in Appendix II reflect the maximum intended potential use in that food code, which is why the use of LegH Prep in Table 5 in the listed food categories are listed as ranges (*i.e.*, minimum and maximum intended use levels). The number of consumers associated with each food code and the amount of that food code consumed by those respective consumers varies, and since the use of LegH Prep is substitutive and not additive, the levels of LegH Prep for each respective food code do not exceed the maximum intended use level for the collective use included under the original food category.

FDA Question 3b

b. We note that Table 5 includes intended uses in cookies, brownies, and nutrition bars at up to 20 mg/g; however, Appendix II does not appear to include any food codes for these food categories. Please clarify whether the use of soy leghemoglobin was considered for these foods in the notifier's dietary exposure assessment

Impossible Foods Inc. Response:

Regarding use of LegH Prep in nutrition bars, we kindly direct you to Appendix II on page 120 of GRN 001202. Under Food Code 53729000, it lists "Nutrition bar or meal replacement bar, NFS" with an amount of 20 mg/g. However, for "cookies and brownies" LegH Prep was not added to these foods nor listed in Appendix II. Therefore, it was not considered in the dietary exposure assessment and as a result, we will remove "cookies and brownies" from Table 5.

FDA Question 3c

c. Some of the food categories listed in Table 5 include a descriptor of "(single code)." Please clarify what the term "single code" refers to.

Impossible Foods Inc. Response:

In the WWEIA database, there are certain categories that only use a single food code. "Single code" refers to coding individual components as a combination, as opposed to individual codes for each component. For example, for sandwiches, a single code is used instead of coding individual ingredients separately. These are represented by "Mixed Dishes – Sandwiches (single code)" and include the categories burgers, frankfurter sandwiches, chicken/turkey sandwiches, egg/breakfast sandwiches, other sandwiches, cheese sandwiches, peanut butter and jelly sandwiches, and seafood sandwiches.[1] These single code categories refer to sandwiches that are represented by a single food code, such as a cheeseburger with tomato and/or catsup on a bun., and are therefore referred to as "single-code sandwiches." Other sandwiches are represented by two or more food codes linked as a "sandwich combination".[2] WWEIA Food Categories are intended for use with dietary intake data are designed to be flexible and can easily be regrouped into smaller or larger food groupings as needed to address specific research questions. Each of the food and beverage items that can be reported in WWEIA and National Health and Nutrition Examination Survey (NHANES) are placed in one of the nine food categories: (1) milk and milk products, (2) meat, poultry, fish, and mixtures, (3) eggs, (4) legumes, nuts, and seeds, (5) grain products, (6) fruits, (7) vegetables, (8) fats, oils, and salad dressings, (9) sugars, sweets, and beverages. This is done by linking each food code contained in the Food and Nutrient Database for Dietary Studies (FNDDS) to one WWEIA category. Food categorization is based on grouping similar foods and beverages together based on usage and nutrient content. This classification scheme includes approximately 165 unique categories (which contain discrete food items such as pizza, grains, cheese, tomatoes, *etc.*), with each category possessing a four-digit number and description.

Each FNDDS food code (an eight-digit number) is linked to a unique category, and each food category contains many different unique food codes. Each category contains food codes that uniquely identifies each food/beverage in the FNDDS, which are assigned according to the classification scheme that associates the first digit with one of nine major food commodity groups and the second digit with a more specific subgroup. The remaining six digits in each food code are used to identify the particular foods within a numerical sequence and are associated with even more specific categorization of those food groups.

[1] https://www.ars.usda.gov/ARSUserFiles/80400530/pdf/1718/Food_Category_List.pdf; site visited November 14th, 2024.

[22] https://www.ars.usda.gov/ARSUserFiles/80400530/pdf/0910/sandwich_sodium_WWEIA.pdf; site visited November 14th, 2024.

Manufacturing

FDA Question 4

4. On page 16, the notifier states that the raw materials used in the fermentation and recovery process are standard ingredients used in the food and enzyme industry and are food-grade, high-quality chemical or pharmaceutical grade that meet established specifications (e.g., Food Chemicals Codex, United States Pharmacopeia, National Formulary). Please provide a statement to confirm that all raw materials are approved for their respective use via a regulation in Part 21 of the U.S. Code of Federal Regulations, are the subject of an effective food contact notification, or are GRAS for the intended use.

Impossible Foods Inc. Response:

We confirm that that all raw materials are approved for their respective use *via* a regulation in Part 21 of the U.S. Code of Federal Regulations, are the subject of an effective food contact notification, or are GRAS for the intended use.

Specifications

FDA Question 5

5. On page 9 of the notice, it states, "LegH Prep is standardized to contain between two percent (2%) and fifteen percent (15%) soy leghemoglobin protein and may be stabilized with suitable food-grade substances such as sodium ascorbate or sodium chloride." On the same page, footnote 1 states, "The original GRAS conclusion stated a specification for LegH protein was 6 – 9% in the LegH Prep." The specification for soy leghemoglobin protein content listed in Table 1, page 10 is 6 – 15% (w/w). Please confirm that 2% is a typographical error and should be "6%".

Impossible Foods Inc. Response:

The specification for LegH protein for the LegH Prep that is the subject of this current notification is 2-15% (w/w). Therefore, the 6-15% in Table 1 is the typographical error and the correct specification for LegH protein in Table 1 should be 2-15% (w/w).

FDA Question 6

6. The notifier states that the recovery process step of the manufacturing process includes an optional spray drying step (page 16) that results in a soy leghemoglobin preparation with different compositional specifications that include reduced moisture and increased protein content. The specifications listed in Appendix I on page 85 (PDF page 87) note that the protein content may exceed 15% if additional water is removed; however, no additional

specifications are provided for the optional form of the ingredient. Please clarify how the specifications for the optional form compare with the specifications provided in Appendix I.

Impossible Foods Inc. Response:

We do not currently manufacture a soy leghemoglobin product with a spray drying step. If we were to spray dry to remove all the water, the protein content would be higher due to lack of moisture. Once spray dried material is rehydrated, it would meet the same specifications of the starting material before water was removed.

FDA Question 7

7. Please provide a statement confirming that the methods of analysis used as part of the notifier's specifications for soy leghemoglobin preparation, including internal methods, are validated for the intended purposes.

Impossible Foods Inc. Response:

We confirm that the methods of analysis used as part of the notifier's specifications for soy leghemoglobin preparation, including internal methods, are validated for the intended purposes.

FDA Question 8

8. The notifier provides the results of batch analyses of soy leghemoglobin preparation in Appendix I of the notice. Please state whether the batches tested were consecutive or non-consecutive.

Impossible Foods Inc. Response:

We confirm that the batches tested were nonconsecutive.

FDA Question 9

9. We note that the specification limits for lead and cadmium are listed as ≤0.4 mg/kg and ≤0.2 mg/kg, respectively. The results from the analyses of five batches of the notified substance demonstrated that lead levels were consistently <0.01 mg/kg and levels of cadmium were consistently <0.001 mg/kg. We note that specifications help to ensure that the ingredient is being manufactured in accordance with good manufacturing practices, and we would like to remind the notifier of FDA's recent "Closer to Zero" initiative that focuses on reducing dietary exposure to lead, arsenic, cadmium, and mercury from food. Further,

we request that specifications for heavy metals be as low as possible and be reflective of the results obtained from the batch analyses. Please consider lowering the specifications for these heavy metals.

Impossible Foods Inc. Response:

These are the same lead and cadmium specifications as listed in the FCC monograph and in 21 CFR 73.520 Soy leghemoglobin. We may consider lowering the specifications as part of our efforts across international markets in the future.

FDA Question 10

10. For consistency with other GRAS Notices, please provide a specification and batch test results for yeasts and molds.

Impossible Foods Inc. Response:

As a result of the biological assessment under the Preventive Controls regulation for Good Manufacturing Practices under 21 CFR §117, Impossible Foods Inc. has production process controls in place that significantly reduce the potential for microbial contamination, in conjunction with testing for specific pathogens. Therefore, analysis for Yeast/Mold counts were concluded by Impossible Foods Inc. as unnecessary as product release specifications and are only evaluated as statistical process controls on an "as needed" basis.

Exposure

FDA Question 11

11. We note that on page 21, the units reported for the mean and 90th *percentile* dietary exposure are in g/day in the text and mg/day in Table 6. Please provide a correction that clarifies the units for these estimates.

Impossible Foods Inc. Response:

We have corrected the units for the mean and 90^{th} *percentile* dietary exposure to "mg/day" in the text to be consistent with the units in Table 6.

FDA Question 12

12. For the estimates of dietary exposure presented in Part 3 of the notice, please clarify the population age group that is considered from the 2017-2018 What We Eat in America (WWEIA) National Health and Nutrition Examination Survey (NHANES). We recommend that the notifier provide an estimate of dietary exposure for the population 2 years and older (eaters-only).

Impossible Foods Inc. Response:

The mean and 90th *percentile* estimated daily intake provided in the notification represents the average of multiple age groups ranging from 1-2 years all the way to 70+ of age. The consumption analysis we conducted based on the WWEIA and NHANES data also provides stratified data for these age groups, and the mean and 90th *percentile* level of consumption for 1–2-year-olds is 20.47 and 40.94 mg/kg/day. This 90th *percentile* level of consumption is below the calculated ADI of 63.97 mg/kg bw/day based on the 75-fold safety factor and the NOAEL of the 90-day study conducted by Impossible Foods Inc. of 4798 mg/kg bw/day. However, even if a 100-fold safety factor was used, which would yield an ADI of 47.98 mg/kg bw/day, the 90th *percentile* level of consumption of 1–2-year-olds would fall below this ADI.

FDA Question 13

13. The notifier states that the intended use of soy leghemoglobin preparation is as a flavor and as a source of iron. Please provide estimates of dietary exposure to iron that would be expected to result from the intended uses of soy leghemoglobin preparation.

Impossible Foods Inc. Response:

The lots representing the levels of iron for LegH Prep are provided below in Table 1.

Table 1. Iron concentrations for LegH Prep lots

			Batch Analysis Results					
Analysis	**Method**	**Specification**	**LH-20-147-148-190FG**	**LH-20-149-150-190FG**	**LH-20-151-152-190FG**	**LH-20-163-164-190FG**	**LH-21-177-178-190FG**	**AVG**
Iron (*w/w*)	ICP MS[3]	N/A	223 ppm	190 ppm	307 ppm	326 ppm	239 ppm	257 ppm

Based on the above lots, the average iron concentration is 257 ppm. Therefore, if the average level of iron in LegH Prep is assumed, and the 90th *percentile* intake of LegH Prep (1788.0 mg/day) is assumed, this would result in the maximum estimated intake of iron of 0.46 mg/day, or 0.00765 mg/kg bw/day for a 60 kg individual.

Toxicology

FDA Question 14

14. The notifier states that soy leghemoglobin preparation is used as a flavor and nutrient component that provides iron. However, there is no discussion of iron included in the notice. Include a discussion regarding the safety of iron relative to the intended uses of soy

[3] Inductively coupled plasma mass spectrometry

leghemoglobin preparation. Ensure that levels of iron are within the limits recommended by the Institute of Medicine (IOM).

Impossible Foods Inc. Response:

As determined above in the response to Question 13, the maximum intake of iron from LegH Prep would be 0.46 mg/day. The Recommended Daily Allowance (RDA) for iron recommended by the IOM, based on age groups and gender, are provided in Table 2.

Table 2. Recommended Dietary Allowances for Iron[4]

Age	Male (mg)	Female (mg)	Pregnancy (mg)	Lactation (mg)
Birth-6 months	0.27	0.27	NP	NP
7-12 months	11	11	NP	NP
1-3 years	7	7	NP	NP
4-8 years	10	10	NP	NP
9-13 years	8	8	NP	NP
14-18 years	11	15	27	10
19-50 years	8	18	27	9
51+ years	8	8	NP	NP

According to the IOM, the Recommended Daily Allowance (RDA) for men of all age groups and postmenopausal women is 8 mg/day and for premenopausal women it is 18 mg/day. The median dietary Intake of iron is approximately 16-18 and 12 mg/day for men and women respectively.

Regarding dietary iron intake from food, in children (2-11 years) the average iron intake from foods is 9.7–14.2 mg/day, in children and teens (12-19 years) the average dietary intake 13.3-15.8 mg/day, and 15.5–16.3 mg/day in men (≥19 years) and 12.0–13.9 mg/day in women (≥19 years).[5] When considering the combined intake of iron from food and supplements, the average iron intake is 12.7–15.1 mg/day in children (2–11 years), 16.3 mg/day in children and teens (12–19 years), and 19.3–20.5 mg/day in men (≥19 years) and 17.0–18.9 mg/day in women (≥19 years). The median dietary iron intake in pregnant women is 14.7 mg/day. The maximum level of iron intake from LegH Prep (0.46 mg/day) is well within the dietary intake of iron already consumed from food and supplements from all groups.

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[4] Institute of Medicine. Food and Nutrition Board. Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc : a Report of the Panel on Micronutrients. Washington, DC: National Academy Press; 2001. https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/#en28; site visited on November 5th, 2024.

[5] U.S. Department of Agriculture, Agricultural Research Service. What We Eat in America, 2017-2020. 2024. https://www.ars.usda.gov/ARSUserFiles/80400530/pdf/1720/tables_1-56_2017-March%202020.pdf; site visited November 5th, 2024.

The Tolerable Upper Daily Intake Level (UL) for iron based on age groups and gender are provided in Table 3.

Table 3.Tolerable Upper Intake Levels (ULs) for Iron

Age	Male (mg)	Female (mg)	Pregnancy (mg)	Lactation (mg)
Birth-6 months	40	40	NP	NP
7-12 months	40	40	NP	NP
1-3 years	40	40	NP	NP
4-8 years	40	40	NP	NP
9-13 years	40	40	NP	NP
14-18 years	45	45	45	45
19+ years	45	45	45	45

According to the IOM, the UL for children (≥13 years) was 40 mg/day, and for adolescents (14-18 years) and adults (19+ years, both sexes) is 45 mg/day. The maximum level of iron intake from LegH Prep (0.46 mg/day) is well within the RDA and UL of iron for both sexes of all stages of life. Taken together, these results demonstrate that exposure to iron from LegH Prep up to the 90th *percentile* level of consumption are within the limits recommended by the IOM and are safe to consume.

FDA Question 15

15. Throughout the notice, soy leghemoglobin preparation is labelled in different ways. The notice includes "LegH Prep", "original LegH Prep", and "LegH Prep 2.0". It is unclear throughout the notice which ingredient is being discussed due to the inconsistently used terms. Please provide clear and consistent explanations for each of the terms used to reference ingredients and maintain consistent use of the terms throughout the notice.

Impossible Foods Inc. Response:

The "original LegH Prep" refers to the LegH Prep that was subject to GRN No. 737, whereas "LegH Prep" or the "LegH Prep described within this notification" refers to the LegH Prep that is the subject of this notification: GRN 001202. The LegH Prep described in this notification (GRN 001202) has not changed in terms of how it is manufactured and has comparable composition, specifications, and stability as the LegH Prep described in GRN No. 737. The only thing that has changed with "LegH Prep described within this notification", with respect to the "original LegH Prep" described in GRN No. 737, is the number and types of meat analog products into which LegH Prep is to be added and the maximum amount that is to be added to these products (≤2.0% for GRN 001202 as opposed to ≤0.8% in GRN No. 737). The LegH Prep described in this notification (GRN 001202) has passed all microbial and heavy metal specifications, and all the compositional specifications, which were also included for the "original LegH Prep" in GRN No. 737. Compositional parameters were also comparable to those in the LegH Prep that was the subject of GRN No. 737 (please see Table 3 on Page 14 of GRN 001202).

LegH Prep 2.0 was the subject of a self-GRAS that was not notified and was composed in 2020. LegH Prep 2.0 was produced by the same production organism, *Pichia pastoris*, that was used for the "original LegH Prep" (GRN No. 737) and the LegH Prep described within this notification (GRN 001202). LegH Prep 2.0 exhibits a higher *P. pastoris* content within the final formulation as compared to the LegH prep described in GRN No. 737 and the current LegH Prep contained in GRN 001202. To ensure complete inactivation of any potential growing *P. pastoris* cells within LegH Prep 2.0, an additional heat treatment step is used in the LegH Prep 2.0 manufacturing process. However, the rest of the manufacturing process utilizes the same raw materials and manufacturing steps as described in GRN No. 737 and this notification (GRN 001202). Impossible Foods Inc. conducted allergenicity studies on LegH Prep 2.0 (Reyes *et al.*, 2021), which also included the methods utilized by the study conducted by Jin *et al.* (2018) in GRN No. 737, to compare proteins and composition of LegH Prep 2.0 to the "original LegH Prep" described in GRN No. 737. Although it is a less purified version of the original LegH Prep and therefore contains more *P. pastoris* components, it is comparable in composition and macronutrient content, contains comparable levels of LegH protein (≥ 65%) and overall protein content, and utilizes the same production organism as described in GRN No. 737 (and as described for the LegH Prep that is the subject of this GRAS notification, GRN 001202). Therefore, this study was utilized and was relevant for establishing the lack of allergenic potential in this LegH Prep (GRN 001202).

FDA Question 16

16. On page 73 (PDF page 75), the notifier states they used "a 75-fold safety factor was deemed acceptable when calculating the ADI". Previous designation as GRAS is not a sufficient reason for using a 75-fold safety factor. Please justify why a 75-fold safety factor was used, and also justify how the 75-fold safety factor could replace (i.e., is acceptable) the default 100-fold safety factor.

Impossible Foods Inc. Response:

The 75-fold safety factor was originally proposed by the FDA during Impossible Foods' Pre-submission consultation with FDA regarding expanding the GRAS and color additive uses of soy leghemoglobin on Nov 15, 2022. The following bullet point is taken directly from the FDA's Memorandum of Meeting:

> "FDA does not make recommendations on what studies should be conducted to support a GRAS conclusion since GRAS conclusions belong to the notifier. If Impossible Foods decides to use a 90-day toxicology study to support the safety of

the expanded GRAS uses, then FDA suggested Impossible Foods consider a safety margin of at least 75-fold."

[Remainder of this page is blank]

1.9 Certification

The undersigned authors of this document— a notice of the GRAS conclusion for the use of soy leghemoglobin preparation (LegH Prep)—hereby certify that, to the best of their knowledge, this GRAS notice is based on a complete, representative, and balanced submission that includes both favorable and unfavorable information, that are pertinent to the evaluation of the safety and GRAS status of LegH Prep under the condition of its intended use.

[redacted]

18 November 2024

Erik Hedrick, Ph.D. – Agent to the Notifier
Associate Scientist, Burdock Group
859 Outer Road
Orlando, FL 32814

Date

From:	Erik Hedrick
To:	Anderson, Ellen
Subject:	[EXTERNAL] RE: Follow-up questions for GRN 001202
Date:	Thursday, February 27, 2025 3:44:34 PM
Attachments:	image004.png image001.png

Good afternoon, Ellen.

Please find below our responses to FDA's additional follow-up questions for GRN 001202:

1. The batch analyses data provided was on the hydrated LegH preparation. The dried preparation was used in the EDI calculations in the GRAS notice to determine how much LegH Prep would need to be added to achieve the desired concentrations of LegH Protein in the selected food categories at the selected amounts in each food category (please refer to pgs. 20-22 of GRN 001202 "Part 3: Dietary Exposure 3.1 Estimated Daily Intake). The NOAEL derived from the 90-day study utilized the dried LegH Prep. Therefore, to achieve the desired amounts of LegH protein within the selected foods provided in Table 5 of GRN 001202, and to get an accurate estimation of the EDI for the LegH Prep, the percentage of LegH protein within the dried solids (which is approximately 17% of the LegH Prep) was used and considered.

   Regarding the discrepancy with the dietary exposure of iron from the 90th *percentile* level of consumption of LegH protein, this was discussed internally, and we utilized the calculations provided to FDA from the notifier's CAP submission for determining the amount of iron derived from LegH protein. LegH protein has a MW of about 16kD, with each LegH molecule containing one heme molecule, which contains one iron molecule.

   Therefore, the amount of iron in LegH is calculated as:

   Fe/LegH = 55.845g/mol/16,000 g/mol = 0.349% = 0.349 g/100g =0.00349g/g = 3.49 mg/g. This corresponds to 1 g of LegH protein containing approximately 3.5 mg iron. Therefore, based on 1.788 g of LegH protein/day at the 90th *percentile* level of consumption (as stated in GRN 001202), the 90th *percentile* level of iron consumption from LegH protein contained within LegH Prep would be 6.258 mg/person/day.

   As stated previously, this level of iron consumption is well under the Tolerable Upper Intake of Level (UL) for iron established by the IOM (i.e., 40-45 mg iron/p/day) and within the Recommended Daily Allowance (RDA) for iron (7-11mg iron/p/day) for all non-infant age groups and sexes. Therefore, the 90th *percentile* dietary exposure to iron from LegH protein contained within LegH Prep is safe to consume for the general population under the intended conditions of use of LegH Prep.

2. The notifier agrees to lower the specification of lead and cadmium to <0.05 ppm, to be in line with the other heavy metal specifications. However, they would like to know when they would have to have these specifications put in place in real time. Based on their current results, the proposed specifications are feasible, but it will require time to formally implement changes at their manufacturing site. The notifier is requesting that they be formally implemented by the end of the year. Please let us know if FDA is amenable to this proposed time to implement the proposed specifications in their manufacturing.

Please let me know if you have any other questions or need further clarification.

Best Regards,

Erik Hedrick, **Ph.D.**
Executive Vice President
Director of Toxicology
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E: ehedrick@burdockgroup.com

From: Anderson, Ellen <Ellen.Anderson@fda.hhs.gov>
Sent: Monday, February 24, 2025 8:41 AM
To: Erik Hedrick <ehedrick@BurdockGroup.com>
Subject: RE: Follow-up questions for GRN 001202

Good morning Erik,

We have two additional follow-up questions for GRN 001202 noted below:

1. Please confirm that the batch analyses data provided in the prior amendment dated November 18, 2024, for the iron content of soy

leghemoglobin preparation is on the dry matter basis and not for the hydrated preparation. If the measured levels of iron content are on the dry matter basis, please provide a rationale for why the iron levels are much lower than expected. The reason for the questions on dietary exposure to iron from the intended use of soy leghemoglobin is that our understanding from previous evaluations is that each molecule of soy leghemoglobin protein (soy legH) contains one molecule of heme and therefore, one molecule of iron. Therefore, based on the respective molecular weights (~16 kiloDaltons* and 55.845 Daltons) and the 90th percentile estimate of dietary exposure to the soy leghemoglobin protein (1788 mg/person (p)/day (d)), we estimated the resulting dietary exposure to iron to be approximately 6 mg/p/d (see below). Conversely, in your response on February 5, 2025, the estimated dietary exposure to iron was determined to be 0.67 to 1.0 mg/p/d based on the average and highest measured levels of iron in the preparation. Please clarify this discrepancy.

$$\frac{1.788 \text{ g soy legH}}{\text{p/d}} \times \frac{1 \text{ mole soy legH}}{16{,}000 \text{ g soy legH}} \times \frac{1 \text{ mole iron}}{1 \text{ mole soy legH}} \times \frac{55.845 \text{ g iron}}{1 \text{ mole iron}} = 0.0062 \frac{\text{g iron}}{\text{p/d}}$$

2. In your amendment dated November 18, 2024, for question 9 regarding specifications for heavy metals in soy leghemoglobin preparation, you responded that the specifications for lead (<0.4 mg/kg) and cadmium (<0.2 mg/kg) are the same as those listed in the color additive regulation (21 CFR 73.520) for soy leghemoglobin and that the notifier may consider lowering the specifications in the future. We anticipate that the notifier may seek to amend this regulation in the future to include the new intended food categories and use levels for use as a color additive. We strongly encourage the notifier to reduce the specifications for these metals as these changes will be needed at that time. We reiterate that specifications help to ensure that an ingredient is being manufactured in accordance with good manufacturing practices, and that FDA has a "Closer to Zero" initiative focused on reducing dietary exposure to lead, arsenic, cadmium, and mercury from food.

*Muhammad Ijaz Ahmad, Shahzad Farooq, Yasmin Alhamoud, Chunbao Li, Hui Zhang, Soy Leghemoglobin: A review of its structure, production, safety aspects, and food applications, Trends in Food Science & Technology, Volume 141, 2023, 104199. https://doi.org/10.1016/j.tifs.2023.104199.

We would appreciate a response at your earliest convenience.

Thank you,
Ellen

From: Anderson, Ellen
Sent: Wednesday, February 5, 2025 4:00 PM
To: 'Erik Hedrick' <ehedrick@BurdockGroup.com>
Subject: RE: [EXTERNAL] RE: Follow-up questions for GRN 001202

Hello Erik,

Thank you, confirming receipt. I've passed this along to our review team. I'll be in touch if additional clarification is needed.

Sincerely,
Ellen

From: Erik Hedrick <ehedrick@BurdockGroup.com>
Sent: Wednesday, February 5, 2025 2:42 PM
To: Anderson, Ellen <Ellen.Anderson@fda.hhs.gov>
Subject: [EXTERNAL] RE: Follow-up questions for GRN 001202

Hello Ellen,

Thank you for your email.

FDA's proposed estimate for the 90th *percentile* dietary exposure to the preparation (LegH Prep), provided in your email below, represents exposure from the ***hydrated*** LegH Prep (*i.e.*, dry solids and water). However, the ***dried solids*** within the LegH Prep (*i.e.*, ***dried*** LegH Prep), which was used in the 90-day study to achieve the NOAEL, contains the protein and iron, as the remainder of the LegH Prep in the ***hydrated*** form is just water. The dried solids represent on average 17% of the hydrated LegH Prep, with the dried solids ranging from 14.6-20.1% of the hydrated LegH Prep. Therefore, using the average concentration of the dried solids from the LegH Prep, the 90th *percentile* dietary exposure to **dried** LegH Prep would be equivalent to **<u>2.6 g/p/day</u>, not 15.3 g/p/day**. Moreover, if we use the range of the concentration of the dry solids (*i.e.*, 14.6-20.1%), the 90th *percentile* dietary exposure to dried LegH Prep would range from **2.23-3.07 g/p/day**. As a result, the average iron content of 257 mg/kg in dried LegH Prep would yield a 90th *percentile* dietary exposure of **0.67 mg iron/p/d,** and the highest measured iron content of 326 mg/kg (batch LH-20-163-164-190FG) would result in a 90th *percentile* dietary exposure of **0.85 mg iron/p/d**. Moreover, if the highest concentration of dried solids (*i.e.*, 20.1%) and highest measured iron content (*i.e.*, 326 mg/kg) are utilized, this would yield a 90th *percentile* dietary exposure to iron of **1.0 mg iron/p/d.**

If the Tolerable Upper Intake of Level (UL) of iron for all ages and sexes, as designated by the IOM, ranging from 40-45 mg/day is considered (see Table 1 below), the highest 90th *percentile* dietary exposure of iron (i.e., 1.0 mg iron/p/day) from dried LegH Prep is well under the UL for iron.

Table 1. Tolerable Upper Intake Levels (ULs) for Iron

Age	Male	Female	Pregnancy	Lactation

	(mg)	(mg)	(mg)	(mg)
Birth-6 months	40	40	NP	NP
7-12 months	40	40	NP	NP
1-3 years	40	40	NP	NP
4-8 years	40	40	NP	NP
9-13 years	40	40	NP	NP
14-18 years	45	45	45	45
19+ years	45	45	45	45

1. mg iron/p/day is also within the Recommended Daily Allowance (RDA) for iron, as recommended by the IOM, for all non-infant age groups and sexes (see Table 2 below).

Table 2. Recommended Dietary Allowances for Iron[1]

Age	Male (mg)	Female (mg)	Pregnancy (mg)	Lactation (mg)
Birth-6 months	0.27	0.27	NP	NP
7-12 months	11	11	NP	NP
1-3 years	7	7	NP	NP
4-8 years	10	10	NP	NP
9-13 years	8	8	NP	NP
14-18 years	11	15	27	10
19-50 years	8	18	27	9
51+ years	8	8	NP	NP

However, even if the 90th *percentile* dietary exposure amounts calculated (by FDA) in your email below are utilized (i.e., ranging from 3.9-5.0 mg iron/p/day), this is still well under the UL for iron established by the IOM (i.e., 40-45 mg iron/p/day) and within the RDA for iron (7-11mg iron/p/day) for all non-infant age groups and sexes. Therefore, the 90th *percentile* dietary exposure to iron from LegH Prep (regardless of what upper limit of estimated iron intake from consumption of the 90th *percentile* level of LegH Prep is utilized) is safe to consume for the general population under the intended conditions of use of LegH Prep.

Please let me know if you require further clarification and/or have any additional questions.

Best Regards,

Erik Hedrick, Ph.D.

Executive Vice President
Director of Toxicology
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From: Anderson, Ellen <Ellen.Anderson@fda.hhs.gov>
Sent: Tuesday, February 4, 2025 10:51 PM
To: Erik Hedrick <ehedrick@BurdockGroup.com>
Subject: Follow-up questions for GRN 001202

Hello Erik,

Our review team respectfully requests the following additional information for GRN 001202:

The estimates of dietary exposure to iron from the intended uses of soy leghemoglobin provided in the amendment dated November 18, 2024, (attached) were reported to be 0.46 mg/person/day (mg/p/d) (0.00765 mg/kg body weight (bw)/d) and were based on the average iron content of five batches of soy leghemoglobin preparation and the estimated 90th percentile dietary exposure to soy leghemoglobin protein (1788 mg/p/d) provided in the notice. We note that the preparation is, on average, 11.68% protein (Table 1, p. 10); therefore, the 90th percentile dietary exposure to the preparation would be equivalent to approximately 15.3 g/p/d. An average iron content of 257 mg/kg in the preparation would lead to a 90th percentile dietary exposure of 3.9 mg iron/p/d and the highest measured iron content of 326 mg/kg (batch LH-20-163-164-190FG) would result in a 90th percentile dietary exposure of 5 mg iron/p/d.

Please confirm whether this interpretation of the information is correct. Also, please compare your updated iron intake estimate, particularly the 90th percentile, with IOM-recommended upper limit of iron intake to ensure that the safety conclusions do not change.

Sincerely,
Ellen
Ellen Anderson (she/her/hers)

Regulatory Review Scientist

Regulatory Management Branch
Division of Food Ingredients
Office of Pre-Market Additive Safety
Office of Food Chemical Safety, Dietary Supplements, and Innovation
Human Foods Program
U.S. Food and Drug Administration
Tel: 240-402-1309
ellen.anderson@fda.hhs.gov

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[1] Institute of Medicine. Food and Nutrition Board. Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc : a Report of the Panel on Micronutrients. Washington, DC: National Academy Press; 2001. https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/#en28; site visited on November 5th, 2024.